DELPHIX CONTINUOUS COMPLIANCE · DISCOVERY POLICY

# Ecuador — Ley Orgánica de Protección de Datos Personales

Discovery and masking of Ecuadorian personal data: cédula and RUC with a valid check digit, quasi-identifiers, the sensitive data of article 4 — which names the cultural identity, the migratory status and the judicial record no neighbouring law does —, health data, credit data and criminal proceedings.

| 82 | 143 | 107 |
|---|---|---|
| DATA CLASSES | CLASSIFIERS | ALGORITHMS |

Discovery policy `EC – LOPDP – v3`

Version 3 of October 12, 2026 · Assignment threshold 60%

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The first sections explain what the policy is for and how it works. The data class chapters then describe every data class: what it is, how its data is found and how it is masked, with examples. The appendices list every algorithm, file and classifier the policy contains.


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# About this discovery policy

This discovery policy finds the personal data in an Ecuadorian database and masks it, so that a non-production copy — for testing, development, training or analysis — no longer identifies anyone.

Masking names and cédulas alone is not enough: a birth date, a sex and a small parroquia already point at one person. The policy works in the layers where the risk is:

- **Direct identifiers** — cédula and RUC, foreigner documents, passport, civil registry, IESS and pension numbers, student code, professional registration, names, contact, address, accounts, cards, devices, cookies, plates, property, free text.
- **Quasi-identifiers** — birth date, age, sex, parroquia and cantón with their DPA code, postal code, ciudadela, marital status, occupation, employer, education.
- **Sensitive data** — those of article 4: etnia, **identidad cultural**, identidad de género, religión, ideología, filiación política, **condición migratoria**, orientación sexual, health, biometric and genetic data, and the data of stateless people and refugees the same article names.
- **Health data** — sensitive by article 4 and a special category by article 25 c): clinical record, insurance, ICD-10, procedures, medicines, HIV, blood group, disability — a special category of its own under article 25 d) —, occupational, reproductive and mental health.
- **Credit data** — the *datos personales crediticios* articles 28 and 29 give a regime of their own, plus debt, income, assets, pensions, social programmes, the Registro Social and housing tenure.
- **Criminal records** — the **pasado judicial** article 4 lists among the sensitive data, and court file numbers.

In Data Control Tower the policy is one discovery policy with its classifiers, the data classes they vote for, the algorithms those data classes name — and the algorithms those rely on — and the lookup files they read. The appendices list each of them with its full configuration.

The assignment threshold is 60%. An unambiguous column name reaches it on its own; an ambiguous one — `NOMBRE`, `DOCUMENTO`, `EXPEDIENTE`, `CARGO` — needs the values to agree. A column with a meaningless name (`CAMPO1`, `OBS`) is still found when its values have a recognizable shape or vocabulary: a cédula, a thirteen-digit RUC, a mobile starting 09, an address with ciudadela, manzana and villa, a plate, a parroquia.

Column names are recognized in Spanish, in the abbreviations of Ecuadorian systems (`NUM_DOC`, `FEC_NAC`, `DPA`, `RUC`, `IESS`, `AMIE`) and in English.

> Before masking, inventory the binary columns (BLOB, images, PDF). Onboarding processes often keep a photo of the cédula — with the fingerprint and the signature — and a validation selfie. Report those columns as residual risk.

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# The LOPDP in brief

The **Ley Orgánica de Protección de Datos Personales** was published in the Fifth Supplement of Official Register 459 of **26 May 2021**, and its **Reglamento General** in Official Register Supplement 435 of **13 November 2023**. The sanctions regime has been applicable since 26 May 2023. The law is the closest in the region to the European model, and it takes its definitions from it.

What matters for a copy of production data:

- **Personal data** is the datum that identifies or makes identifiable a natural person, directly or indirectly (art. 4). The **titular** is always a natural person: a company is not a data subject.
- **Sensitive data** is the longest list in the region (art. 4): etnia, identidad de género, **identidad cultural**, religión, ideología, filiación política, **pasado judicial**, **condición migratoria**, orientación sexual, salud, datos biométricos, datos genéticos, and the data of stateless people and refugees who need international protection — closing with an open clause for anything whose improper processing may give rise to discrimination.
- **Special categories** (art. 25) are four: sensitive data, the data of **children and adolescents**, health data, and the data of **persons with disabilities and of their substitutes**. Article 26 forbids processing sensitive data unless one of the grounds it lists applies.
- **Anonymization** is the application of measures that prevent the identification or **re-identification** of a natural person without disproportionate effort (art. 4). **Pseudonymization** is processing after which the data can no longer be attributed to a subject without using additional information, provided that information is kept apart and under technical and organizational measures (art. 4) — the European pair, word for word.
- **Where they appear.** The law does **not apply** to anonymized data, so long as the subject cannot be identified; as soon as the data stops being dissociated or anonymous, its processing falls back under the law (art. 2 c). Article 37 lists **anonymization, pseudonymization or encryption** first among the security measures. Article 31.2 asks that health data be **previously anonymized or pseudonymized whenever possible**.
- **Security.** A breach is notified to the Authority and to ARCOTEL as soon as possible and at the latest within **five days**; a processor notifies the controller within two (art. 43). Articles 39 to 42 ask for protection by design and by default, risk analysis and an impact assessment.

**Credit data have a regime of their own.** Article 4 defines the *datos personales crediticios* as those that make up the economic behaviour of a natural person; article 28 allows processing them to analyse solvency and **forbids any secondary purpose**, and forbids communicating them once five years have passed since the obligation became due.

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# Anonymization, pseudonymization and non-production copies

Ecuador writes the two concepts the way the European regulation does. **Anonymization** prevents identification **and re-identification** without disproportionate effort, and takes the data out of the law (art. 2 c). **Pseudonymization** accepts that the data can be attributed to its subject again *with additional information*, and for that reason requires that information to be kept apart and under technical and organizational measures.

The algorithms of this policy are deterministic for the Delphix masking key: the same cédula masks the same way in every table, which keeps joins working. That key is exactly the *additional information* article 4 speaks of. The masked copy is **pseudonymized**, not anonymized, and remains personal data for whoever holds the key and production.

- Keep the engine key apart from the copy and under technical and organizational measures — that is what article 4 asks for pseudonymization to count, and article 37 counts it as a security measure.
- For health data, article 31.2 asks for anonymization or pseudonymization *whenever possible*: a masked copy of a clinical database is the ordinary way to comply with it.
- If what you need is anonymized data — out of the law by article 2 c) —, measure the re-identification risk of the quasi-identifiers that remain; the standard of article 4 is higher than this policy meets.

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# How sensitive data is discovered

A discovery job looks at every column of its rule set and asks each classifier of the discovery policy how confident it is that the column belongs to its data class. The policy uses the four classifier frameworks of Delphix:

- **Column name** (Path) — regular expressions on the column name. Each rule has a match strength: the confidence it gives when it matches.
- **Data type** (Type) — accepts the column type and length, or rules the data class out. It never assigns a data class on its own.
- **Value format** (Regex) — tests a sample of the values against patterns, some with a check digit. The confidence follows the share of values that match.
- **Known values** (List) — looks the sampled values up in lists; for names, word by word.

Evidence from the name and from the values is combined per data class, and the column is assigned to the data class with the highest confidence when it reaches the assignment threshold of the discovery policy — **60%** here. Values that do not match a format or a list count against their data class by the classifier's reject strength.

These are the names Data Control Tower gives the objects: data classes, classifiers and a discovery policy, run by a discovery job. A Continuous Compliance Engine used on its own calls the same objects domains, classifiers and a profile set, run by a profiling job; everything in this document applies to both.

Databases pad values: a CHAR column fills them with spaces up to its length, and a numeric code kept as text often comes with zeros on the left. Discovery tests values as they come, so every format pattern first removes the spaces around the value, patterns of digits also accept zeros on the left, lists of single words are looked up word by word, and lists of names made of several words (places, categories) are held as a single value-format pattern. A column whose name says nothing — `CAMPO1`, `FLD_07`, the short names of packaged systems — is found by its values alone wherever their format is distinctive: documents with a check digit, e-mail addresses, telephone numbers, cards, places, names.

Column-name rules are shown as their alternatives. They ignore case and must not be glued to other letters: `rut` finds `RUT_CLIENTE` and `CLIENTE_RUT` but not `RUTA`. In them, `_?` is an optional underscore, `(a|b)` either form, `s?` an optional letter and `[a-z_]*` any continuation.

> Discovery proposes; it does not decide. Review the data classes assigned in the rule set before running a masking job, and look for columns the policy left unassigned.

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# How data is masked

Each data class names one algorithm, which a masking job applies to every column the rule set assigns to the data class. Algorithms can be composed: a **Regex Decompose** splits a value into parts and gives each part its own action; a **String Algorithm Chain** applies algorithms one after another. The tree under each data class shows everything its algorithm uses; Appendix A has every configuration in full.

The algorithms are deterministic for their key: the same value masks the same way in every table, database and job, so joins between masked systems keep working — as long as every engine and compliance node uses the same keys.

Where an algorithm that comes with Delphix already does the job — CM Numeric for amounts, Credit Card, the blood group lookup — the policy names it instead of creating one of its own. Those algorithms appear in the trees by their `dlpx-core:` name, the one an engine uses; Data Control Tower lists them without the prefix ( `CM Numeric` , `CreditCard` , `BloodType SL` ). They are on every engine and in every Data Control Tower, and are not listed in Appendix A.

Padding stays where it is: the spaces around a value are kept, and zeros on the left of a number are not masked — `0012345678` becomes `00` followed by what `12345678` becomes. A value kept in a zero-filled CHAR column still joins the same value kept as a number.

> The masked examples in this document were produced with one particular key. With other keys the masked values differ; their shape does not.

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# Essential and extended packs

The policy can be adopted whole or in part. The **extended pack** is everything this document describes. The **essential pack** is the minimum for the law — identity documents, names, contact details, address and birth date. To adopt only the essential pack, create just its data classes, the classifiers that vote for them and the algorithms and files those use, and put only those classifiers in the discovery policy.

| PACK | DATA CLASSES | CLASSIFIERS | ALGORITHMS | FILES |
|---|---|---|---|---|
| **Essential** | 12 | 25 | 17 | 5 |
| **Extended** | 82 | 143 | 107 | 58 |

The essential pack holds these data classes, marked **Essential** in the tables and chapters:

- Cédula de identidad — `EC_L1_CEDULA`
- RUC — `EC_L1_RUC`
- Foreigner documents — `EC_L1_DOCUMENTO_EXTRANJERO`
- Passport — `EC_L1_PASAPORTE`
- Given name — `EC_L1_NOMBRE`
- Surname — `EC_L1_APELLIDO`
- Full name — `EC_L1_NOMBRE_COMPLETO`
- E-mail — `EC_L1_EMAIL`
- Phone — `EC_L1_TELEFONO`
- Address — `EC_L1_DIRECCION`
- Address complement — `EC_L1_DIRECCION_COMPLEMENTO`
- Birth date — `EC_L2_FECHA_NACIMIENTO`

> With fewer data classes competing, a column that a data class of the extended pack would claim by its name can be proposed for an essential data class instead — a numeric code, for instance, can look like a telephone number. Review the discovery result.

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# Data classes at a glance

Every data class of the policy, how it is found and the algorithm that masks it.

| DATA CLASS | FOUND BY | ALGORITHM |
|---|---|---|
| **DIRECT IDENTIFIERS** | | |
| **Cédula de identidad** ESSENTIAL<br>EC_L1_CEDULA | COLUMN NAME · DATA TYPE · VALUE FORMAT | EC_DOCUMENTO |
| **RUC** ESSENTIAL<br>EC_L1_RUC | COLUMN NAME · VALUE FORMAT | EC_RUC |
| **Foreigner documents** ESSENTIAL<br>EC_L1_DOCUMENTO_EXTRANJERO | COLUMN NAME | EC_CM_ALFANUM |
| **Passport** ESSENTIAL<br>EC_L1_PASAPORTE | COLUMN NAME · VALUE FORMAT | EC_CM_ALFANUM |
| **Civil registry records**<br>EC_L1_ACTA | COLUMN NAME | EC_CM_ALFANUM |
| **IESS, ISSFA and ISSPOL**<br>EC_L1_SEGURO_SOCIAL | COLUMN NAME | EC_CM_ALFANUM |
| **Student code**<br>EC_L1_CODIGO_ESTUDIANTE | COLUMN NAME | EC_CM_ALFANUM |
| **Professional registration**<br>EC_L1_REGISTRO_PROFESIONAL | COLUMN NAME | EC_CM_ALFANUM |
| **Given name** ESSENTIAL<br>EC_L1_NOMBRE | COLUMN NAME · KNOWN VALUES | EC_NOMBRES |
| **Surname** ESSENTIAL<br>EC_L1_APELLIDO | COLUMN NAME · KNOWN VALUES | EC_APELLIDOS |
| **Full name** ESSENTIAL<br>EC_L1_NOMBRE_COMPLETO | COLUMN NAME · KNOWN VALUES | EC_NOMBRE_COMPLETO |
| **Business name**<br>EC_L1_RAZON_SOCIAL | COLUMN NAME | EC_RAZON_SOCIAL |
| **E-mail** ESSENTIAL<br>EC_L1_EMAIL | COLUMN NAME · DATA TYPE · VALUE FORMAT | EC_EMAIL |
| **Phone** ESSENTIAL<br>EC_L1_TELEFONO | COLUMN NAME · VALUE FORMAT | EC_TELEFONO |
| **Address** ESSENTIAL<br>EC_L1_DIRECCION | COLUMN NAME · DATA TYPE · VALUE FORMAT | EC_DIRECCION |
| **Address complement** ESSENTIAL<br>EC_L1_DIRECCION_COMPLEMENTO | COLUMN NAME | EC_CM_ALFANUM |
| **Bank account**<br>EC_L1_CUENTA_BANCARIA | COLUMN NAME | EC_CM_DIGITOS |
| **Payment card**<br>EC_L1_TARJETA | COLUMN NAME · VALUE FORMAT | EC_TARJETA |
| **Digital wallet**<br>EC_L1_BILLETERA | COLUMN NAME · VALUE FORMAT | EC_CM_ALFANUM |

| DATA CLASS | FOUND BY | ALGORITHM |
|---|---|---|
| **IP address** `EC_L1_IP` | COLUMN NAME · VALUE FORMAT | `EC_IP` |
| **Device identifier** `EC_L1_DISPOSITIVO` | COLUMN NAME · VALUE FORMAT | `EC_CM_HEX` |
| **Cookies and tracking identifiers** `EC_L1_COOKIE` | COLUMN NAME | `EC_CM_ALFANUM` |
| **Vehicle plate** `EC_L1_PLACA` | COLUMN NAME · VALUE FORMAT | `EC_CM_DIGITOS_LETRAS` |
| **Vehicle and licence** `EC_L1_VEHICULO` | COLUMN NAME · VALUE FORMAT | `EC_CM_ALFANUM` |
| **Property** `EC_L1_INMUEBLE` | COLUMN NAME | `EC_CM_ALFANUM` |
| **Customer and contract numbers** `EC_L1_CONTRATO` | COLUMN NAME | `EC_CM_ALFANUM` |
| **User and social media** `EC_L1_USUARIO` | COLUMN NAME | `EC_CM_ALFANUM` |
| **Credentials** `EC_L1_CREDENCIAL` | COLUMN NAME · VALUE FORMAT | `EC_REDACTAR` |
| **Geolocation** `EC_L1_GEOLOCALIZACION` | COLUMN NAME · VALUE FORMAT | `EC_COORDENADA` |
| **Free text** `EC_L1_TEXTO_LIBRE` | COLUMN NAME · DATA TYPE · VALUE FORMAT | `EC_TEXTO_LIBRE` |
| **QUASI-IDENTIFIERS** | | |
| **Birth date** ESSENTIAL `EC_L2_FECHA_NACIMIENTO` | COLUMN NAME · DATA TYPE | `EC_FECHA_NACIMIENTO` |
| **Birth year** `EC_L2_ANIO_NACIMIENTO` | COLUMN NAME · DATA TYPE | `EC_ANIO` |
| **Age** `EC_L2_EDAD` | COLUMN NAME · DATA TYPE | `EC_EDAD` |
| **Personal event dates** `EC_L2_FECHA_EVENTO` | COLUMN NAME · DATA TYPE | `EC_FECHA_EVENTO` |
| **Sex** `EC_L2_SEXO` | COLUMN NAME · KNOWN VALUES | `EC_SEXO` |
| **Parroquia and cantón** `EC_L2_PARROQUIA` | COLUMN NAME · VALUE FORMAT | `EC_PARROQUIA` |
| **DPA code** `EC_L2_DPA` | COLUMN NAME · VALUE FORMAT | `EC_DPA` |
| **Ciudadela, urbanización and barrio** `EC_L2_CIUDADELA` | COLUMN NAME | `EC_CIUDADELA` |
| **Postal code** `EC_L2_CODIGO_POSTAL` | COLUMN NAME · DATA TYPE | `EC_CODIGO_POSTAL` |
| **Marital status** `EC_L2_ESTADO_CIVIL` | COLUMN NAME · VALUE FORMAT | `EC_ESTADO_CIVIL` |
| **Occupation** `EC_L2_OCUPACION` | COLUMN NAME | `EC_OCUPACION_O_CODIGO` |

| DATA CLASS | FOUND BY | ALGORITHM |
|---|---|---|
| **Employer**<br>EC_L2_EMPLEADOR | COLUMN NAME | EC_EMPLEADOR |
| **Education**<br>EC_L2_NIVEL_INSTRUCCION | COLUMN NAME VALUE FORMAT | EC_NIVEL_INSTRUCCION |
| **School**<br>EC_L2_INSTITUCION_EDUCATIVA | COLUMN NAME | EC_INSTITUCION_EDUCATIVA |
| **Dependants and children**<br>EC_L2_PERSONAS_A_CARGO | COLUMN NAME DATA TYPE | EC_CONTEO_LIMITADO |
| **SENSITIVE DATA (ART. 4)** | | |
| **Ethnicity**<br>EC_L3_ETNIA | COLUMN NAME VALUE FORMAT | EC_AUTOIDENTIFICACION |
| **Cultural identity**<br>EC_L3_IDENTIDAD_CULTURAL | COLUMN NAME VALUE FORMAT | EC_IDENTIDAD_CULTURAL |
| **Language**<br>EC_L3_LENGUA | COLUMN NAME VALUE FORMAT | EC_LENGUA |
| **Migratory status**<br>EC_L3_CONDICION_MIGRATORIA | COLUMN NAME VALUE FORMAT | EC_CONDICION_MIGRATORIA |
| **Nationality**<br>EC_L3_NACIONALIDAD | COLUMN NAME KNOWN VALUES | EC_NACIONALIDAD_PAIS |
| **Religion**<br>EC_L3_RELIGION | COLUMN NAME VALUE FORMAT | EC_RELIGION |
| **Ideology**<br>EC_L3_IDEOLOGIA | COLUMN NAME VALUE FORMAT | EC_IDEOLOGIA |
| **Political affiliation**<br>EC_L3_FILIACION_POLITICA | COLUMN NAME VALUE FORMAT | EC_PARTIDO |
| **Trade union membership**<br>EC_L3_AFILIACION_SINDICAL | COLUMN NAME VALUE FORMAT | EC_CATEGORIA_SUPRIMIDA |
| **Biometric data**<br>EC_L3_BIOMETRICO | COLUMN NAME DATA TYPE | EC_REDACTAR |
| **Genetic data**<br>EC_L3_GENETICO | COLUMN NAME VALUE FORMAT | EC_REDACTAR |
| **Sexual orientation**<br>EC_L3_ORIENTACION_SEXUAL | COLUMN NAME VALUE FORMAT | EC_ORIENTACION_SEXUAL |
| **Gender identity**<br>EC_L3_IDENTIDAD_GENERO | COLUMN NAME VALUE FORMAT | EC_IDENTIDAD_GENERO |
| **Victims of violence**<br>EC_L3_VICTIMA | COLUMN NAME | EC_CATEGORIA_SUPRIMIDA |
| **HEALTH DATA** | | |
| **Health identifiers**<br>EC_SALUD_IDENTIFICADOR | COLUMN NAME | EC_CM_ALFANUM |
| **Health insurance**<br>EC_SALUD_COBERTURA | COLUMN NAME VALUE FORMAT | EC_COBERTURA_SALUD |
| **Diagnosis and ICD-10**<br>EC_SALUD_DIAGNOSTICO | COLUMN NAME VALUE FORMAT | EC_DIAGNOSTICO |

| DATA CLASS | FOUND BY | ALGORITHM |
|---|---|---|
| **Procedures**<br>EC_SALUD_PROCEDIMIENTO | COLUMN NAME | EC_PROCEDIMIENTO |
| **Medicines**<br>EC_SALUD_MEDICAMENTO | COLUMN NAME VALUE FORMAT | EC_MEDICAMENTO |
| **Clinical text**<br>EC_SALUD_TEXTO_CLINICO | COLUMN NAME DATA TYPE | EC_TEXTO_LIBRE |
| **Test results**<br>EC_SALUD_RESULTADO_EXAMEN | COLUMN NAME | EC_CM_ALFANUM |
| **HIV and sexually transmitted infections**<br>EC_SALUD_VIH | COLUMN NAME | EC_CATEGORIA_SUPRIMIDA |
| **Blood group and Rh**<br>EC_SALUD_GRUPO_SANGUINEO | COLUMN NAME VALUE FORMAT | dlpx-core:BloodType SL |
| **Disability**<br>EC_SALUD_DISCAPACIDAD | COLUMN NAME VALUE FORMAT | EC_DISCAPACIDAD |
| **Occupational health**<br>EC_SALUD_OCUPACIONAL | COLUMN NAME VALUE FORMAT | EC_APTITUD_LABORAL |
| **Sexual and reproductive health**<br>EC_SALUD_REPRODUCTIVA | COLUMN NAME | EC_CATEGORIA_SUPRIMIDA |
| **Mental health**<br>EC_SALUD_MENTAL | COLUMN NAME | EC_CATEGORIA_SUPRIMIDA |
| **CREDIT AND SOCIOECONOMIC DATA** | | |
| **Credit history**<br>EC_FIN_HISTORIAL_CREDITICIO | COLUMN NAME | EC_HISTORIAL_CREDITICIO |
| **Debt**<br>EC_FIN_DEUDA | COLUMN NAME DATA TYPE | dlpx-core:CM Numeric |
| **Income and pay**<br>EC_FIN_INGRESOS | COLUMN NAME DATA TYPE | dlpx-core:CM Numeric |
| **Assets**<br>EC_FIN_PATRIMONIO | COLUMN NAME DATA TYPE | dlpx-core:CM Numeric |
| **Pension and benefits**<br>EC_FIN_PENSION | COLUMN NAME DATA TYPE | dlpx-core:CM Numeric |
| **Social programmes**<br>EC_FIN_PROGRAMA_SOCIAL | COLUMN NAME VALUE FORMAT | EC_PROGRAMA_SOCIAL |
| **Registro Social**<br>EC_FIN_REGISTRO_SOCIAL | COLUMN NAME VALUE FORMAT | EC_REGISTRO_SOCIAL |
| **Housing tenure**<br>EC_FIN_VIVIENDA | COLUMN NAME VALUE FORMAT | EC_TENENCIA_VIVIENDA |
| **JUDICIAL RECORD** | | |
| **Judicial record**<br>EC_PENAL_ANTECEDENTES | COLUMN NAME | EC_ANTECEDENTES |
| **Court file**<br>EC_PENAL_EXPEDIENTE | COLUMN NAME VALUE FORMAT | EC_EXPEDIENTE_JUDICIAL |

PART 8

# Direct identifiers

Data that identifies a person on its own. The cédula is the key of almost every Ecuadorian record, and a natural person's RUC is that same cédula followed by 001. Both come out valid, because applications validate them, and both mask the same way everywhere. A RUC that belongs to a company is left as it is: article 4 makes only a natural person a data subject.

## 8.1 Cédula de identidad

EC_L1_CEDULA

ESSENTIAL | COLUMN NAME | DATA TYPE | VALUE FORMAT | REGEX DECOMPOSE

### WHAT IT IS

The ten-digit number of the Registro Civil, `1710034065`, the key of almost every Ecuadorian record. The first two digits are the province of issue — 01 to 24, or 30 for a citizen registered abroad —, the third says the holder is a natural person, and the tenth is a check digit.

The same data class takes RUCs: a document column often holds both.

### HOW IT IS FOUND

**COLUMN NAME** EC_L1_CEDULA – Path

90% `c[eé]dula(_?(de_?)?(identidad|ciudadan[ií]a))?` `ci` `n(o|ro|um|umero)_?(c[eé]dula|documento|doc|identificaci[oó]n|id(ent)?)` `doc_?identidad` `nro_?doc` `num_?doc` `identificacion` `(cliente|cli|paciente|pac|usuario|afiliado|asegurado|beneficiario|benef|titular|empleado|emp|trabajador|colaborador|alumno|estudiante|deudor|codeudor|garante|solicitante|propietario|arrendatario|inquilino|conductor|responsable|representante|apoderado|persona|ciudadano|elector|votante|contribuyente|proveedor|socio|conyuge|conviviente|padre|madre|hijo|familiar|jubilado|pensionista|aportante|derechohabiente)_?(cedula|doc|documento)` `cedula_?(cliente|cli|paciente|pac|usuario|afiliado|asegurado|beneficiario|benef|titular|empleado|emp|trabajador|colaborador|alumno|estudiante|deudor|codeudor|garante|solicitante|propietario|arrendatario|inquilino|conductor|responsable|representante|apoderado|persona|ciudadano|elector|votante|contribuyente|proveedor|socio|conyuge|conviviente|padre|madre|hijo|familiar|jubilado|pensionista|aportante|derechohabiente)`

60% `documento` `doc`

**DATA TYPE** EC_L1_CEDULA – Type

Accepts text (length 10 or more) or number (length 10 or more); any other type rules the data class out.

**VALUE FORMAT** EC_L1_CEDULA – Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `0*(?:(0[1-9]\|1\d\|2[0-4]\|30)[0-5]\d{7})` | 90% | LUHN check digit |
| `0*(?:\d{10})` | 50% | |

A value that matches none counts 30% against the data class.

### HOW IT IS MASKED

Another valid cédula, with the province of issue and the natural-person digit kept and the check digit recomputed. A value shaped like a RUC is masked as a RUC.

`EC_DOCUMENTO` REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: `EC_CM_ALFANUM` .

> `EC_RUC` REGEX DECOMPOSE
>
> Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_CM_DIGITOS` .
>
> > `EC_CEDULA_LUHN` PAYMENT CARD
> >
> > Replaces the card number with another that passes the Luhn check.
> >
> > `EC_CM_DIGITOS` CHARACTER MAPPING
> >
> > Maps each character to another of its group (1 groups), keeping length and format.
>
> `EC_CEDULA` REGEX DECOMPOSE
>
> Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_CM_DIGITOS` .
>
> `EC_CM_ALFANUM` CHARACTER MAPPING
>
> Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 1710034065 | → | **1718687906** |
| 1710034065001 | → | **1718687906001** |
| 1790010937001 | → | **1790010937001** |

---

## 8.2 RUC

EC_L1_RUC

ESSENTIAL COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

The thirteen-digit taxpayer number of the SRI. For a natural person it is the cédula followed by the establishment code `001` ; the third digit decides — 0 to 5 a natural person, **6** a public body, **9** a private company.

HOW IT IS FOUND

COLUMN NAME EC_L1_RUC — Path

90% `ruc[a-z0-9_]*` `n(o|ro|um|umero)_?ruc` `registro_?[uú]nico_?(de_?)?contribuyentes?` `(cliente|cli|paciente|pac|usuario|afiliado|asegurado|beneficiario|benef|titular|empleado|emp|trabajador|colaborador|alumno|estudiante|deudor|codeudor|garante|solicitante|propietario|arrendatario|inquilino|conductor|responsable|representante|apoderado|persona|ciudadano|elector|votante|contribuyente|proveedor|socio|conyuge|conviviente|padre|madre|hijo|familiar|jubilado|pensionista|aportante|derechohabiente)_?ruc` `ruc_?(cliente|cli|paciente|pac|usuario|afiliado|asegurado|beneficiario|benef|titular|empleado|emp|trabajador|colaborador|alumno|estudiante|deudor|codeudor|garante|solicitante|propietario|arrendatario|inquilino|conductor|responsable|representante|apoderado|persona|ciudadano|elector|votante|contribuyente|proveedor|socio|conyuge|conviviente|padre|madre|hijo|familiar|jubilado|pensionista|aportante|derechohabiente)` `id_?tributario` `tax_?id`

VALUE FORMAT EC_L1_RUC — Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `0*(?:(0[1-9]\|1\d\|2[0-4]\|30)[0-9]\d{7}00\d)` | 85% | |
| `0*(?:\d{13})` | 50% | |

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

A natural person's RUC is masked as the cédula it carries, so both agree. A RUC whose third digit is 6 or 9 belongs to a legal person, which article 4 does not make a data subject: it is left exactly as it is.

`EC_RUC` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_CM_DIGITOS`.

> `EC_CEDULA_LUHN` PAYMENT CARD
>
> Replaces the card number with another that passes the Luhn check.
>
> `EC_CM_DIGITOS` CHARACTER MAPPING
>
> Maps each character to another of its group (1 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 1710034065001 | → | **1718687906001** |
| 1790010937001 | → | **1790010937001** |
| 1760013210001 | → | **1760013210001** |

---

## 8.3 Foreigner documents

EC_L1_DOCUMENTO_EXTRANJERO

ESSENTIAL · COLUMN NAME · CHARACTER MAPPING

WHAT IT IS

The cédula de extranjería, the refugee or applicant card, the visa and the migration procedure numbers. The **condición migratoria** these carry is sensitive data by article 4 and has a data class of its own.

HOW IT IS FOUND

> COLUMN NAME EC_L1_DOCUMENTO_EXTRANJERO — Path
>
> 85% `c[eé]dula_?(de_?)?extranjer[ií]a` `carn[eé]_?(de_?)?(extranjer[ií]a|refugiado|solicitante)` `documento_?(de_?)?extranjer[oa]` `doc_?extranjero` `visa(_?(n(o|ro|um|umero)|tipo|categor[ií]a))?` `n(o|ro|um|umero)_?(visa|refugiado|extranjer[ií]a)` `permiso_?(de_?)?(residencia|trabajo)` `amparo_?(legal|migratorio)` `n(o|ro|um|umero)_?(de_?)?(tr[aá]mite_?migratorio|movimiento_?migratorio)`

HOW IT IS MASKED

Character by character, keeping length and format.

`EC_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| R123456 | → | **Q801473** |

---

## 8.4 Passport

EC_L1_PASAPORTE

ESSENTIAL · COLUMN NAME · VALUE FORMAT · CHARACTER MAPPING

WHAT IT IS

Ecuadorian passport numbers and foreign ones.

HOW IT IS FOUND

COLUMN NAME EC_L1_PASAPORTE – Path

90% `pasaporte[a-z0-9_]*` `n(o|ro|um|umero)_?pasaporte` `passport[a-z0-9_]*`

VALUE FORMAT EC_L1_PASAPORTE – Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `[A-Z]{1,3}\d{6,8}` | 40% | |

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Character by character, keeping length and format.

`EC_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| A1234567 | → | **U2438404** |

## 8.5 Civil registry records

EC_L1_ACTA

COLUMN NAME CHARACTER MAPPING

WHAT IT IS

The acta, tomo and folio of the birth, marriage, union of fact and death records the Dirección General de Registro Civil, Identificación y Cedulación keeps.

HOW IT IS FOUND

COLUMN NAME EC_L1_ACTA – Path

85% `acta_?(de_?)?(nacimiento|matrimonio|defunci[oó]n|uni[oó]n_?de_?hecho)(_?(n(o|ro|um|umero)|folio|tomo))?` `n(o|ro|um|umero)_?acta` `partida_?(de_?)?(nacimiento|matrimonio|defunci[oó]n)` `tomo(_?(acta|registro))?` `folio(_?(acta|registro))?` `n(o|ro|um|umero)_?(de_?)?inscripci[oó]n_?civil` `registro_?civil`

HOW IT IS MASKED

Character by character, keeping length and format.

`EC_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 123/1985 | → | **091/7273** |

## 8.6 IESS, ISSFA and ISSPOL

EC_L1_SEGURO_SOCIAL

COLUMN NAME | CHARACTER MAPPING

### WHAT IT IS

The IESS affiliation and employer numbers, the labour history, and the numbers of the armed forces and police funds.

### HOW IT IS FOUND

> COLUMN NAME EC_L1_SEGURO_SOCIAL — Path
>
> 85% `iess[a-z0-9_]*` `n(o|ro|um|umero)_?(iess|afiliaci[oó]n|afiliado|asegurado|patronal)` `c[oó]digo_?(de_?)?(afiliado|asegurado|patronal)` `issfa` `isspol` `seguro_?social_?campesino` `n(o|ro|um|umero)_?(de_?)?(historia_?laboral|aportaciones)` `biess`

### HOW IT IS MASKED

Character by character, keeping length and format.

`EC_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 123456789 | → | **110858568** |

## 8.7 Student code

EC_L1_CODIGO_ESTUDIANTE

COLUMN NAME | CHARACTER MAPPING

### WHAT IT IS

The student code, the enrolment number and the university card.

### HOW IT IS FOUND

> COLUMN NAME EC_L1_CODIGO_ESTUDIANTE — Path
>
> 80% `c[oó]digo_?(de_?)?(estudiante|alumno|matr[ií]cula|educando)` `cod_?(alumno|estudiante)` `n(o|ro|um|umero)_?(de_?)?(matr[ií]cula|carn[eé]_?estudiantil)` `carn[eé]_?(de_?)?(estudiante|universitario)` `id_?(alumno|estudiante)` `c[oó]digo_?amie_?(del_?)?estudiante`

### HOW IT IS MASKED

Character by character, keeping length and format.

`EC_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 202012345 | → | **239159952** |

## 8.8 Professional registration

EC_L1_REGISTRO_PROFESIONAL

COLUMN NAME · CHARACTER MAPPING

### WHAT IT IS

The SENESCYT registration number of a degree, published together with the name, and the professional licences.

### HOW IT IS FOUND

> **COLUMN NAME** EC_L1_REGISTRO_PROFESIONAL — Path
>
> **85%** `registro_?(senescyt|profesional|de_?t[ií]tulo)` `n(o|ro|um|umero)_?(de_?)?(registro_?senescyt|t[ií]tulo|colegiatura)` `senescyt` `c[oó]digo_?(de_?)?t[ií]tulo` `matr[ií]cula_?profesional` `licencia_?profesional`

### HOW IT IS MASKED

Character by character, keeping length and format.

`EC_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `1005-12-1234567` | → | **`0014-51-8129032`** |

## 8.9 Given name

EC_L1_NOMBRE

ESSENTIAL · COLUMN NAME · KNOWN VALUES · REGEX DECOMPOSE

### WHAT IT IS

The given name, one or two: `Narcisa`, `Luis Alberto`, `María de los Ángeles`.

### HOW IT IS FOUND

> **COLUMN NAME** EC_L1_NOMBRE — Path
>
> **85%** `primer_?nombre` `segundo_?nombre` `nombres?_?(de_?)?pila` `pri(mer)?_?nom` `seg(undo)?_?nom` `nombre_?1` `nombre_?2` `first_?names?` `given_?names?` `fname` `middle_?names?` `nombres` *(except when followed by `completos?`, `y_?apellidos?`, `apellidos?`)*
>
> **50%** `nombre` *(except when followed by list E1)* `nom` *(except when followed by list E2)* `name` *(except when followed by list E3)*

> **KNOWN VALUES** EC_L1_NOMBRE — List
>
| LIST | VALUES | STRENGTH | FIRST VALUES |
|---|---|---|---|
| `ec-detectar-nombres.txt` | 395 | 70% | José, Jose, María, Maria, Juan, Carlos |
>
> Values are split into words, and each word is looked up.
>
> A value that matches none counts 30% against the data class.

### HOW IT IS MASKED

Word by word, replacing each name with one from the list. `de`, `del` and `la` stay in place. More than four words are masked character by character.

`EC_NOMBRES` REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: `EC_CM_ALFANUM`.

> `EC_PALABRA_NOMBRE` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `EC_NOMBRE`.

`EC_NOMBRE` NAME

Replaces the name with one of the 338 names in `ec-nombres.txt`.

`EC_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| María de los Ángeles | → | **Washington de los Patricia** |
| LUIS ALBERTO | → | **SISA ELENA** |

## 8.10 Surname

EC_L1_APELLIDO

ESSENTIAL · COLUMN NAME · KNOWN VALUES · REGEX DECOMPOSE

WHAT IT IS

The surnames — the father's and the mother's —, in separate columns or together: `Zambrano Cedeño`. The policy carries the Ecuadorian ones, from Zambrano and Loor on the coast to Guamán and Quishpe in the sierra.

HOW IT IS FOUND

COLUMN NAME EC_L1_APELLIDO – Path

85% `apellidos?` `primer_?apellido` `segundo_?apellido` `apellido_?(1|2|paterno|materno|casada)` `ape_?(1|2|pat|mat)` `ap_?(paterno|materno)` `last_?names?` `surnames?` `family_?names?` `lname`

KNOWN VALUES EC_L1_APELLIDO – List

| LIST | VALUES | STRENGTH | FIRST VALUES |
|---|---|---|---|
| `ec-detectar-apellidos.txt` | 454 | 70% | Zambrano, Vera, García, Garcia, Mero, Loor |

Values are split into words, and each word is looked up.

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Word by word, replacing each surname with one from the list; particles stay in place.

`EC_APELLIDOS` REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: `EC_CM_ALFANUM`.

`EC_PALABRA_APELLIDO` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `EC_APELLIDO`.

`EC_APELLIDO` NAME

Replaces the name with one of the 366 names in `ec-apellidos.txt`.

`EC_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Zambrano Cedeño | → | **Morales Calle** |

## 8.11 Full name

EC_L1_NOMBRE_COMPLETO

ESSENTIAL | COLUMN NAME | KNOWN VALUES | REGEX DECOMPOSE

### WHAT IT IS

Given names and surnames in one column: `Luis Alberto Zambrano Cedeño`. Also those of the mother, father, guardian, spouse, partner, emergency contact and reference.

### HOW IT IS FOUND

**COLUMN NAME** EC_L1_NOMBRE_COMPLETO — Path

**90%**

`nombres?_?(completos?|y_?apellidos?|apellidos?)` `apellidos?_?(y_?)?nombres?` `nombre_?(del_?|de_?la_?)?(cliente|cli|paciente|pac|usuario|afiliado|asegurado|beneficiario|benef|titular|empleado|trabajador|colaborador|alumno|estudiante|deudor|codeudor|garante|solicitante|propietario|arrendatario|inquilino|conductor|responsable|representante|apoderado|persona|ciudadano|contribuyente|madre|padre|conyuge|conviviente|contacto|referencia|victima|denunciante|procesado|testigo|medico|docente|socio|heredero|donante|tutor|jubilado|pensionista|derechohabiente|customer|patient|employee|member|holder|mother|father|spouse|contact|person|student|driver|owner)` `(cliente|cli|paciente|pac|usuario|afiliado|asegurado|beneficiario|benef|titular|empleado|trabajador|colaborador|alumno|estudiante|deudor|codeudor|garante|solicitante|propietario|arrendatario|inquilino|conductor|responsable|representante|apoderado|persona|ciudadano|contribuyente|madre|padre|conyuge|conviviente|contacto|referencia|victima|denunciante|procesado|testigo|medico|docente|socio|heredero|donante|tutor|jubilado|pensionista|derechohabiente|customer|patient|employee|member|holder|mother|father|spouse|contact|person|student|driver|owner)_?(nombre|nom)` `nom_?(cliente|cli|paciente|pac|usuario|afiliado|asegurado|beneficiario|benef|titular|empleado|trabajador|colaborador|alumno|estudiante|deudor|codeudor|garante|solicitante|propietario|arrendatario|inquilino|conductor|responsable|representante|apoderado|persona|ciudadano|contribuyente|madre|padre|conyuge|conviviente|contacto|referencia|victima|denunciante|procesado|testigo|medico|docente|socio|heredero|donante|tutor|jubilado|pensionista|derechohabiente|customer|patient|employee|member|holder|mother|father|spouse|contact|person|student|driver|owner)` `nombre_?(madre|padre|tutor|conyuge|conviviente|contacto_?emergencia|referencia)` `full_?name` `person_?name`

**60%**

`madre` `padre` `tutor` `conyuge` `conviviente` `representante_?legal` `apoderado` `garante` `beneficiario` `heredero` `titular` `referencia_?(personal|familiar)`

**50%**

`nombre` *(except when followed by list E1)* `nom` *(except when followed by list E2)* `name` *(except when followed by list E3)*

**KNOWN VALUES** EC_L1_NOMBRE_COMPLETO — List

| LIST | VALUES | STRENGTH | FIRST VALUES |
|---|---|---|---|
| `ec-detectar-nombres.txt` | 395 | 65% | José, Jose, María, Maria, Juan, Carlos |
| `ec-detectar-apellidos.txt` | 454 | 65% | Zambrano, Vera, García, Garcia, Mero, Loor |

Values are split into words, and each word is looked up.

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

In Ecuadorian order: two words are a given name and a surname; three, a given name and two surnames; four, two and two. `APELLIDOS, NOMBRES` with a comma is masked keeping the comma.

`EC_NOMBRE_COMPLETO` REGEX DECOMPOSE

Tries 7 patterns in order and gives each part of the value its own action; when none matches: `EC_CM_ALFANUM`.

> `EC_APELLIDOS` REGEX DECOMPOSE
>
> Tries 4 patterns in order and gives each part of the value its own action; when none matches: `EC_CM_ALFANUM`.
>
> > `EC_PALABRA_APELLIDO` REGEX DECOMPOSE
> >
> > Matches one pattern and gives each part of the value its own action; when none matches: `EC_APELLIDO`.
> >
> > > `EC_APELLIDO` NAME
> > >
> > > Replaces the name with one of the 366 names in `ec-apellidos.txt`.
> >
> > `EC_CM_ALFANUM` CHARACTER MAPPING
> >
> > Maps each character to another of its group (7 groups), keeping length and format.
>
> `EC_NOMBRES` REGEX DECOMPOSE
>
> Tries 4 patterns in order and gives each part of the value its own action; when none matches: `EC_CM_ALFANUM`.
>
> > `EC_PALABRA_NOMBRE` REGEX DECOMPOSE
> >
> > Matches one pattern and gives each part of the value its own action; when none matches: `EC_NOMBRE`.
> >
> > > `EC_NOMBRE` NAME
> > >
> > > Replaces the name with one of the 338 names in `ec-nombres.txt`.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `Luis Alberto Zambrano Cedeño` | → | **`Sisa Elena Morales Calle`** |
| `Narcisa Zambrano` | → | **`Soledad Morales`** |
| `ZAMBRANO CEDEÑO, LUIS ALBERTO` | → | **`MORALES CALLE, SISA ELENA`** |

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## 8.12 Business name

`EC_L1_RAZON_SOCIAL`

COLUMN NAME · SECURE LOOKUP

WHAT IT IS

The business name a natural person trades under, beside the RUC that names her. The company itself is not a data subject; the person behind it is.

HOW IT IS FOUND

> COLUMN NAME `EC_L1_RAZON_SOCIAL — Path`
>
> 85% `raz[oó]n_?social` `razonsocial` `nombre_?(comercial|de_?la_?empresa|empresa)` `denominaci[oó]n_?social` `nombre_?legal` `business_?name` `company_?name` `legal_?name`

HOW IT IS MASKED

Replaced by a fictitious business name.

`EC_RAZON_SOCIAL` SECURE LOOKUP

Replaces the value with one of the 20 values in `ec-razones-sociales.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `Corporación Comercial Ecuatoriana S.A.` | → | **`Transportes Mitad del Mundo S.A.`** |

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## 8.13 E-mail

EC_L1_EMAIL

ESSENTIAL | COLUMN NAME | DATA TYPE | VALUE FORMAT | REGEX DECOMPOSE

### WHAT IT IS

Personal and corporate e-mail addresses.

### HOW IT IS FOUND

**COLUMN NAME** EC_L1_EMAIL – Path

85% `e_?mail[a-z0-9_]*` `mail[a-z0-9_]*` `correo(_?electr[oó]nico)?[a-z0-9_]*` `dir_?correo` `direcci[oó]n_?(de_?)?correo` `direcci[oó]n_?electr[oó]nica`

**DATA TYPE** EC_L1_EMAIL – Type

Accepts text (length 5 or more); any other type rules the data class out.

**VALUE FORMAT** EC_L1_EMAIL – Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `[\p{L}\p{N}._%+'\-]+@[\p{L}\p{N}.\-]+\.[A-Za-z]{2,24}` | 100% | |

A value that matches none counts 20% against the data class.

### HOW IT IS MASKED

User and domain are mapped character by character, and the top-level domain — `.com`, `.com.ec` — becomes `.test`, reserved for tests: a masked address can never deliver.

`EC_EMAIL` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `EC_CM_ALFANUM`.

> `EC_CM_ALFANUM` CHARACTER MAPPING
>
> Maps each character to another of its group (7 groups), keeping length and format.

### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `narcisa.zambrano@gmail.com` | → | **`dipcola.gojmyeju@pgoeb.test`** |

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## 8.14 Phone

EC_L1_TELEFONO

ESSENTIAL | COLUMN NAME | VALUE FORMAT | REGEX DECOMPOSE

### WHAT IT IS

Mobiles of ten digits starting `09`, and landlines with a one-digit area code after the zero — 02 Pichincha, 04 Guayas, 07 Azuay and El Oro — and seven digits.

### HOW IT IS FOUND

**COLUMN NAME** EC_L1_TELEFONO – Path

85% `tel` `tel[eé]fono[a-z0-9_]*` `tel_?(fijo|casa|oficina|trabajo|contacto|celular|m[oó]vil|convencional)` `celular[a-z0-9_]*` `cel` `m[oó]vil` `whats_?app` `convencional` `n(o|ro|um|umero)_?(tel|tel[eé]fono|celular|cel|contacto)` `fax` `phone[a-z0-9_]*` `mobile[a-z0-9_]*`

**VALUE FORMAT** EC_L1_TELEFONO — Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `0*(?:(\+?593[\s\-]?)?0?9\d{2}[\s\-]?\d{3}[\s\-]?\d{3})` | 90% | |
| `\(?0?[2-7]\)?[\s\-]?\d{3}[\s\-]?\d{4}` | 50% | |

A value that matches none counts 30% against the data class.

#### HOW IT IS MASKED

`+593` , the 9 of a mobile and the area code of a landline stay; the rest of the digits are mapped and the spacing of the original is kept, so the number still says mobile or landline and which region.

`EC_TELEFONO` **REGEX DECOMPOSE**

Tries 3 patterns in order and gives each part of the value its own action; when none matches: `EC_CM_DIGITOS` .

> `EC_CM_DIGITOS` **CHARACTER MAPPING**
>
> Maps each character to another of its group (1 groups), keeping length and format.

#### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `+593 99 123 4567` | → | **`+593 93 743 6352`** |
| `0991234567` | → | **`0937436352`** |
| `02 234-5678` | → | **`02 495-1769`** |

---

## 8.15 Address

EC_L1_DIRECCION

ESSENTIAL · COLUMN NAME · DATA TYPE · VALUE FORMAT · SECURE LOOKUP

#### WHAT IT IS

The address with the main street and the cross street — `Av. Amazonas N34-120 y Av. República` —, or the ciudadela with manzana, villa and solar.

#### HOW IT IS FOUND

**COLUMN NAME** EC_L1_DIRECCION — Path

**80%** `direcci[oó]n[a-z0-9_]*` *(except when preceded by* `mac` , `ip` , `e_?mail` , `web` , `url` , `correo` *; except when followed by list E4)* `domicilio[a-z0-9_]*` `dom_?(residencia|particular|laboral|cliente)` `residencia` `lugar_?(de_?)?residencia` `calle(_?(principal|secundaria|y_?n[uú]mero))?` `address[a-z0-9_]*` *(except when preceded by* `mac` , `ip` , `e_?mail` , `web` , `url` *; except when followed by list E5)*

**DATA TYPE** EC_L1_DIRECCION — Type

Accepts text (length 6 or more); any other type rules the data class out.

VALUE FORMAT EC_L1_DIRECCION — Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `(?i)(av\|avenida\|calle\|ca\|pasaje\|psje\|callej[oó]n\|malec[oó]n)\.?\s+[a-záéíóúñ0-9.' ]{2,40}\s*(n\d{1,2}-\d{1,3}\|\d{1,5}).*` | 80% | |
| `(?i).*\b(mz\|manzana)\.?\s*[a-z0-9]{1,3}\s*(villa\|solar\|lote\|lt)\.?\s*\d{1,3}.*` | 80% | |
| `(?i).*\b(cdla\|ciudadela\|urb\|urbanizaci[oó]n\|barrio\|conjunto\|lotizaci[oó]n\|cooperativa)\.?\s+[a-záéíóúñ0-9.' ]{2,40}.*` | 70% | |
| `(?i)k(m\|il[oó]metro)\.?\s*\d{1,3}\s+(v[ií]a\|carretera\|autopista).*` | 70% | |

A value that matches none counts 20% against the data class.

HOW IT IS MASKED

The whole line is replaced by a fictitious Ecuadorian address of the same form. Ciudadela, parroquia and postal code have data classes of their own for when they live in separate columns.

`EC_DIRECCION` SECURE LOOKUP

Replaces the value with one of the 4,000 values in `ec-direcciones.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Av. Amazonas N34-120 y Av. República | → | **Malecón Malecón Simón Bolívar 2125 y Gaspar de Villarroel** |

## 8.16 Address complement

EC_L1_DIRECCION_COMPLEMENTO

ESSENTIAL COLUMN NAME CHARACTER MAPPING

WHAT IT IS

Manzana, villa, solar, lote, interior, flat, floor, block, stage or building in columns of their own.

HOW IT IS FOUND

COLUMN NAME EC_L1_DIRECCION_COMPLEMENTO — Path

70% `manzana` `mz` `solar` `villa(_?n(o|ro|um|umero))?` `lote` `lt` `interior` `departamento_?(n(o|ro|um|umero)|interior)` `dpto_?(n(o|ro|um|umero))?` `piso` `block` `bloque` `etapa` `edificio` `n(o|ro|um|umero)_?(casa|puerta|dpto|interior)` `calle_?secundaria` `intersecci[oó]n` `referencia_?(de_?)?(direcci[oó]n|domicilio)` `complemento(_?direcci[oó]n)?`

HOW IT IS MASKED

Character by character, keeping length and format.

`EC_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Mz. B Villa 12 | → | **Tk. T Tujpu 18** |

## 8.17 Bank account

EC_L1_CUENTA_BANCARIA

COLUMN NAME | CHARACTER MAPPING

### WHAT IT IS

Savings, current and payroll account numbers. The bank code is not personal data and has no data class.

### HOW IT IS FOUND

COLUMN NAME EC_L1_CUENTA_BANCARIA — Path

85% `cuenta_?(bancaria|banco|corriente|ahorros?|sueldo|n[oó]mina|abono|dep[oó]sito|destino|origen)` `n(o|ro|um|umero)_?(de_?)?(cuenta|cta)` `nro_?cta` `num_?cta` `cta_?(cte|corriente|ahorros?|bancaria)` `iban` `account_?(no|num|number)` `bank_?account`

### HOW IT IS MASKED

Digits are mapped; separators stay.

`EC_CM_DIGITOS` CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 2100123456 | → | **1786723060** |

## 8.18 Payment card

EC_L1_TARJETA

COLUMN NAME | VALUE FORMAT | REGEX DECOMPOSE

### WHAT IT IS

Credit and debit card numbers, recognized by the Luhn algorithm.

### HOW IT IS FOUND

COLUMN NAME EC_L1_TARJETA — Path

85% `tarjeta(_?(cr[eé]dito|d[eé]bito|n(o|ro|um|umero)))?` `n(o|ro|um|umero)_?tarjeta` `pan` `card_?(no|num|number)` `credit_?card` `tc_?(numero|num|nro)`

VALUE FORMAT EC_L1_TARJETA — Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `0*(?:3[47]\d{13}\|[1-9]\d{12,13}\|[1-9]\d{15,18})` | 90% | LUHN check digit; after removing `[\s\-]` |

A value that matches none counts 30% against the data class.

### HOW IT IS MASKED

Credit Card, which comes with Delphix: another number that passes the Luhn check, keeping the first six digits — the issuer, not the holder. Only values shaped like a card reach it; the others pass unchanged instead of failing the row.

`EC_TARJETA` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

> `dlpx-core:Credit Card` PAYMENT CARD
>
> Comes with Delphix and is used as it ships. Replaces the card number with another that passes the Luhn check, keeping the first six digits — the issuer, not the holder.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 4539 5787 6362 1486 | → | **4539 5722 0015 3609** |

## 8.19 Digital wallet

EC_L1_BILLETERA

COLUMN NAME · VALUE FORMAT · CHARACTER MAPPING

WHAT IT IS

DeUna, Peigo, PayPhone and other wallet identifiers, and the bitcoin addresses some systems keep.

HOW IT IS FOUND

COLUMN NAME EC_L1_BILLETERA — Path

85% `billetera(_?(digital|electr[oó]nica|m[oó]vil))?` `wallet(_?(id|address|direccion))?` `de_?una` `peigo` `payphone` `dinero_?electr[oó]nico` `c[oó]digo_?qr` `qr_?(id|codigo)` `direcci[oó]n_?` `(bitcoin|btc|wallet)` `btc_?(address|direccion)`

VALUE FORMAT EC_L1_BILLETERA — Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `(bc1\|[13])[a-zA-HJ-NP-Z0-9]{25,59}` | 70% | |

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Character by character, keeping the shape. A masked bitcoin address no longer passes its own checksum, which is what a test copy wants.

`EC_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| bc1qar0srrr7xfkvy5l643lydnw9re59gtzzwf5mdq | → | **mh2wis8tsdm0pdjdv8v511yqxqg4ze53qkjbvp7dgf** |

## 8.20 IP address

EC_L1_IP

COLUMN NAME · VALUE FORMAT · REGEX DECOMPOSE

WHAT IT IS

IPv4 and IPv6 addresses from access logs and transactions.

HOW IT IS FOUND

COLUMN NAME EC_L1_IP — Path

85% `ip` `ip_?(address|addr|origen|destino|cliente|usuario|acceso|login|remota|publica)` `direcci[oó]n_?ip` `dir_?ip` `ipv[46]` `remote_?addr(ess)?` `client_?ip` `host_?ip`

VALUE FORMAT EC_L1_IP — Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `0*(?:((25[0-5]\|2[0-4]\d\|1?\d?\d)\.){3}(25[0-5]\|2[0-4]\d\|1?\d?\d))` | 100% | |
| `0*(?:[0-9A-Fa-f]{1,4}(:[0-9A-Fa-f]{0,4}){2,7})` | 90% | |

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Each IPv4 octet is mapped to 1–254; an IPv6 address is mapped keeping its hex characters.

`EC_IP` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `EC_CM_HEX`.

> `EC_OCTETO` NUMERIC MAPPING
>
> Maps the number to another between 1 and 254.
>
> `EC_CM_HEX` CHARACTER MAPPING
>
> Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `186.42.163.10` | → | **`153.35.221.16`** |

## 8.21 Device identifier

EC_L1_DISPOSITIVO

COLUMN NAME VALUE FORMAT CHARACTER MAPPING

WHAT IT IS

IMEI, IMSI, ICCID, MAC address, advertising and session identifiers.

HOW IT IS FOUND

COLUMN NAME EC_L1_DISPOSITIVO — Path

80% `imei` `imsi` `iccid` `mac` `mac_?address` `direcci[oó]n_?mac` `device_?(id|uuid|serial)` `id_?(dispositivo|equipo|celular)` `serial_?(equipo|celular|dispositivo)` `advertising_?id` `idfa` `gaid` `session_?id` `id_?sesi[oó]n`

VALUE FORMAT EC_L1_DISPOSITIVO — Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `0*(?:([0-9A-Fa-f]{2}[:\-]){5}[0-9A-Fa-f]{2})` | 100% | |
| `0*(?:\d{15})` | 80% | LUHN check digit |

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Character by character, with hex digits kept hex: a MAC stays a MAC.

`EC_CM_HEX` CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `3c:22:fb:9a:10:e4` | → | **`5e:83:bb:3c:31:f9`** |

## 8.22 Cookies and tracking identifiers

EC_L1_COOKIE

COLUMN NAME | CHARACTER MAPPING

### WHAT IT IS

The identifier a site stores in the browser and the campaign parameters that travel with it. Article 4 defines the **elaboración de perfiles** these feed: any processing that evaluates or predicts a person's professional performance, economic situation, health, preferences, interests, location or movement.

### HOW IT IS FOUND

COLUMN NAME — EC_L1_COOKIE — Path

80% `cookie(_?(id|value|valor|nombre))?` `cookies` `_?ga` `_?gid` `fbp` `fbclid` `utm_?(source|medium|campaign|term|content)` `id_?(navegador|browser|visitante)` `visitor_?id` `tracking_?id` `client_?id`

### HOW IT IS MASKED

Character by character, keeping length and format.

`EC_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `GA1.2.123456789.1700000000` | → | **`ZE0.5.891552838.7889916184`** |

---

## 8.23 Vehicle plate

EC_L1_PLACA

COLUMN NAME | VALUE FORMAT | CHARACTER MAPPING

### WHAT IT IS

The plate with three letters and three or four digits, `PCA–1234`; the first letter is the province — A Azuay, G Guayas, P Pichincha, M Manabí.

### HOW IT IS FOUND

COLUMN NAME — EC_L1_PLACA — Path

85% `placa(_?(veh[ií]culo|carro|moto|automotor))?` `n(o|ro|um|umero)_?placa` `matr[ií]cula_?(veh[ií]culo|automotor)` `license_?plate` `plate(_?(no|number))?`

VALUE FORMAT — EC_L1_PLACA — Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `[A-Z]{3}[\s\-]?\d{3,4}` | 70% | |
| `[A-Z]{2}[\s\-]?\d{3,4}` | 40% | |

A value that matches none counts 30% against the data class.

### HOW IT IS MASKED

Letters to letters and digits to digits, position by position: the plate format stays.

`EC_CM_DIGITOS_LETRAS` CHARACTER MAPPING

Maps each character to another of its group (2 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| PCA–1234 | → | **MQP–8512** |
| GSB–123 | → | **WKG–104** |

## 8.24 Vehicle and licence

EC_L1_VEHICULO

COLUMN NAME · VALUE FORMAT · CHARACTER MAPPING

WHAT IT IS

The VIN or chassis number, the engine number, the registration and the driving licence.

HOW IT IS FOUND

COLUMN NAME — EC_L1_VEHICULO — Path

85% `vin` `chasis` `n(o|ro|um|umero)_?(chasis|motor|serie|vin)` `motor_?n(o|ro|um|umero)` `matr[ií]cula(_?veh[ií]culo)?` `licencia_?(de_?)?conducir(_?(n(o|ro|um|umero)|tipo))?` `n(o|ro|um|umero)_?licencia` `puntos_?licencia`

VALUE FORMAT — EC_L1_VEHICULO — Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `[A-HJ-NPR-Z0-9]{17}` | 70% | |

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Character by character, keeping length and format.

`EC_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 8AJHA8CD3J1234567 | → | **8EWKE1QN1T0772894** |

## 8.25 Property

EC_L1_INMUEBLE

COLUMN NAME · CHARACTER MAPPING

WHAT IT IS

The cadastral key, the predial code and the entry in the municipal Registro de la Propiedad: they lead to the home and its owner.

HOW IT IS FOUND

COLUMN NAME — EC_L1_INMUEBLE — Path

80% `clave_?catastral` `c[oó]digo_?(catastral|predial|de_?predio)` `n(o|ro|um|umero)_?(de_?)?(predio|inmueble|catastral)` `catastro` `ficha_?catastral` `inscripci[oó]n_?(registral|de_?la_?propiedad)` `registro_?(de_?la_?)?propiedad` `repertorio` `n(o|ro|um|umero)_?(de_?)?medidor`

HOW IT IS MASKED

Character by character, keeping length and format.

`EC_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 12–345–678 | → | **46–916–122** |

## 8.26 Customer and contract numbers

EC_L1_CONTRATO

COLUMN NAME CHARACTER MAPPING

WHAT IT IS

Contract, policy, loan, application, member, subscriber, supply, meter, case or administrative file numbers, and customer or employee codes.

HOW IT IS FOUND

COLUMN NAME EC_L1_CONTRATO – Path

70% `n(o|ro|um|umero)_?(contrato|p[oó]liza|cr[eé]dito|pr[eé]stamo|solicitud|expediente_?(administrativo|interno)|tr[aá]mite|referencia|cliente|socio|suscriptor|suministro|medidor|servicio|caso|ticket|beneficio)` `contrato_?(n(o|ro|um|umero))` `c[oó]digo_?(cliente|socio|empleado|afiliado|suministro)` `id_?(cliente|socio|empleado|afiliado)` `n(o|ro)_?empleado` `c[oó]digo_?(de_?)?(empleado|trabajador|n[oó]mina)`

HOW IT IS MASKED

Character by character, keeping length and format. The mapping is deterministic, so joins keep working.

`EC_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| CT–123456 | → | **NG–557411** |

## 8.27 User and social media

EC_L1_USUARIO

COLUMN NAME CHARACTER MAPPING

WHAT IT IS

Login, alias and social media profiles.

HOW IT IS FOUND

COLUMN NAME EC_L1_USUARIO – Path

70% `usuario` `user_?name` `username` `login` `alias` `apodo` `nick(_?name)?` `perfil_?(instagram|facebook|twitter|tiktok|linkedin|x)` `instagram` `facebook` `twitter` `tiktok` `linkedin` `red_?social` `handle`

HOW IT IS MASKED

Character by character, keeping length and format.

`EC_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `@narcisa_zambrano` | → | **`@dipcola_gojmyeju`** |

## 8.28 Credentials

EC_L1_CREDENCIAL

COLUMN NAME · VALUE FORMAT · REGEX DECOMPOSE

WHAT IT IS

Passwords and their hashes, PIN, security and OTP codes, CVV, tokens, API keys, the key of an electronic signature, private keys and seed phrases.

HOW IT IS FOUND

COLUMN NAME EC_L1_CREDENCIAL — Path

90% `contrase[nñ]a` `clave` *(except when followed by* `catastral`, `primaria`, `for[aá]nea`, `valor`, `producto` *)* `password` `passwd` `pwd` `pass` `hash_?(clave|contrasena|password)` `pin` `c[oó]digo_?(seguridad|verificaci[oó]n|acceso|otp)` `cvv` `cvc` `token(_?(acceso|refresh|api))?` `access_?token` `refresh_?token` `api_?key` `llave_?(api|privada)` `secreto` `secret` `otp` `firma_?electr[oó]nica_?(clave|pin)` `pregunta_?secreta` `respuesta_?secreta` `frase_?semilla` `seed_?phrase`

VALUE FORMAT EC_L1_CREDENCIAL — Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `\$2[abxy]?\$\d{2}\$[./A-Za-z0-9]{53}` | 95% | |
| `\$(1\|5\|6)\$(rounds=\d+\$)?[./A-Za-z0-9]{1,16}\$[./A-Za-z0-9]{22,86}` | 95% | |
| `\$argon2(id\|i\|d)\$v=\d+\$m=\d+,t=\d+,p=\d+\$\S+` | 95% | |
| `pbkdf2_sha(1\|256\|512)\$\d+\$\S+\$\S+` | 95% | |
| `\{(SSHA\|SHA\|SSHA256\|SSHA512\|SMD5\|MD5\|CRYPT)\}\S+` | 95% | |

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Every character becomes `X`. A credential has no use in a test copy, and a hash masked character by character could still be cracked.

`EC_REDACTAR` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `Clave#2024` | → | **`XXXXXXXXX`** |

## 8.29 Geolocation

EC_L1_GEOLOCALIZACION

COLUMN NAME | VALUE FORMAT | REGEX DECOMPOSE

WHAT IT IS

Latitude and longitude, together or in separate columns, from apps, visits, addresses and deliveries.

HOW IT IS FOUND

COLUMN NAME EC_L1_GEOLOCALIZACION – Path

85% `lat` `latitud` `lon` `lng` `longitud` `coordenadas?` `coord_?[xy]` `geo_?(lat|lon|lng|loc|localizacion|posicion)` `georreferenciaci[oó]n` `geolocalizaci[oó]n` `gps(_?(ubicaci[oó]n|posici[oó]n|coordenadas))?` `ubicaci[oó]n_?gps`

50% `long`

VALUE FORMAT EC_L1_GEOLOCALIZACION – Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `-?[0-5]\.\d{3,}\s*,\s*-(7[5-9]\|8[01]\|9[01])\.\d{3,}` | 90% | |
| `-(7[5-9]\|8[01])\.\d{4,}` | 50% | |
| `-?[0-5]\.\d{4,}` | 40% | |

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

The integer part and first decimal stay — about 11 km — and the other decimals are mapped: the place is generalized, not moved.

`EC_COORDENADA` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

> `EC_CM_DIGITOS` CHARACTER MAPPING
>
> Maps each character to another of its group (1 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `-0.180653,-78.467834` | → | **`-0.134127,-78.499081`** |

## 8.30 Free text

EC_L1_TEXTO_LIBRE

COLUMN NAME | DATA TYPE | VALUE FORMAT | FREE TEXT REDACTION

WHAT IT IS

Remarks, complaint descriptions, accounts of events and messages where anything gets written — cédula, RUC, phone and names in the middle of a sentence.

HOW IT IS FOUND

COLUMN NAME EC_L1_TEXTO_LIBRE – Path

60% `observaci[oó]n(es)?` `obs` `comentarios?` `notas?` `anotaci[oó]n(es)?` `glosa` `sustento` `descripci[oó]n_?(queja|reclamo|solicitud|caso|hechos|novedad|atenci[oó]n|denuncia)` `detalle_?(reclamo|solicitud|caso|atenci[oó]n)` `hechos` `relato` `justificaci[oó]n` `mensaje` *(except when preceded by list E6; except when followed by list E7)* `texto(_?libre)?` `resumen_?(caso|atenci[oó]n)` `queja` `reclamo` `denuncia_?texto` `remarks?` `comments?` `notes?` `free_?text` `memo`

**DATA TYPE** `EC_L1_TEXTO_LIBRE – Type`

Accepts text (length 30 or more); any other type rules the data class out.

**VALUE FORMAT** `EC_L1_TEXTO_LIBRE – Regex`

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `0*(?:(0[1-9]\|1\d\|2[0-4]\|30)[0-5]\d{7})` | 70% | anywhere in the value |
| `[\w.+\-]+@[\w\-]+\.[A-Za-z]{2,}` | 70% | anywhere in the value |
| `0*(?:0?9\d{2}[\s\-]?\d{3}[\s\-]?\d{3})` | 60% | anywhere in the value |

A value that matches none does not count against the data class.

### HOW IT IS MASKED

Free Text Redaction: cédulas, RUCs, e-mail, mobiles, cards, plates and IPs become `[DATO]`, and the names of the lists `[NOMBRE]`.

`EC_TEXTO_LIBRE` **FREE TEXT REDACTION**

Replaces the text matching 8 patterns with `[DATO]`, and the 1,668 terms of `ec-texto-nombres.txt` with `[NOMBRE]`.

### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `Cliente Zambrano, cédula 1710034065, correo n.zambrano@gmail.com, cel 0991234567.` | → | **`Cliente [NOMBRE], cédula [DATO], correo [DATO], cel [DATO].`** |

PART 9

# Quasi-identifiers

Data that identifies nobody on its own, but does in combination. Ecuador is divided into about 1,040 parroquias, and 913 of them had fewer than 20,000 inhabitants in the 2022 census. A birth date, a sex and one of those parroquias point at a handful of people. These attributes are generalized or shifted rather than replaced, and the data keeps its analytical value.

## 9.1 Birth date

EC_L2_FECHA_NACIMIENTO

ESSENTIAL · COLUMN NAME · DATA TYPE · DATE SHIFT

### WHAT IT IS

The birth date, as a date, timestamp or text. It also decides whether the row belongs to a child or adolescent, a special category under art. 25 b).

### HOW IT IS FOUND

**COLUMN NAME** EC_L2_FECHA_NACIMIENTO – Path

90% `(fecha|fec|fch|f)_?(de_?)?nac(imiento)?` `fnac` `fechanac(imiento)?` `cumplea[ñn]os` `dob` `date_?of_?birth` `birth_?date` `birthday`

**DATA TYPE** EC_L2_FECHA_NACIMIENTO – Type

Accepts date, TIMESTAMP or text (length 6 or more); any other type rules the data class out.

### HOW IT IS MASKED

Shifted up to 60 days either way. Ages and school years change for few people; the date stays plausible and the same birth date always moves the same way.

`EC_FECHA_NACIMIENTO` DATE SHIFT

Moves the date by -60 to 60 days.

### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 1985-07-20 | → | **1985-08-05T00:00** |

## 9.2 Birth year

EC_L2_ANIO_NACIMIENTO

COLUMN NAME · DATA TYPE · REGEX DECOMPOSE

### WHAT IT IS

The birth year in a column of its own.

### HOW IT IS FOUND

**COLUMN NAME** EC_L2_ANIO_NACIMIENTO – Path

90% `a[ñn]o_?(de_?)?nac(imiento)?` `anio_?nac(imiento)?` `year_?of_?birth` `birth_?year` `yob`

**DATA TYPE** EC_L2_ANIO_NACIMIENTO – Type

Accepts number (length up to 4) or text (length up to 4); any other type rules the data class out.

### HOW IT IS MASKED

The decade stays and the last digit is mapped: `1985` becomes a year from 1980 to 1989.

`EC_ANIO` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

> `EC_CM_DIGITOS` CHARACTER MAPPING
>
> Maps each character to another of its group (1 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 1985 | → | **1980** |

---

## 9.3 Age

EC_L2_EDAD

COLUMN NAME · DATA TYPE · REGEX DECOMPOSE

WHAT IT IS

Age in years, also age group or five-year band.

HOW IT IS FOUND

> COLUMN NAME EC_L2_EDAD — Path
>
> 85% `edad(_?(actual|a[nñ]os|anios|paciente|afiliado|cliente|al_?(ingreso|diagn[oó]stico|fallecimiento)))?` `grupo_?etario` `rango_?(de_?)?edad` `quinquenio` `age(_?(years|group))?`

> DATA TYPE EC_L2_EDAD — Type
>
> Accepts number (length up to 6) or text (length up to 6); any other type rules the data class out.

HOW IT IS MASKED

The tens digit stays and the unit is mapped (37 becomes 30–39). A hundred or more becomes 90: rare ages identify.

`EC_EDAD` REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.

> `EC_CM_DIGITOS` CHARACTER MAPPING
>
> Maps each character to another of its group (1 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 37 | → | **35** |
| 104 | → | **90** |

---

## 9.4 Personal event dates

EC_L2_FECHA_EVENTO

COLUMN NAME · DATA TYPE · DATE SHIFT

WHAT IT IS

Dates that say something about the person: death, marriage, hiring and leaving, retirement, admission, diagnosis, arrest, sentence, sick leave, cédula issue.

HOW IT IS FOUND

COLUMN NAME EC_L2_FECHA_EVENTO – Path

80% `(fecha|fec|fch|f)_?(de_?)?(defunci[oó]n|fallecimiento|muerte|matrimonio|divorcio|ingreso|salida|egreso|alta|baja|retiro|contrataci[oó]n|despido|jubilaci[oó]n|hospitalizaci[oó]n|internamiento|diagn[oó]stico|atenci[oó]n|consulta|parto|detenci[oó]n|sentencia|expedici[oó]n(_?(cedula|documento))?|caducidad_?c[eé]dula|vacunaci[oó]n|cirug[ií]a|accidente|reposo) fec_?(exp|expedicion|defuncion|ingreso|salida|baja) date_?of_?(death|marriage|hire|admission|discharge)`

DATA TYPE EC_L2_FECHA_EVENTO – Type

Accepts date, TIMESTAMP or text (length 6 or more); any other type rules the data class out.

HOW IT IS MASKED

Shifted up to 30 days either way.

`EC_FECHA_EVENTO` DATE SHIFT

Moves the date by -30 to 30 days.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 2021-03-15 | → | **2021-03-05T00:00** |

## 9.5 Sex

EC_L2_SEXO

COLUMN NAME KNOWN VALUES REGEX DECOMPOSE

WHAT IT IS

Sex as the system records it: `F` / `M`, `Femenino` / `Masculino`, `Mujer` / `Hombre` or the codes 1 and 2.

HOW IT IS FOUND

COLUMN NAME EC_L2_SEXO – Path

85% `sexo(_?(biol[oó]gico|al_?nacer|paciente|afiliado|cliente))?` `(tp|tipo|cod|cd|id)_?sexo` `g[eé]nero` *(except when followed by* `identidad`, `autopercibido` *)* `sex` `gender` *(except when followed by* `identity` *)*

KNOWN VALUES EC_L2_SEXO – List

| LIST | VALUES | STRENGTH | FIRST VALUES |
|---|---|---|---|
| `ec-detectar-sexo.txt` | 6 | 80% | femenino, masculino, mujer, hombre, female, male |
| `ec-detectar-sexo-letra.txt` | 3 | 60% | f, m, h |

Values are split into words, and each word is looked up.

A value that matches none counts 50% against the data class.

HOW IT IS MASKED

Replaced within its own vocabulary: `F` stays `F` or `M`, `1` stays `1` or `2`. The **identidad de género** of art. 4 has its own data class among the sensitive data.

`EC_SEXO` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`EC_SEXO_FEMENINO_MASCULINO` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-sexo-femenino-masculino.txt`, chosen by a keyed hash.

`EC_SEXO_MUJER_HOMBRE` **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-sexo-mujer-hombre.txt`, chosen by a keyed hash.

`EC_SEXO_FEMALE_MALE` **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-sexo-female-male.txt`, chosen by a keyed hash.

`EC_SEXO_F_M` **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-sexo-f-m.txt`, chosen by a keyed hash.

`EC_SEXO_H_M` **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-sexo-h-m.txt`, chosen by a keyed hash.

`EC_SEXO_1_2` **SECURE LOOKUP**

Replaces the value with one of the 2 values in `ec-sexo-1-2.txt`, chosen by a keyed hash.

**EXAMPLES**

| VALUE | | MASKED |
|---|---|---|
| F | → | **M** |
| Femenino | → | **Femenino** |

## 9.6 Parroquia and cantón

`EC_L2_PARROQUIA`

**COLUMN NAME** **VALUE FORMAT** **DATA CLEANSING**

**WHAT IT IS**

The parroquia or cantón of residence, birth or care, by name — alone or with the cantón or the provincia, `Facundo Vela – Bolívar`.

**HOW IT IS FOUND**

**COLUMN NAME** `EC_L2_PARROQUIA – Path`

**85%** `parroquia(_?(residencia|nacimiento|domicilio))?` *(except when followed by* `cod`, `codigo`, `id`, `eclesiastica`*)* `cant[oó]n(_?(residencia|nacimiento|domicilio))?` *(except when followed by* `cod`, `codigo`, `id`*)* `canton` *(except when followed by* `cod`, `codigo`, `id`*)* `ciudad(_?(residencia|nacimiento|domicilio|cliente))?` `(nom|nombre|desc)_?(parroquia|canton|ciudad)` `lugar_?(de_?)?nacimiento` `city` `town`

**VALUE FORMAT** `EC_L2_PARROQUIA – Values`

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 980 known values, ignoring case — Guaranda, Facundo Vela, Julio E. Moreno Catanahuan Grande, San Lorenzo, San Simón Yacoto, Santa Fe Santa Fe, … | 80% | |

A value that matches none counts 40% against the data class.

**HOW IT IS MASKED**

A parroquia with fewer than 20,000 inhabitants becomes the nearest parroquia of its cantón with at least 20,000 — or of its provincia when the cantón has none. The 93 cantones under 20,000 follow the same rule inside their provincia. The 24 provincias stay.

`EC_PARROQUIA` **DATA CLEANSING**

Replaces each value found in `ec-parroquias-generalizadas.txt` (25,214 pairs) by its counterpart; values not in the file stay as they are.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Chavezpamba | → | **Pomasqui** |
| Quito | → | **Quito** |
| Facundo Vela – Bolívar | → | **Guaranda – Bolívar** |
| Isabela | → | **Santa Cruz** |

## 9.7 DPA code

EC_L2_DPA

COLUMN NAME · VALUE FORMAT · REGEX DECOMPOSE

WHAT IT IS

The código de la División Política Administrativa: two digits for the provincia, four for the cantón and six for the parroquia — the same six digits the postal code uses.

HOW IT IS FOUND

COLUMN NAME EC_L2_DPA – Path

85% `dpa[a-z0-9_]*` `(cod|codigo|cd|id)_?(dpa|parroquia|canton|cant|ciudad|lugar)(_?(inei|inec|censo|res|residencia|nac|nacimiento))?` `(cod|codigo)_?(inec|geogr[aá]fico|divisi[oó]n_?pol[ií]tica)` `codparroquia` `codcanton`

VALUE FORMAT EC_L2_DPA – Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `0*(?:(0[1-9]\|1\d\|2[0-4]\|90)\d{2}(\d{2})?)` | 40% | |

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

The same generalization as the name: a small parroquia's code becomes the code of the parroquia that replaces its name. The code of a large one does not change.

`EC_DPA` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

> `EC_DPA_TABLA` DATA CLEANSING
>
> Replaces each value found in `ec-dpa-generalizado.txt` (1,441 pairs) by its counterpart; values not in the file stay as they are.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 170158 | → | **170177** |
| 170150 | → | **170150** |

## 9.8 Ciudadela, urbanización and barrio

EC_L2_CIUDADELA

COLUMN NAME · SECURE LOOKUP

WHAT IT IS

The ciudadela, urbanización, cooperativa, conjunto, recinto, comuna or barrio of residence — the unit below the parroquia, and the one a delivery address really uses.

HOW IT IS FOUND

COLUMN NAME EC_L2_CIUDADELA — Path

80% `ciudadela` `cdla` `urbanizaci[oó]n` `urb` `barrio(_?(residencia|domicilio))?` `lotizaci[oó]n` `cooperativa_?(de_?)?vivienda` `conjunto_?(habitacional|residencial)` `recinto` `comuna` `sector(_?(vivienda|domicilio|residencia))?` `(nom|nombre)_?(ciudadela|barrio|urbanizacion)` `neighbou?rhood`

HOW IT IS MASKED

Replaced by a common Ecuadorian one, unrelated to the original.

`EC_CIUDADELA` SECURE LOOKUP

Replaces the value with one of the 108 values in `ec-ciudadelas.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Cdla. Kennedy | → | **Cdla. Los Almendros** |

## 9.9 Postal code

EC_L2_CODIGO_POSTAL

COLUMN NAME DATA TYPE REGEX DECOMPOSE

WHAT IT IS

The six-digit postal code in use since 2007: it is the DPA code, so the first two digits are the provincia.

HOW IT IS FOUND

COLUMN NAME EC_L2_CODIGO_POSTAL — Path

85% `c[oó]digo_?postal` `cod_?postal` `codpostal` `zip(_?code)?` `postal_?code` `cp`

DATA TYPE EC_L2_CODIGO_POSTAL — Type

Accepts text (length 5 to 6) or number (length up to 6); any other type rules the data class out.

HOW IT IS MASKED

The first two digits stay and the others become zeros: `170150` becomes `170000`.

`EC_CODIGO_POSTAL` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 170150 | → | **170000** |

## 9.10 Marital status

EC_L2_ESTADO_CIVIL

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

Single, married, in a union of fact — which the Constitution puts on the same footing as marriage —, separated, divorced or widowed, in words or as a code.

HOW IT IS FOUND

COLUMN NAME EC_L2_ESTADO_CIVIL – Path

85% `estado_?civil` `est_?civil` `(cod|tipo|id)_?estado_?civil` `situaci[oó]n_?conyugal` `uni[oó]n_?de_?hecho` `marital_?status` `civil_?status`

VALUE FORMAT EC_L2_ESTADO_CIVIL – Values

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 20 known values, ignoring case — Soltero a, Casado a, Unión de hecho, Separado a, Divorciado a, Viudo a, … | 70% | |

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

Replaced by another status from the list; a code stays a code.

`EC_ESTADO_CIVIL` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_ESTADO_CIVIL_LISTA`.

`EC_BANDERA` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`EC_BANDERA_SI_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-si-no.txt`, chosen by a keyed hash.

`EC_BANDERA_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-true-false.txt`, chosen by a keyed hash.

`EC_BANDERA_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-yes-no.txt`, chosen by a keyed hash.

`EC_BANDERA_S_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-s-n.txt`, chosen by a keyed hash.

`EC_BANDERA_Y_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-y-n.txt`, chosen by a keyed hash.

`EC_BANDERA_0_1` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-0-1.txt`, chosen by a keyed hash.

`EC_ESTADO_CIVIL_CODIGO` SECURE LOOKUP

Replaces the value with one of the 6 values in `ec-estado-civil-codigos.txt`, chosen by a keyed hash.

`EC_ESTADO_CIVIL_LISTA` SECURE LOOKUP

Replaces the value with one of the 6 values in `ec-estado-civil.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Unión libre | → | **Divorciado/a** |

## 9.11 Occupation

EC_L2_OCUPACION

COLUMN NAME REGEX DECOMPOSE

WHAT IT IS

Occupation, profession or trade, in words or by the CIUO code. A rare occupation in a small parroquia identifies on its own.

HOW IT IS FOUND

COLUMN NAME EC_L2_OCUPACION — Path

80% `ocupaci[oó]n` `profesi[oó]n` `oficio` `ciuo` `(cod|codigo)_?(ocupacion|profesion|ciuo)` `puesto_?(de_?)?trabajo` `categor[ií]a_?ocupacional` `cargo_?(actual|empleado)` `occupation` `profession` `job_?title`

50% `cargo` `puesto`

HOW IT IS MASKED

Text is replaced by a common occupation; a code is mapped digit by digit.

`EC_OCUPACION_O_CODIGO` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `EC_OCUPACION`.

> `EC_CM_DIGITOS` CHARACTER MAPPING
>
> Maps each character to another of its group (1 groups), keeping length and format.
>
> `EC_OCUPACION` SECURE LOOKUP
>
> Replaces the value with one of the 38 values in `ec-ocupaciones.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Gerente de operaciones | → | **Contador** |
| 2221 | → | **1980** |

---

## 9.12 Employer

EC_L2_EMPLEADOR

COLUMN NAME SECURE LOOKUP

WHAT IT IS

The name of the company or institution where the person works.

HOW IT IS FOUND

COLUMN NAME EC_L2_EMPLEADOR — Path

90% `empleador(_?(nombre|raz[oó]n_?social|ruc))?` `(nombre|nom)_?(empleador|empresa_?donde_?trabaja|patrono)` `empresa_?(donde_?)?(trabaja|labora)` `lugar_?(de_?)?trabajo` `centro_?(de_?)?(trabajo|labores)` `entidad_?empleadora` `employer(_?name)?` `workplace`

HOW IT IS MASKED

Replaced by a fictitious company.

`EC_EMPLEADOR` SECURE LOOKUP

Replaces the value with one of the 24 values in `ec-empleadores.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Corporación Comercial Ecuatoriana S.A. | → | **Cooperativa Agrícola La Unión** |

---

## 9.13 Education

EC_L2_NIVEL_INSTRUCCION

COLUMN NAME | VALUE FORMAT | REGEX DECOMPOSE

### WHAT IT IS

The level of instruction attained — from none to a postgraduate degree — in the categories the census uses, in words or as a code.

### HOW IT IS FOUND

**COLUMN NAME** EC_L2_NIVEL_INSTRUCCION — Path

80% `escolaridad` `nivel_?(educativo|de_?estudios|acad[eé]mico|alcanzado|de_?instrucci[oó]n)` `instrucci[oó]n(_?formal)?` `grado_?(de_?)?(estudio|instrucci[oó]n|escolaridad)` `m[aá]ximo_?nivel` `a[nñ]os_?(de_?)?estudio` `education(_?level)?`

**VALUE FORMAT** EC_L2_NIVEL_INSTRUCCION — Values

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 20 known values, ignoring case — Ninguno, Centro de alfabetización, Educación inicial, Educación general básica, Bachillerato, Superior no universitaria, … | 60% | |

A value that matches none counts 40% against the data class.

### HOW IT IS MASKED

Replaced by another level from the list; a code stays a code.

`EC_NIVEL_INSTRUCCION` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_NIVEL_INSTRUCCION_LISTA` .

> `EC_BANDERA` REGEX DECOMPOSE
>
> Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `EC_BANDERA_SI_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-si-no.txt` , chosen by a keyed hash.
> >
> > `EC_BANDERA_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-true-false.txt` , chosen by a keyed hash.
> >
> > `EC_BANDERA_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-yes-no.txt` , chosen by a keyed hash.
> >
> > `EC_BANDERA_S_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-s-n.txt` , chosen by a keyed hash.
> >
> > `EC_BANDERA_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-y-n.txt` , chosen by a keyed hash.
> >
> > `EC_BANDERA_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-0-1.txt` , chosen by a keyed hash.
>
> `EC_NIVEL_INSTRUCCION_CODIGO` SECURE LOOKUP
>
> Replaces the value with one of the 8 values in `ec-nivel-instruccion-codigos.txt` , chosen by a keyed hash.
>
> `EC_NIVEL_INSTRUCCION_LISTA` SECURE LOOKUP
>
> Replaces the value with one of the 8 values in `ec-nivel-instruccion.txt` , chosen by a keyed hash.

### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `Educación superior en curso` | → | **`Bachillerato`** |

## 9.14 School

EC_L2_INSTITUCION_EDUCATIVA

COLUMN NAME | SECURE LOOKUP

### WHAT IT IS

The unidad educativa, institute or university, by name or by the Ministry of Education AMIE code.

### HOW IT IS FOUND

> **COLUMN NAME** EC_L2_INSTITUCION_EDUCATIVA — Path
>
> 75% `unidad_?educativa` `instituci[oó]n_?educativa` `centro_?(educativo|de_?estudios)` `(nombre|nom)_?(institucion_?educativa|escuela|colegio|universidad)` `escuela` `colegio` `instituto` `universidad` `c[oó]digo_?amie` `amie` `school(_?name)?` `university`

### HOW IT IS MASKED

Replaced by a fictitious school.

`EC_INSTITUCION_EDUCATIVA` SECURE LOOKUP

Replaces the value with one of the 12 values in `ec-instituciones-educativas.txt`, chosen by a keyed hash.

### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Universidad Central del Ecuador | → | **Escuela Politécnica del Austro** |

---

## 9.15 Dependants and children

EC_L2_PERSONAS_A_CARGO

COLUMN NAME | DATA TYPE | REGEX DECOMPOSE

### WHAT IT IS

Number of dependants, children, derechohabientes or household members.

### HOW IT IS FOUND

> **COLUMN NAME** EC_L2_PERSONAS_A_CARGO — Path
>
> 80% `personas_?a_?cargo` `cargas_?familiares` `n(o|ro|um|umero)_?(de_?)?(hijos|dependientes|cargas|derechohabientes|personas_?a_?cargo)` `cant(idad)?_?(hijos|dependientes|personas_?hogar|miembros)` `hijos` `dependientes` `derechohabientes` `personas_?(en_?el_?)?hogar` `tama[nñ]o_?(del_?)?hogar` `miembros_?(del_?)?hogar` `number_?of_?children` `dependents`

> **DATA TYPE** EC_L2_PERSONAS_A_CARGO — Type
>
> Accepts number (length up to 4) or text (length up to 4); any other type rules the data class out.

### HOW IT IS MASKED

Values from 0 to 5 stay; above 5 they become 5. Very large households are rare, and a rare count identifies.

`EC_CONTEO_LIMITADO` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 9 | → | **5** |

PART 10

# Sensitive data (art. 4)

The list of article 4 is the longest in the region, and three of its items appear in no neighbouring law: the **identidad cultural**, the **condición migratoria** and the **pasado judicial**. Health has a group of its own below, and the judicial record the PENAL group; everything else is here, together with the trade union membership and the victim status the open clause of the same article reaches.

## 10.1 Ethnicity

EC_L3_ETNIA

COLUMN NAME | VALUE FORMAT | REGEX DECOMPOSE

### WHAT IT IS

Self-identification as mestizo, montubio, afroecuatoriano, negro, mulato, indígena or blanco — the categories the 2022 census asked. Article 4 names the etnia first among the sensitive data.

### HOW IT IS FOUND

COLUMN NAME — EC_L3_ETNIA — Path

95% `etnia` `origen_?([eé]tnico|racial)` `grupo_?[eé]tnico` `pertenencia_?[eé]tnica` `autoidentificaci[oó]n(_?[eé]tnica)?` `autodefinici[oó]n` `(cod|codigo|tipo|id)_?(etnia|autoidentificacion)` `raza` `ind[ií]gena` `montubio` `afroecuatoriano` `afrodescendiente` `ethnicity` `race`

VALUE FORMAT — EC_L3_ETNIA — Values

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 18 known values, ignoring case — Mestizo, Montubio, Afroecuatoriano, Negro, Mulato, Indígena, … | 90% | |
| One of 10 known values, ignoring case — s, n, sí, no, 1, 2, … | 30% | |

A value that matches none counts 40% against the data class.

### HOW IT IS MASKED

A flag stays a flag, a code is replaced by another code and any other value by a category from the list.

`EC_AUTOIDENTIFICACION` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_AUTOIDENTIFICACION_LISTA` .

`EC_BANDERA` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`EC_BANDERA_SI_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-si-no.txt` , chosen by a keyed hash.

`EC_BANDERA_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-true-false.txt` , chosen by a keyed hash.

`EC_BANDERA_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-yes-no.txt` , chosen by a keyed hash.

`EC_BANDERA_S_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-s-n.txt` , chosen by a keyed hash.

`EC_BANDERA_Y_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-y-n.txt` , chosen by a keyed hash.

`EC_BANDERA_0_1` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-0-1.txt` , chosen by a keyed hash.

`EC_AUTOIDENTIFICACION_CODIGO` SECURE LOOKUP

Replaces the value with one of the 8 values in `ec-autoidentificacion-codigos.txt`, chosen by a keyed hash.

`EC_AUTOIDENTIFICACION_LISTA` SECURE LOOKUP

Replaces the value with one of the 8 values in `ec-autoidentificacion.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Indígena | → | **Otro** |
| S | → | **S** |

## 10.2 Cultural identity

EC_L3_IDENTIDAD_CULTURAL

COLUMN NAME · VALUE FORMAT · REGEX DECOMPOSE

WHAT IT IS

The **identidad cultural** article 4 names apart from the etnia: the indigenous nationality or pueblo a person belongs to. The Constitution recognizes fourteen nationalities and eighteen kichwa pueblos, and naming one of them names a few thousand people. No other law of the region lists this.

HOW IT IS FOUND

COLUMN NAME EC_L3_IDENTIDAD_CULTURAL — Path

90% `identidad_?cultural` `nacionalidad_?(ind[ií]gena|originaria)` `pueblo_?(ind[ií]gena|originario|al_?que_?pertenece)` `pertenece_?(a_?)?` `(pueblo|nacionalidad|comunidad)` `comunidad_?(ind[ií]gena|ancestral)` `nacionalidad_?(ecuatoriana)?` *(except when followed by* `pais`, `extranjera`, `documento`*)* `indigenous_?people`

VALUE FORMAT EC_L3_IDENTIDAD_CULTURAL — Values

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 54 known values, ignoring case — Kichwa, Shuar, Achuar, Awá, Chachi, Épera, … | 85% | |

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

Replaced by another nationality or pueblo from the list; a flag stays a flag.

`EC_IDENTIDAD_CULTURAL` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_IDENTIDAD_CULTURAL_LISTA`.

`EC_BANDERA` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`EC_BANDERA_SI_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-si-no.txt`, chosen by a keyed hash.

`EC_BANDERA_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-true-false.txt`, chosen by a keyed hash.

`EC_BANDERA_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-yes-no.txt`, chosen by a keyed hash.

`EC_BANDERA_S_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-s-n.txt`, chosen by a keyed hash.

`EC_BANDERA_Y_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-y-n.txt`, chosen by a keyed hash.

`EC_BANDERA_0_1` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-0-1.txt`, chosen by a keyed hash.

`EC_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

`EC_IDENTIDAD_CULTURAL_LISTA` SECURE LOOKUP

Replaces the value with one of the 30 values in `ec-identidad-cultural.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Pueblo Manta-Huancavilca | → | **Siekopai** |

## 10.3 Language

EC_L3_LENGUA

COLUMN NAME | VALUE FORMAT | REGEX DECOMPOSE

### WHAT IT IS

The mother or ancestral language. Thirteen indigenous languages are spoken in Ecuador beside Spanish, and some by a few hundred people: the language says who a person is.

### HOW IT IS FOUND

COLUMN NAME EC_L3_LENGUA – Path

90% `lengua(_?(ind[ií]gena|materna|nativa|originaria|ancestral|de_?se[ñn]as))?` `idioma(_?(ind[ií]gena|materno|nativo|ancestral))?` `habla_?(lengua|idioma)` `mother_?tongue` `native_?language`

VALUE FORMAT EC_L3_LENGUA – Values

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 25 known values, ignoring case — Castellano, Kichwa, Shuar chicham, Achuar chicham, Awapit, Cha palaa, … | 80% | |

A value that matches none counts 40% against the data class.

### HOW IT IS MASKED

Replaced by a language from the list; a flag stays a flag.

`EC_LENGUA` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_LENGUA_LISTA`.

`EC_BANDERA` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`EC_BANDERA_SI_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-si-no.txt`, chosen by a keyed hash.

`EC_BANDERA_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-true-false.txt`, chosen by a keyed hash.

`EC_BANDERA_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-yes-no.txt`, chosen by a keyed hash.

`EC_BANDERA_S_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-s-n.txt`, chosen by a keyed hash.

`EC_BANDERA_Y_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-y-n.txt`, chosen by a keyed hash.

`EC_BANDERA_0_1` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-0-1.txt`, chosen by a keyed hash.

`EC_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

`EC_LENGUA_LISTA` SECURE LOOKUP

Replaces the value with one of the 16 values in `ec-lenguas.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Kichwa de la sierra | → | **Otra** |

## 10.4 Migratory status

EC_L3_CONDICION_MIGRATORIA

COLUMN NAME · VALUE FORMAT · REGEX DECOMPOSE

WHAT IT IS

The **condición migratoria** article 4 lists among the sensitive data, and the stateless and refugee status the same article names: residence category, refugee or asylum application, regular or irregular status, human mobility. In Ecuador this column separates the nationals from the Venezuelan and Colombian communities.

HOW IT IS FOUND

COLUMN NAME EC_L3_CONDICION_MIGRATORIA — Path

95% `condici[oó]n_?migratoria` `estatus_?migratorio` `situaci[oó]n_?migratoria` `categor[ií]a_?migratoria` `movilidad_?humana` `refugiad[oa]` `solicitante_?(de_?)?(refugio|asilo)` `asilad[oa]` `ap[aá]trida` `protecci[oó]n_?internacional` `migrante` `desplazad[oa]` `retornad[oa]` `deportad[oa]` `migratory_?status` `refugee`

VALUE FORMAT EC_L3_CONDICION_MIGRATORIA — Values

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 19 known values, ignoring case — Ecuatoriano, Residente permanente, Residente temporal, Visitante temporal, Refugiado, Solicitante de refugio, … | 80% | |

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

Replaced by another status from the list; a flag stays a flag.

`EC_CONDICION_MIGRATORIA` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_CONDICION_MIGRATORIA_LISTA`.

`EC_BANDERA` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`EC_BANDERA_SI_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-si-no.txt`, chosen by a keyed hash.

`EC_BANDERA_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-true-false.txt`, chosen by a keyed hash.

`EC_BANDERA_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-yes-no.txt`, chosen by a keyed hash.

`EC_BANDERA_S_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-s-n.txt`, chosen by a keyed hash.

`EC_BANDERA_Y_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-y-n.txt`, chosen by a keyed hash.

`EC_BANDERA_0_1` SECURE LOOKUP

> Replaces the value with one of the 2 values in `ec-bandera-0-1.txt`, chosen by a keyed hash.

`EC_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

`EC_CONDICION_MIGRATORIA_LISTA` SECURE LOOKUP

Replaces the value with one of the 10 values in `ec-condicion-migratoria.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Residente en trámite | → | **En situación irregular** |

## 10.5 Nationality

EC_L3_NACIONALIDAD

COLUMN NAME KNOWN VALUES SECURE LOOKUP

WHAT IT IS

The nationality of a country and the country of birth or origin. Article 4 does not name it, but it travels with the migratory status it does name.

HOW IT IS FOUND

COLUMN NAME EC_L3_NACIONALIDAD — Path

85% `nacionalidad_?(pais|extranjera|documento)` `pa[ií]s_?(de_?)?(nacimiento|origen|nacionalidad|procedencia)` `ciudadan[ií]a` `nationality` `citizenship` `country_?of_?birth`

KNOWN VALUES EC_L3_NACIONALIDAD — List

| LIST | VALUES | STRENGTH | FIRST VALUES |
|---|---|---|---|
| `ec-detectar-nacionalidades.txt` | 32 | 70% | Ecuatoriana, Venezolana, Colombiana, Peruana, Cubana, Haitiana |

Values are split into words, and each word is looked up.

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

Replaced by a nationality from the list: the Ecuadorian one and the largest foreign communities.

`EC_NACIONALIDAD_PAIS` SECURE LOOKUP

Replaces the value with one of the 14 values in `ec-nacionalidades.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Panameña | → | **Boliviana** |

## 10.6 Religion

EC_L3_RELIGION

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

The religion, church or congregation the person belongs to.

HOW IT IS FOUND

COLUMN NAME EC_L3_RELIGION — Path

95% `religi[oó]n` `creencia_?(religiosa|espiritual)` `convicci[oó]n_?(religiosa|espiritual)` `credo` `confesi[oó]n_?religiosa` `culto` `iglesia(_?(a_?la_?que_?pertenece|nombre))?` `denominaci[oó]n_?religiosa` `(cod|codigo|tipo)_?religion` `religion` `church`

VALUE FORMAT EC_L3_RELIGION — Values

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 24 known values, ignoring case — Católica, Evangélica, Testigo de Jehová, Iglesia de Jesucristo de los Santos de los Últimos Días, Adventista, Protestante, … | 90% | |

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

A flag stays a flag and any other value is replaced by a religion from the list.

`EC_RELIGION` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_RELIGION_LISTA`.

`EC_BANDERA` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`EC_BANDERA_SI_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-si-no.txt`, chosen by a keyed hash.

`EC_BANDERA_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-true-false.txt`, chosen by a keyed hash.

`EC_BANDERA_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-yes-no.txt`, chosen by a keyed hash.

`EC_BANDERA_S_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-s-n.txt`, chosen by a keyed hash.

`EC_BANDERA_Y_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-y-n.txt`, chosen by a keyed hash.

`EC_BANDERA_0_1` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-0-1.txt`, chosen by a keyed hash.

`EC_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

`EC_RELIGION_LISTA` SECURE LOOKUP

Replaces the value with one of the 10 values in `ec-religiones.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Ortodoxa | → | **Otra** |

## 10.7 Ideology

EC_L3_IDEOLOGIA

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

The **ideología** article 4 names on its own, beside the filiación política: leaning, sympathy, voting intention, and the philosophical or moral convictions of the same family.

HOW IT IS FOUND

COLUMN NAME EC_L3_IDEOLOGIA — Path

90% `ideolog[ií]a(_?pol[ií]tica)?` `opini[oó]n_?pol[ií]tica` `orientaci[oó]n_?pol[ií]tica` `tendencia_?pol[ií]tica` `preferencia_?(pol[ií]tica|partidaria|electoral)` `intenci[oó]n_?(de_?)?voto` `simpat[ií]a_?pol[ií]tica` `convicci[oó]n_?(filos[oó]fica|moral)` `political_?(opinion|orientation|view)` `voting_?intention`

VALUE FORMAT EC_L3_IDEOLOGIA — Values

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 16 known values, ignoring case — Izquierda, Centroizquierda, Centro, Centroderecha, Derecha, Independiente, … | 70% | |

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

Replaced by a position from the list — left to right, independent.

`EC_IDEOLOGIA` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_IDEOLOGIA_LISTA` .

`EC_BANDERA` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`EC_BANDERA_SI_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-si-no.txt` , chosen by a keyed hash.

`EC_BANDERA_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-true-false.txt` , chosen by a keyed hash.

`EC_BANDERA_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-yes-no.txt` , chosen by a keyed hash.

`EC_BANDERA_S_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-s-n.txt` , chosen by a keyed hash.

`EC_BANDERA_Y_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-y-n.txt` , chosen by a keyed hash.

`EC_BANDERA_0_1` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-0-1.txt` , chosen by a keyed hash.

`EC_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

`EC_IDEOLOGIA_LISTA` SECURE LOOKUP

Replaces the value with one of the 7 values in `ec-ideologias.txt` , chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `Oficialista` | → | **`Centro`** |

## 10.8 Political affiliation

EC_L3_FILIACION_POLITICA

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

The political organization the person belongs to, from the register of the Consejo Nacional Electoral.

HOW IT IS FOUND

COLUMN NAME EC_L3_FILIACION_POLITICA — Path

95% `partido(_?pol[ií]tico)?(_?(afiliaci[oó]n|nombre))?` `organizaci[oó]n_?pol[ií]tica` `movimiento_?pol[ií]tico` `filiaci[oó]n_?(pol[ií]tica|partidaria)` `afiliaci[oó]n_?(pol[ií]tica|partidaria|partido)` `militancia(_?pol[ií]tica)?` `militante` `(cod|codigo|nombre|nom)_?partido` `political_?party` `party_?membership`

VALUE FORMAT EC_L3_FILIACION_POLITICA — Values

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 19 known values, ignoring case — Revolución Ciudadana, Acción Democrática Nacional, Movimiento CREO, Partido Social Cristiano, Pachakutik, Izquierda Democrática, … | 80% | |

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

Replaced by another organization from the list; a flag stays a flag.

`EC_PARTIDO` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_PARTIDO_LISTA`.

`EC_BANDERA` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`EC_BANDERA_SI_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-si-no.txt`, chosen by a keyed hash.

`EC_BANDERA_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-true-false.txt`, chosen by a keyed hash.

`EC_BANDERA_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-yes-no.txt`, chosen by a keyed hash.

`EC_BANDERA_S_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-s-n.txt`, chosen by a keyed hash.

`EC_BANDERA_Y_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-y-n.txt`, chosen by a keyed hash.

`EC_BANDERA_0_1` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-0-1.txt`, chosen by a keyed hash.

`EC_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

`EC_PARTIDO_LISTA` SECURE LOOKUP

Replaces the value with one of the 13 values in `ec-partidos.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `Movimiento Suma` | → | **Avanza** |
| `S` | → | **S** |

## 10.9 Trade union membership

EC_L3_AFILIACION_SINDICAL

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

The union, the comité de empresa, the membership and the dues deducted on the payroll. The law does not name it, but the open clause of art. 4 reaches it: it is data whose improper processing discriminates.

HOW IT IS FOUND

COLUMN NAME EC_L3_AFILIACION_SINDICAL — Path

95% `sindicato(_?(nombre|afiliado|afiliaci[oó]n|codigo))?` `sindicalizado` `afiliado_?sindicato` `afiliaci[oó]n_?sindical` `cuota_?sindical` `descuento_?sindical` `aporte_?sindical` `asociaci[oó]n_?(sindical|de_?trabajadores)` `comit[eé]_?de_?empresa` `contrato_?colectivo` `fuero_?sindical` `dirigente_?sindical` `trade_?union` `union_?member(ship)?`

VALUE FORMAT EC_L3_AFILIACION_SINDICAL — Values

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 11 known values, ignoring case — FUT, CEOSL, CEDOCUT, CTE, UGTE, UNE, … | 60% | |

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

A flag stays a flag; any other value becomes `Sin información`.

`EC_CATEGORIA_SUPRIMIDA` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_SUPRIMIR`.

`EC_BANDERA` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`EC_BANDERA_SI_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-si-no.txt`, chosen by a keyed hash.

`EC_BANDERA_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-true-false.txt`, chosen by a keyed hash.

`EC_BANDERA_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-yes-no.txt`, chosen by a keyed hash.

`EC_BANDERA_S_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-s-n.txt`, chosen by a keyed hash.

`EC_BANDERA_Y_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-y-n.txt`, chosen by a keyed hash.

`EC_BANDERA_0_1` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-0-1.txt`, chosen by a keyed hash.

`EC_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

`EC_SUPRIMIR` SECURE LOOKUP

Replaces the value with one of the 1 values in `ec-sin-informacion.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| UNE | → | **Sin información** |
| S | → | **S** |

## 10.10 Biometric data

EC_L3_BIOMETRICO

COLUMN NAME DATA TYPE REGEX DECOMPOSE

WHAT IT IS

Fingerprint templates — the cédula's fingerprint is the country's identification pattern —, face, iris, voice and digitized signature. Article 4 defines the biometric datum as the one that permits or confirms the unique identification of a person.

HOW IT IS FOUND

**COLUMN NAME** EC_L3_BIOMETRICO — Path

90% `biometr[ií]a[a-z_]*` `biom[eé]trico[a-z_]*` `huella(_?(dactilar|digital))?(_?(template|plantilla|imagen|hash))?` `huellas` `dactilar` `dactiloscop[ií]a` `template_?(facial|huella|biometrico)` `plantilla_?(facial|biom[eé]trica)` `rostro_?(hash|template)` `reconocimiento_?facial` `firma_?(biom[eé]trica|digitalizada|electr[oó]nica)` `iris` `voz_?(biom[eé]trica|template)` `fingerprints?` `face_?(template|hash|encoding)` `biometric[a-z_]*`

**DATA TYPE** EC_L3_BIOMETRICO — Type

Accepts text; any other type rules the data class out.

HOW IT IS MASKED

Every character becomes `X`. A masked template authenticates nobody.

`EC_REDACTAR` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| A1B2C3D4E5F6 | → | XXXXXXXXXXXX |

## 10.11 Genetic data

EC_L3_GENETICO

**COLUMN NAME** **VALUE FORMAT** **REGEX DECOMPOSE**

WHAT IT IS

Results of genetic and DNA tests, paternity tests, genetic profiles, mutations, karyotype and newborn screening. Article 4 defines the genetic datum as the one that gives unique information on the physiology or health of an individual.

HOW IT IS FOUND

**COLUMN NAME** EC_L3_GENETICO — Path

90% `adn` `dna` `gen[eé]tic[oa]s?` `datos?_?gen[eé]ticos?` `genoma` `genotip[a-z_]*` `prueba_?(gen[eé]tica|de_?adn|de_?paternidad|de_?filiaci[oó]n)` `perfil_?gen[eé]tico` `mutaci[oó]n` `brca[12]?` `cariotipo` `tamizaje_?neonatal` `genetic[a-z_]*`

**VALUE FORMAT** EC_L3_GENETICO — Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `[ACGTUN]{24,}` | | 90% |

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Every character becomes `X`.

`EC_REDACTAR` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| BRCA1 positivo | → | XXXXXXXXXXXXXX |

## 10.12 Sexual orientation

EC_L3_ORIENTACION_SEXUAL

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

Self-declared sexual orientation or preference, and the sexual life of the same sentence of article 4.

HOW IT IS FOUND

COLUMN NAME EC_L3_ORIENTACION_SEXUAL — Path

95% `orientaci[oó]n_?sexual` `preferencias?_?sexuales?` `(cod|codigo|tipo)_?orientacion_?sexual` `vida_?sexual` `sexual_?orientation`

VALUE FORMAT EC_L3_ORIENTACION_SEXUAL — Values

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 14 known values, ignoring case — Heterosexual, Gay, Lesbiana, Bisexual, Pansexual, Asexual, … | 80% | |

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

Replaced by a category from the list.

`EC_ORIENTACION_SEXUAL` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_ORIENTACION_SEXUAL_LISTA`.

> `EC_BANDERA` REGEX DECOMPOSE
>
> Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `EC_BANDERA_SI_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-si-no.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-true-false.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-yes-no.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_S_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-s-n.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-y-n.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-0-1.txt`, chosen by a keyed hash.
>
> `EC_CM_ALFANUM` CHARACTER MAPPING
>
> Maps each character to another of its group (7 groups), keeping length and format.
>
> `EC_ORIENTACION_SEXUAL_LISTA` SECURE LOOKUP
>
> Replaces the value with one of the 8 values in `ec-orientaciones-sexuales.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Homosexual | → | Pansexual |

## 10.13 Gender identity

EC_L3_IDENTIDAD_GENERO

COLUMN NAME | VALUE FORMAT | REGEX DECOMPOSE

### WHAT IT IS

The **identidad de género** article 4 names second in its list, the social name and the intersex condition.

### HOW IT IS FOUND

COLUMN NAME EC_L3_IDENTIDAD_GENERO — Path

95% `identidad_?(de_?)?g[eé]nero` `(cod|codigo|tipo)_?identidad_?genero` `g[eé]nero_?(autopercibido|identitario)` `expresi[oó]n_?de_?g[eé]nero` `transg[eé]nero` `persona_?trans` `intersexual` `nombre_?social` `pronombres?` `gender_?identity`

VALUE FORMAT EC_L3_IDENTIDAD_GENERO — Values

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 17 known values, ignoring case — Mujer cisgénero, Hombre cisgénero, Mujer trans, Hombre trans, Persona no binaria, Intersexual, … | 70% | |

A value that matches none counts 40% against the data class.

### HOW IT IS MASKED

Replaced by a category from the list.

`EC_IDENTIDAD_GENERO` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_IDENTIDAD_GENERO_LISTA`.

> `EC_BANDERA` REGEX DECOMPOSE
>
> Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `EC_BANDERA_SI_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-si-no.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-true-false.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-yes-no.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_S_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-s-n.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-y-n.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-0-1.txt`, chosen by a keyed hash.
>
> `EC_CM_ALFANUM` CHARACTER MAPPING
>
> Maps each character to another of its group (7 groups), keeping length and format.
>
> `EC_IDENTIDAD_GENERO_LISTA` SECURE LOOKUP
>
> Replaces the value with one of the 8 values in `ec-identidades-genero.txt`, chosen by a keyed hash.

### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Transgénero | → | **Otra** |

## 10.14 Victims of violence

EC_L3_VICTIMA

COLUMN NAME | REGEX DECOMPOSE

### WHAT IT IS

Being a victim of gender, domestic or sexual violence, of trafficking or of enforced disappearance, the type of violence and the protection measures. The Ley Orgánica Integral para Prevenir y Erradicar la Violencia contra las Mujeres protects the identity of the victim.

### HOW IT IS FOUND

COLUMN NAME EC_L3_VICTIMA — Path

90% `v[ií]ctima(_?(de_?)?(violencia(_?(de_?g[eé]nero|intrafamiliar|f[ií]sica|sexual|psicol[oó]gica|econ[oó]mica))?|delito|trata|abuso|desaparici[oó]n))?` `violencia_?(de_?)?g[eé]nero` `violencia_?(intrafamiliar|dom[eé]stica|sexual|psicol[oó]gica)` `tipo_?(de_?)?violencia` `medida_?(de_?)?protecci[oó]n` `boleta_?de_?auxilio` `desaparici[oó]n_?forzada` `trata_?(de_?)?personas` `denunciante_?violencia`

### HOW IT IS MASKED

A flag stays a flag; any other value becomes `Sin información`.

`EC_CATEGORIA_SUPRIMIDA` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_SUPRIMIR`.

> `EC_BANDERA` REGEX DECOMPOSE
>
> Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `EC_BANDERA_SI_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-si-no.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-true-false.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-yes-no.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_S_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-s-n.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-y-n.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-0-1.txt`, chosen by a keyed hash.
>
> `EC_CM_ALFANUM` CHARACTER MAPPING
>
> Maps each character to another of its group (7 groups), keeping length and format.
>
> `EC_SUPRIMIR` SECURE LOOKUP
>
> Replaces the value with one of the 1 values in `ec-sin-informacion.txt`, chosen by a keyed hash.

### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `Violencia intrafamiliar` | → | **`Sin información`** |
| `S` | → | **`S`** |

PART 11

# Health data

Health data is sensitive by article 4 and a special category by article 25 c). Article 31.2 asks in so many words that it be **previously anonymized or pseudonymized whenever possible**, and article 25 d) adds the data of persons with disabilities and of their substitutes as a category of its own.

## 11.1 Health identifiers

EC_SALUD_IDENTIFICADOR

COLUMN NAME | CHARACTER MAPPING

### WHAT IT IS

Clinical record, admission, care, appointment and prescription numbers, and the RDACAA identifier of an outpatient visit.

### HOW IT IS FOUND

COLUMN NAME EC_SALUD_IDENTIFICADOR — Path

85% `n(o|ro|um|umero)_?(historia(_?cl[ií]nica)?|hc|hcu|expediente(_?cl[ií]nico)?|ingreso|atenci[oó]n|consulta|cita|referencia_?m[eé]dica|receta)` `historia_?cl[ií]nica(_?[uú]nica)?` `hc_?(n(o|ro|um|umero))?` `hcu` `n(o|ro|um|umero)_?(asegurado|afiliado_?iess)` `c[oó]digo_?(paciente|historia|atenci[oó]n)` `id_?paciente` `rdacaa`

55% `expediente` `atenci[oó]n` `ingreso` `consulta`

### HOW IT IS MASKED

Character by character, keeping length and format.

`EC_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| HC-2024-00123 | → | TW-3389-01317 |

## 11.2 Health insurance

EC_SALUD_COBERTURA

COLUMN NAME | VALUE FORMAT | REGEX DECOMPOSE

### WHAT IT IS

The IESS general insurance, the Seguro Social Campesino, ISSFA, ISSPOL, the public system or a private insurer, by name or code. Which one a person has says a great deal about them.

### HOW IT IS FOUND

COLUMN NAME EC_SALUD_COBERTURA — Path

85% `cobertura(_?(m[eé]dica|de_?salud|social))?` `seguro_?(m[eé]dico|salud)` `r[eé]gimen(_?(de_?)?salud)?` `tipo_?(de_?)?seguro` `proveedor_?(de_?)?salud` `iess(_?(afiliado|cobertura))?` `issfa` `isspol` `seguro_?campesino` `msp` `aseguradora_?(m[eé]dica|salud)` `plan_?(m[eé]dico|de_?salud)` `medicina_?prepagada`

VALUE FORMAT EC_SALUD_COBERTURA — Values

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 16 known values, ignoring case — IESS Seguro General, Seguro Social Campesino, ISSFA, ISSPOL, Ministerio de Salud Pública, Seguro privado con hospitalización, … | 80% | |

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

A name is replaced by another cover from the list; a code is mapped digit by digit; a flag stays a flag.

`EC_COBERTURA_SALUD` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_COBERTURA` .

> `EC_BANDERA` REGEX DECOMPOSE
>
> Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `EC_BANDERA_SI_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-si-no.txt` , chosen by a keyed hash.
> >
> > `EC_BANDERA_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-true-false.txt` , chosen by a keyed hash.
> >
> > `EC_BANDERA_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-yes-no.txt` , chosen by a keyed hash.
> >
> > `EC_BANDERA_S_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-s-n.txt` , chosen by a keyed hash.
> >
> > `EC_BANDERA_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-y-n.txt` , chosen by a keyed hash.
> >
> > `EC_BANDERA_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-0-1.txt` , chosen by a keyed hash.
>
> `EC_CM_DIGITOS` CHARACTER MAPPING
>
> Maps each character to another of its group (1 groups), keeping length and format.
>
> `EC_COBERTURA` SECURE LOOKUP
>
> Replaces the value with one of the 8 values in `ec-cobertura-salud.txt` , chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| IESS | → | **ISSPOL** |
| 12345 | → | **31564** |

## 11.3 Diagnosis and ICD-10

EC_SALUD_DIAGNOSTICO

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

Diagnoses in words or by ICD-10 code, with a dot ( `F33.1` ) or without ( `F331` ); causes of death, of sick leave and of admission; allergies and comorbidities.

### HOW IT IS FOUND

**COLUMN NAME** EC_SALUD_DIAGNOSTICO — Path

85% `diagn[oó]stico[a-z_]*` `dx(_?(principal|secundario|ingreso|egreso)[0-9]?)?` `cie_?10` `cie(_?(10|11|principal))?` `(cod|codigo)_?(diagnostico|dx|cie)` `impresi[oó]n_?diagn[oó]stica` `enfermedad(es)?[a-z_]*` `patolog[ií]a` `comorbilidad(es)?` `antecedentes?_?(m[eé]dicos|patol[oó]gicos|cl[ií]nicos|personales)` `causa_?(de_?)?(muerte|defunci[oó]n|incapacidad|hospitalizaci[oó]n|consulta)` `motivo_?(de_?)?(consulta|reposo|hospitalizaci[oó]n)` `alergias?` `diagnos(is|es)` `icd_?(10|11)?`

**VALUE FORMAT** EC_SALUD_DIAGNOSTICO — Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `[A-TV-Za-tv-z]\d{2}(\.\d{1,2}\|\d\|X)?` | 50% | |

A value that matches none counts 30% against the data class.

### HOW IT IS MASKED

An ICD-10 code becomes another common code in the same format, and text a common diagnosis: an HIV or mental disorder diagnosis does not survive as such.

`EC_DIAGNOSTICO` **REGEX DECOMPOSE**

Tries 3 patterns in order and gives each part of the value its own action; when none matches: `EC_DIAGNOSTICO_TEXTO`.

> `EC_CIE10_DECIMAL` **SECURE LOOKUP**
>
> Replaces the value with one of the 33 values in `ec-cie10-decimal.txt`, chosen by a keyed hash.
>
> `EC_CIE10` **SECURE LOOKUP**
>
> Replaces the value with one of the 37 values in `ec-cie10.txt`, chosen by a keyed hash.
>
> `EC_BANDERA` **REGEX DECOMPOSE**
>
> Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `EC_BANDERA_SI_NO` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `ec-bandera-si-no.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_TRUE_FALSE` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `ec-bandera-true-false.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_YES_NO` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `ec-bandera-yes-no.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_S_N` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `ec-bandera-s-n.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_Y_N` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `ec-bandera-y-n.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_0_1` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `ec-bandera-0-1.txt`, chosen by a keyed hash.
>
> `EC_DIAGNOSTICO_TEXTO` **SECURE LOOKUP**
>
> Replaces the value with one of the 30 values in `ec-diagnosticos.txt`, chosen by a keyed hash.

### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| F33.1 | → | **K21.9** |
| B24X | → | **R51** |

## 11.4 Procedures

EC_SALUD_PROCEDIMIENTO

COLUMN NAME | REGEX DECOMPOSE

WHAT IT IS

Procedures, tests and surgeries, in words or by code. A viral load or a pregnancy test reveals the diagnosis.

HOW IT IS FOUND

COLUMN NAME EC_SALUD_PROCEDIMIENTO — Path

85% `procedimiento(_?(m[eé]dico|quir[uú]rgico|realizado|autorizado))?` `(cod|codigo)_?(procedimiento|prestaci[oó]n|servicio_?salud|cpt)` `cpt` `prestaci[oó]n_?(m[eé]dica|de_?salud)` `examen_?(ordenado|realizado)` `cirug[ií]a(_?realizada)?` `intervenci[oó]n_?quir[uú]rgica` `medical_?procedure`

HOW IT IS MASKED

A code is mapped digit by digit; text is replaced by a common procedure.

`EC_PROCEDIMIENTO` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `EC_PROCEDIMIENTO_TEXTO`.

> `EC_CM_DIGITOS` CHARACTER MAPPING
>
> Maps each character to another of its group (1 groups), keeping length and format.
>
> `EC_PROCEDIMIENTO_TEXTO` SECURE LOOKUP
>
> Replaces the value with one of the 10 values in `ec-procedimientos.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 901234 | → | **928120** |

---

## 11.5 Medicines

EC_SALUD_MEDICAMENTO

COLUMN NAME | VALUE FORMAT | REGEX DECOMPOSE

WHAT IT IS

Prescribed or dispensed medicines, by name or active ingredient. An antiretroviral, an antipsychotic or a contraceptive says what the diagnosis column was masked for.

HOW IT IS FOUND

COLUMN NAME EC_SALUD_MEDICAMENTO — Path

85% `medicamentos?(_?(recetado|prescrito|entregado|despachado|nombre|uso))?` `f[aá]rmacos?` `principio_?activo` `denominaci[oó]n_?com[uú]n_?internacional` `dci` `receta(_?m[eé]dica)?` `prescripci[oó]n(_?medicamento)?` `posolog[ií]a` `tratamiento_?farmacol[oó]gico` `(cod|codigo)_?medicamento` `cuadro_?nacional_?(de_?)?medicamentos` `medications?` `drugs?_?prescribed`

VALUE FORMAT EC_SALUD_MEDICAMENTO — Values

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 46 known values, ignoring case — Paracetamol, Ibuprofeno, Diclofenaco, Metformina, Glibenclamida, Losartán, … | 70% | |

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Replaced by a common medicine from the list; a flag stays a flag.

`EC_MEDICAMENTO` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_MEDICAMENTO_LISTA` .

> `EC_BANDERA` REGEX DECOMPOSE
>
> Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `EC_BANDERA_SI_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-si-no.txt` , chosen by a keyed hash.
> >
> > `EC_BANDERA_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-true-false.txt` , chosen by a keyed hash.
> >
> > `EC_BANDERA_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-yes-no.txt` , chosen by a keyed hash.
> >
> > `EC_BANDERA_S_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-s-n.txt` , chosen by a keyed hash.
> >
> > `EC_BANDERA_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-y-n.txt` , chosen by a keyed hash.
> >
> > `EC_BANDERA_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-0-1.txt` , chosen by a keyed hash.
>
> `EC_CM_ALFANUM` CHARACTER MAPPING
>
> Maps each character to another of its group (7 groups), keeping length and format.
>
> `EC_MEDICAMENTO_LISTA` SECURE LOOKUP
>
> Replaces the value with one of the 22 values in `ec-medicamentos.txt` , chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `Sertralina 50 mg` | → | **`Complejo B`** |

---

## 11.6 Clinical text

`EC_SALUD_TEXTO_CLINICO`

COLUMN NAME | DATA TYPE | FREE TEXT REDACTION

WHAT IT IS

History, present illness, progress notes, physical examination, treatment plan, discharge summaries and nursing notes.

HOW IT IS FOUND

> COLUMN NAME `EC_SALUD_TEXTO_CLINICO – Path`
>
> 85% `anamnesis` `evoluci[oó]n(_?(m[eé]dica|cl[ií]nica|enfermer[ií]a))?` `enfermedad_?actual` `examen_?f[ií]sico` `plan_?(de_?)?(manejo|tratamiento)` `indicaciones_?m[eé]dicas` `conducta(_?m[eé]dica)?` `epicrisis` `resumen_?(de_?)?(alta|egreso|atenci[oó]n)` `nota_?(m[eé]dica|de_?enfermer[ií]a|cl[ií]nica)` `notas?_?(m[eé]dicas|cl[ií]nicas)` `triaje(_?texto)?` `informe_?(patolog[ií]a|radiolog[ií]a|m[eé]dico)` `clinical_?notes?` `discharge_?summary`

> DATA TYPE `EC_SALUD_TEXTO_CLINICO – Type`
>
> Accepts text (length 30 or more); any other type rules the data class out.

HOW IT IS MASKED

The same Free Text Redaction as free text: documents, contacts and names from the list are redacted. The clinical content stays, because that is what makes the copy useful.

`EC_TEXTO_LIBRE` FREE TEXT REDACTION

Replaces the text matching 8 patterns with `[DATO]`, and the 1,668 terms of `ec-texto-nombres.txt` with `[NOMBRE]`.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `Paciente Zambrano, cédula 1710034065, consulta por cefalea.` | → | **`Paciente [NOMBRE], cédula [DATO], consulta por cefalea.`** |

## 11.7 Test results

EC_SALUD_RESULTADO_EXAMEN

COLUMN NAME | CHARACTER MAPPING

WHAT IT IS

Laboratory and test results — glucose, pregnancy, toxicology, blood alcohol —, also as numeric values.

HOW IT IS FOUND

COLUMN NAME EC_SALUD_RESULTADO_EXAMEN – Path

80% `resultado_?(del_?)?` `(examen|prueba|laboratorio|pcr|serolog[ií]a|biopsia|glucosa|citolog[ií]a|papanicolaou)` `(examen|prueba)_?(resultado|laboratorio)` `glucemia` `hemoglobina(_?glicosilada)?` `hba1c` `colesterol` `imc` `presi[oó]n_?arterial` `prueba_?(de_?)?embarazo` `toxicol[oó]gico` `alcoholemia` `lab_?results?` `test_?results?`

50% `resultado` `prueba`

HOW IT IS MASKED

Character by character, keeping the format: a number stays a number.

`EC_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `Reactivo` | → | **`Quilqaju`** |

## 11.8 HIV and sexually transmitted infections

EC_SALUD_VIH

COLUMN NAME | REGEX DECOMPOSE

WHAT IT IS

Serological status, viral load and antiretroviral treatment, hepatitis, tuberculosis and sexually transmitted infections. The Ley de Prevención y Atención Integral del VIH/SIDA orders confidentiality and forbids demanding a test to get or keep a job.

HOW IT IS FOUND

> COLUMN NAME EC_SALUD_VIH — Path
>
> 90% `vih(_?(estado|resultado|prueba|diagn[oó]stico|positivo))?` `hiv` `sida` `aids` `serolog[ií]a_?(vih|hiv)` `estado_?serol[oó]gico` `carga_?viral` `cd4` `tarv` `antirretroviral(es)?` `its` `ets` `infecci[oó]n(es)?_?(de_?)?transmisi[oó]n_?sexual` `s[ií]filis` `hepatitis_?[bc]` `tuberculosis` `tb`

HOW IT IS MASKED

A flag stays a flag; any other value becomes `Sin información`.

`EC_CATEGORIA_SUPRIMIDA` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_SUPRIMIR`.

> `EC_BANDERA` REGEX DECOMPOSE
>
> Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `EC_BANDERA_SI_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-si-no.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-true-false.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-yes-no.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_S_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-s-n.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-y-n.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-0-1.txt`, chosen by a keyed hash.
>
> `EC_CM_ALFANUM` CHARACTER MAPPING
>
> Maps each character to another of its group (7 groups), keeping length and format.
>
> `EC_SUPRIMIR` SECURE LOOKUP
>
> Replaces the value with one of the 1 values in `ec-sin-informacion.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Reactivo | → | **Sin información** |
| S | → | **S** |

---

## 11.9 Blood group and Rh

EC_SALUD_GRUPO_SANGUINEO

COLUMN NAME VALUE FORMAT SECURE LOOKUP

WHAT IT IS

Blood group and Rh factor, which the cédula itself carries.

HOW IT IS FOUND

> COLUMN NAME EC_SALUD_GRUPO_SANGUINEO — Path
>
> 90% `grupo_?sangu[ií]neo` `tipo_?(de_?)?sangre` `rh` `factor_?rh` `gs_?rh` `grupo_?rh` `abo(_?rh)?` `blood_?(type|group)`

VALUE FORMAT EC_SALUD_GRUPO_SANGUINEO — Values

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 18 known values, ignoring case — A, A-, B, B-, AB, AB-, … | 90% | |

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

Replaced by another group from the list.

`dlpx-core:BloodType SL` SECURE LOOKUP

Comes with Delphix and is used as it ships. Replaces the blood group with one of the eight groups, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 0+ | → | **0−** |

## 11.10 Disability

EC_SALUD_DISCAPACIDAD

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

Disability status, type and percentage, the CONADIS card and the substitute. Article 25 d) makes the data of persons with disabilities **and of their substitutes** a special category of its own — no other law of the region does.

HOW IT IS FOUND

COLUMN NAME EC_SALUD_DISCAPACIDAD — Path

90% `discapacidad[a-z_]*` `(tipo|cod|codigo|categoria|porcentaje|grado)_?discapacidad` `persona_?con_?discapacidad` `pcd` `conadis` `carn[eé]_?(de_?)?(conadis|discapacidad)` `certificado_?(de_?)?discapacidad` `condici[oó]n_?(de_?)?discapacidad` `sustituto(_?de_?persona_?con_?discapacidad)?` `necesidades_?(educativas_?)?especiales` `movilidad_?reducida` `disabilit(y|ies)`

VALUE FORMAT EC_SALUD_DISCAPACIDAD — Values

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 23 known values, ignoring case — Física, Auditiva, Visual, Intelectual, Psicosocial, Del lenguaje, … | 60% | |

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

A flag stays a flag and any other value is replaced by a category from the list.

`EC_DISCAPACIDAD` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_DISCAPACIDAD_LISTA`.

`EC_BANDERA` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`EC_BANDERA_SI_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-si-no.txt`, chosen by a keyed hash.

`EC_BANDERA_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-true-false.txt`, chosen by a keyed hash.

`EC_BANDERA_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-yes-no.txt`, chosen by a keyed hash.

`EC_BANDERA_S_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-s-n.txt`, chosen by a keyed hash.

`EC_BANDERA_Y_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-y-n.txt`, chosen by a keyed hash.

`EC_BANDERA_0_1` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-0-1.txt`, chosen by a keyed hash.

`EC_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

`EC_DISCAPACIDAD_LISTA` SECURE LOOKUP

Replaces the value with one of the 8 values in `ec-discapacidades.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Trastorno del espectro autista | → | **Ninguna** |

## 11.11 Occupational health

EC_SALUD_OCUPACIONAL

COLUMN NAME · VALUE FORMAT · REGEX DECOMPOSE

WHAT IT IS

The fitness opinion of occupational medical exams, work accidents and occupational diseases, sick leave and absenteeism.

HOW IT IS FOUND

COLUMN NAME EC_SALUD_OCUPACIONAL — Path

85% `aptitud_?(laboral|m[eé]dica)` `certificado_?(de_?)?aptitud` `examen_?(m[eé]dico_?)?(ocupacional|preocupacional|de_?ingreso|de_?retiro|peri[oó]dico)` `accidente_?(de_?)?trabajo` `enfermedad_?(laboral|profesional|ocupacional)` `riesgos?_?(del_?)?trabajo` `incapacidad(_?(m[eé]dica|laboral|d[ií]as))?` `reposo_?m[eé]dico` `d[ií]as_?(de_?)?(incapacidad|reposo)` `licencia_?(m[eé]dica|por_?enfermedad|de_?maternidad)` `ausentismo` `comit[eé]_?(de_?)?seguridad`

VALUE FORMAT EC_SALUD_OCUPACIONAL — Values

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 5 known values, ignoring case — apto, no apto, apto con restricciones, pendiente, apto con recomendaciones | 60% | |

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

An opinion is replaced by another from the list — fit, unfit —; a flag stays a flag; the diagnosis falls under its own data class.

`EC_APTITUD_LABORAL` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_APTITUD_LABORAL_LISTA`.

`EC_BANDERA` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`EC_BANDERA_SI_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-si-no.txt`, chosen by a keyed hash.

`EC_BANDERA_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-true-false.txt`, chosen by a keyed hash.

`EC_BANDERA_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-yes-no.txt`, chosen by a keyed hash.

`EC_BANDERA_S_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-s-n.txt`, chosen by a keyed hash.

`EC_BANDERA_Y_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-y-n.txt`, chosen by a keyed hash.

`EC_BANDERA_0_1` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-0-1.txt`, chosen by a keyed hash.

`EC_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

`EC_APTITUD_LABORAL_LISTA` SECURE LOOKUP

Replaces the value with one of the 4 values in `ec-aptitud-laboral.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| No apto temporalmente | → | **Pendiente** |

## 11.12 Sexual and reproductive health

EC_SALUD_REPRODUCTIVA

COLUMN NAME REGEX DECOMPOSE

WHAT IT IS

Pregnancy, last menstrual period, prenatal care, childbirth, family planning, fertility.

HOW IT IS FOUND

COLUMN NAME EC_SALUD_REPRODUCTIVA — Path

90% `embarazo` `embarazada` `gestante` `gestaci[oó]n` `edad_?gestacional` `semanas_?(de_?)?gestaci[oó]n` `fum` `fecha_?[uú]ltima_?(regla|menstruaci[oó]n)` `control_?prenatal` `prenatal` `parto(_?tipo)?` `ces[aá]rea` `aborto` `m[eé]todo_?(de_?)?planificaci[oó]n` `planificaci[oó]n_?familiar` `anticoncepci[oó]n` `anticonceptivo` `fertilidad` `salud_?(sexual|reproductiva)` `pregnan(t|cy)` `contracepti(on|ve)`

HOW IT IS MASKED

Flags keep their shape; any other value becomes `Sin información`.

`EC_CATEGORIA_SUPRIMIDA` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_SUPRIMIR`.

`EC_BANDERA` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`EC_BANDERA_SI_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-si-no.txt`, chosen by a keyed hash.

`EC_BANDERA_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-true-false.txt`, chosen by a keyed hash.

`EC_BANDERA_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-yes-no.txt`, chosen by a keyed hash.

`EC_BANDERA_S_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-s-n.txt`, chosen by a keyed hash.

`EC_BANDERA_Y_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-y-n.txt`, chosen by a keyed hash.

`EC_BANDERA_0_1` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-0-1.txt`, chosen by a keyed hash.

`EC_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

`EC_SUPRIMIR` SECURE LOOKUP

Replaces the value with one of the 1 values in `ec-sin-informacion.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 12 semanas de gestación | → | **Sin información** |
| S | → | **S** |

## 11.13 Mental health

EC_SALUD_MENTAL

COLUMN NAME REGEX DECOMPOSE

WHAT IT IS

Mental disorders, psychiatric or psychological care, suicide attempts and substance use. Article 4 defines *datos relativos a la salud* as covering physical **and mental** health.

HOW IT IS FOUND

COLUMN NAME EC_SALUD_MENTAL — Path

90% `salud_?mental` `trastorno[a-z_]*` `psiqui[aá]tri[a-z_]*` `psicol[oó]gi[a-z_]*` `depresi[oó]n` `ansiedad` `suicid[a-z_]*` `intento_?(de_?)?suicidio` `autolesi[oó]n` `consumo_?(de_?)?(sustancias|alcohol|drogas)` `adicci[oó]n(es)?` `alcoholismo` `tabaquismo` `mental_?health`

HOW IT IS MASKED

Flags keep their shape; any other value becomes `Sin información`.

`EC_CATEGORIA_SUPRIMIDA` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_SUPRIMIR`.

`EC_BANDERA` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`EC_BANDERA_SI_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-si-no.txt`, chosen by a keyed hash.

`EC_BANDERA_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-true-false.txt`, chosen by a keyed hash.

`EC_BANDERA_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-yes-no.txt`, chosen by a keyed hash.

`EC_BANDERA_S_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-s-n.txt`, chosen by a keyed hash.

`EC_BANDERA_Y_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-y-n.txt`, chosen by a keyed hash.

`EC_BANDERA_0_1` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-0-1.txt`, chosen by a keyed hash.

`EC_CM_ALFANUM` CHARACTER MAPPING

> Maps each character to another of its group (7 groups), keeping length and format.
>
> `EC_SUPRIMIR` **SECURE LOOKUP**
>
> Replaces the value with one of the 1 values in `ec-sin-informacion.txt`, chosen by a keyed hash.

**EXAMPLES**

| VALUE | | MASKED |
|---|---|---|
| `Trastorno bipolar` | → | **`Sin información`** |

PART 12

# Credit and socioeconomic data

Article 4 defines the **datos personales crediticios** as those that make up the economic behaviour of a natural person, and articles 28 and 29 give them a regime of their own: only to analyse solvency, never for a secondary purpose, and not communicated once five years have passed. The rest of the group is not credit data, but it identifies: a debt, an asset, a pension, a social programme or a Registro Social quintile narrows a person down as fast as a postcode.

## 12.1 Credit history

EC_FIN_HISTORIAL_CREDITICIO

COLUMN NAME | REGEX DECOMPOSE

### WHAT IT IS

The bureau report, the score, days in arrears and the risk category the Superintendencia de Bancos uses — A-1 to E. Article 4 defines these as *datos personales crediticios*, and article 28 allows them only for analysing solvency, with **no secondary purpose** and no communication after five years.

### HOW IT IS FOUND

COLUMN NAME — EC_FIN_HISTORIAL_CREDITICIO – Path

90% `historial_?(crediticio|de_?cr[eé]dito)` `bur[oó]_?(de_?)?cr[eé]dito` `central_?(de_?)?riesgos?` `equifax` `credit_?report` `score(_?(crediticio|interno))?` `puntaje_?(de_?)?cr[eé]dito` `calificaci[oó]n_?(de_?)?(riesgo|cr[eé]dito|cartera|deudor)` `categor[ií]a_?(de_?)?riesgo` `d[ií]as_?(de_?)?(mora|atraso)` `estado_?(de_?)?(cr[eé]dito|cartera|obligaci[oó]n)` `morosidad` `moroso` `cobranza_?judicial` `credit_?score` `credit_?rating`

### HOW IT IS MASKED

A score gets other digits of the same length; a flag stays a flag; a category is replaced by another from the list.

`EC_HISTORIAL_CREDITICIO` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_ESTADO_CREDITO`.

> `EC_CM_DIGITOS` CHARACTER MAPPING
>
> Maps each character to another of its group (1 groups), keeping length and format.
>
> `EC_BANDERA` REGEX DECOMPOSE
>
> Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `EC_BANDERA_SI_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-si-no.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-true-false.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-yes-no.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_S_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-s-n.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-y-n.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-0-1.txt`, chosen by a keyed hash.
>
> `EC_ESTADO_CREDITO` SECURE LOOKUP

> Replaces the value with one of the 15 values in `ec-estados-credito.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Reportado en el buró de crédito | → | **Refinanciado** |
| 745 | → | **471** |

## 12.2 Debt

EC_FIN_DEUDA

COLUMN NAME · DATA TYPE · NUMERIC MAPPING

WHAT IT IS

Credit and portfolio balances, amounts in arrears, instalments, disbursed amounts and limits.

HOW IT IS FOUND

COLUMN NAME — EC_FIN_DEUDA – Path

85%

`saldo_?(deuda|cr[eé]dito|capital|obligaci[oó]n|cartera|en_?mora|adeudado)` `monto_?(de_?la_?)?(deuda|cuota|obligaci[oó]n|cr[eé]dito|pr[eé]stamo|mora|desembolsado)` `deuda(_?total)?` `l[ií]mite_?(de_?)?(cr[eé]dito|tarjeta)` `cupo_?(de_?)?cr[eé]dito` `cuota_?mensual` `loan_?(amount|balance)` `outstanding_?balance` `debt`

DATA TYPE — EC_FIN_DEUDA – Type

Accepts number; any other type rules the data class out.

HOW IT IS MASKED

Masked by CM Numeric, which comes with Delphix: each digit becomes another, and the amount keeps its number of digits, its sign and its decimal places — the same order of magnitude, with no link to the real value.

`dlpx-core:CM Numeric` NUMERIC MAPPING

Comes with Delphix and is used as it ships. Maps each digit to another, keeping the number of digits, the sign, the decimal places and the zeros on the left — an amount keeps its order of magnitude.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 8400 | → | **5981** |

## 12.3 Income and pay

EC_FIN_INGRESOS

COLUMN NAME · DATA TYPE · NUMERIC MAPPING

WHAT IT IS

Salary, monthly or household income, fees and rents. The unified basic wage was 470 dollars a month in 2025.

HOW IT IS FOUND

COLUMN NAME EC_FIN_INGRESOS – Path

85% `sueldo(_?(b[aá]sico|mensual|bruto|neto|unificado))?` `salario(_?(b[aá]sico|mensual|bruto|neto))?` `remuneraci[oó]n(_?(b[aá]sica|mensual|unificada))?` `rmu` `ingresos?(_?(mensuales?|totales?|familiares?|declarados?|netos?))?` `honorarios` `renta(_?(bruta|neta|mensual|anual|imponible))?` `ingreso_?(base|familiar|percapita)` `salary` `income` `wages?`

DATA TYPE EC_FIN_INGRESOS – Type

Accepts number; any other type rules the data class out.

HOW IT IS MASKED

Masked by CM Numeric, which comes with Delphix: each digit becomes another, and the amount keeps its number of digits, its sign and its decimal places — the same order of magnitude, with no link to the real value.

`dlpx-core:CM Numeric` NUMERIC MAPPING

Comes with Delphix and is used as it ships. Maps each digit to another, keeping the number of digits, the sign, the decimal places and the zeros on the left — an amount keeps its order of magnitude.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 1200 | → | 1191 |

---

## 12.4 Assets

EC_FIN_PATRIMONIO

COLUMN NAME DATA TYPE NUMERIC MAPPING

WHAT IT IS

Net worth, assets, valuations and cadastral values, account and fund balances, term deposits and the sworn declaration of assets.

HOW IT IS FOUND

COLUMN NAME EC_FIN_PATRIMONIO – Path

85% `patrimonio(_?(neto|l[ií]quido))?` `activos?_?(totales?)?` `valor_?(del_?)?(inmueble|predio|veh[ií]culo|bienes|aval[uú]o|comercial)` `aval[uú]o(_?catastral)?` `saldo_?(cuenta|ahorros?|dep[oó]sito|inversi[oó]n|fondo)` `dep[oó]sito_?a_?plazo` `inversiones` `declaraci[oó]n_?(juramentada|patrimonial)` `net_?worth` `account_?balance`

DATA TYPE EC_FIN_PATRIMONIO – Type

Accepts number; any other type rules the data class out.

HOW IT IS MASKED

Masked by CM Numeric, which comes with Delphix: each digit becomes another, and the amount keeps its number of digits, its sign and its decimal places — the same order of magnitude, with no link to the real value.

`dlpx-core:CM Numeric` NUMERIC MAPPING

Comes with Delphix and is used as it ships. Maps each digit to another, keeping the number of digits, the sign, the decimal places and the zeros on the left — an amount keeps its order of magnitude.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 85000 | → | **78007** |

## 12.5 Pension and benefits

EC_FIN_PENSION

COLUMN NAME · DATA TYPE · NUMERIC MAPPING

WHAT IT IS

The amount of the pension, subsidy or benefit, and child support.

HOW IT IS FOUND

COLUMN NAME EC_FIN_PENSION – Path

90% `pensi[oó]n(_?(monto|mensual|valor|jubilar))?` `monto_?(pensi[oó]n|jubilaci[oó]n|beneficio|subsidio)` `valor_?(pensi[oó]n|subsidio|beneficio)` `jubilaci[oó]n(_?monto)?` `pensi[oó]n_?(de_?)?alimentos` `pensi[oó]n_?alimenticia` `sut[eé]` `pension_?amount`

DATA TYPE EC_FIN_PENSION – Type

Accepts number; any other type rules the data class out.

HOW IT IS MASKED

Masked by CM Numeric, which comes with Delphix: each digit becomes another, and the amount keeps its number of digits, its sign and its decimal places — the same order of magnitude, with no link to the real value.

`dlpx-core:CM Numeric` NUMERIC MAPPING

Comes with Delphix and is used as it ships. Maps each digit to another, keeping the number of digits, the sign, the decimal places and the zeros on the left — an amount keeps its order of magnitude.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 520 | → | **528** |

## 12.6 Social programmes

EC_FIN_PROGRAMA_SOCIAL

COLUMN NAME · VALUE FORMAT · REGEX DECOMPOSE

WHAT IT IS

The Bono de Desarrollo Humano, the Bono Joaquín Gallegos Lara, the pensions for older adults and for persons with disabilities, the Crédito de Desarrollo Humano, by name or as a flag. Being in one says the household is poor.

### HOW IT IS FOUND

**COLUMN NAME** EC_FIN_PROGRAMA_SOCIAL — Path

**90%** `programa_?social` `beneficiario_?(programa|subsidio|bono|pensi[oó]n)` `bono(_?(desarrollo_?humano|joaqu[ií]n_?gallegos|1000_?d[ií]as|contingencia))?` `bdh` `pensi[oó]n_?(adultos?_?mayores|toda_?una_?vida)` `cr[eé]dito_?(de_?)?desarrollo_?humano` `registro_?social` `subsidio(_?(gas|electricidad|tipo))?` `transferencia_?monetaria` `ayuda_?(social|estatal)` `social_?programme`

**VALUE FORMAT** EC_FIN_PROGRAMA_SOCIAL — Values

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 13 known values, ignoring case — Bono de Desarrollo Humano, Bono Joaquín Gallegos Lara, Pensión para Adultos Mayores, Pensión para Personas con Discapacidad, Crédito de Desarrollo Humano, Bono 1000 Días, … | 80% | |
| One of 10 known values, ignoring case — s, n, sí, no, 1, 2, … | 30% | |

A value that matches none counts 40% against the data class.

### HOW IT IS MASKED

A flag stays a flag and any other value is replaced by a programme from the list.

`EC_PROGRAMA_SOCIAL` **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_PROGRAMA_SOCIAL_LISTA`.

> `EC_BANDERA` **REGEX DECOMPOSE**
>
> Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `EC_BANDERA_SI_NO` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `ec-bandera-si-no.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_TRUE_FALSE` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `ec-bandera-true-false.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_YES_NO` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `ec-bandera-yes-no.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_S_N` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `ec-bandera-s-n.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_Y_N` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `ec-bandera-y-n.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_0_1` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `ec-bandera-0-1.txt`, chosen by a keyed hash.
>
> `EC_CM_ALFANUM` **CHARACTER MAPPING**
>
> Maps each character to another of its group (7 groups), keeping length and format.
>
> `EC_PROGRAMA_SOCIAL_LISTA` **SECURE LOOKUP**
>
> Replaces the value with one of the 7 values in `ec-programas-sociales.txt`, chosen by a keyed hash.

### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Bono de contingencia | → | **Pensión para Personas con Discapacidad** |

## 12.7 Registro Social

EC_FIN_REGISTRO_SOCIAL

COLUMN NAME | VALUE FORMAT | REGEX DECOMPOSE

### WHAT IT IS

The score and quintile the Registro Social gives every household, which decides who gets a subsidy, and the socioeconomic level.

### HOW IT IS FOUND

COLUMN NAME EC_FIN_REGISTRO_SOCIAL – Path

90% `registro_?social` `[ií]ndice_?(de_?)?registro_?social` `puntaje_?(registro_?social|rs)` `clasificaci[oó]n_?socioecon[oó]mica` `nivel_?socioecon[oó]mico` `nse` `condici[oó]n_?(de_?)?pobreza` `pobreza` `quintil(_?(de_?)?(ingreso|riqueza))?` `decil` `estrato_?socioecon[oó]mico`

VALUE FORMAT EC_FIN_REGISTRO_SOCIAL – Values

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 18 known values, ignoring case — Quintil 1, Quintil 2, Quintil 3, Quintil 4, Quintil 5, Pobreza extrema, … | 60% | |

A value that matches none counts 40% against the data class.

### HOW IT IS MASKED

A flag stays a flag, a code is replaced by another code and any other value by a class from the list.

`EC_REGISTRO_SOCIAL` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_REGISTRO_SOCIAL_LISTA`.

`EC_BANDERA` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`EC_BANDERA_SI_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-si-no.txt`, chosen by a keyed hash.

`EC_BANDERA_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-true-false.txt`, chosen by a keyed hash.

`EC_BANDERA_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-yes-no.txt`, chosen by a keyed hash.

`EC_BANDERA_S_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-s-n.txt`, chosen by a keyed hash.

`EC_BANDERA_Y_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-y-n.txt`, chosen by a keyed hash.

`EC_BANDERA_0_1` SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-0-1.txt`, chosen by a keyed hash.

`EC_REGISTRO_SOCIAL_CODIGO` SECURE LOOKUP

Replaces the value with one of the 5 values in `ec-registro-social-codigos.txt`, chosen by a keyed hash.

`EC_REGISTRO_SOCIAL_LISTA` SECURE LOOKUP

Replaces the value with one of the 9 values in `ec-registro-social.txt`, chosen by a keyed hash.

### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `Pobreza extrema` | → | **`Quintil 1`** |
| 2 | → | **4** |

## 12.8 Housing tenure

EC_FIN_VIVIENDA

COLUMN NAME | VALUE FORMAT | REGEX DECOMPOSE

### WHAT IT IS

Housing tenure in the categories the census uses: owned outright or being paid for, donated or inherited, lent, by services, rented, or held in anticresis.

### HOW IT IS FOUND

> **COLUMN NAME** EC_FIN_VIVIENDA — Path
>
> **90%** `tenencia_?(de_?)?(la_?)?vivienda` `tipo_?(de_?)?tenencia` `vivienda_?(propia|arrendada|tipo_?tenencia)` `condici[oó]n_?(de_?)?(la_?)?vivienda` `ocupaci[oó]n_?vivienda` `anticresis` `housing_?tenure`

> **VALUE FORMAT** EC_FIN_VIVIENDA — Values
>
| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 14 known values, ignoring case — Propia y totalmente pagada, Propia y la está pagando, Propia regalada donada heredada, Prestada o cedida, Por servicios, Arrendada, … | 60% | |

> A value that matches none counts 30% against the data class.

### HOW IT IS MASKED

Replaced by another tenure from the list.

`EC_TENENCIA_VIVIENDA` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_TENENCIA_VIVIENDA_LISTA` .

> `EC_BANDERA` REGEX DECOMPOSE
>
> Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `EC_BANDERA_SI_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-si-no.txt` , chosen by a keyed hash.
> >
> > `EC_BANDERA_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-true-false.txt` , chosen by a keyed hash.
> >
> > `EC_BANDERA_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-yes-no.txt` , chosen by a keyed hash.
> >
> > `EC_BANDERA_S_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-s-n.txt` , chosen by a keyed hash.
> >
> > `EC_BANDERA_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-y-n.txt` , chosen by a keyed hash.
> >
> > `EC_BANDERA_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-0-1.txt` , chosen by a keyed hash.
>
> `EC_CM_ALFANUM` CHARACTER MAPPING
>
> Maps each character to another of its group (7 groups), keeping length and format.
>
> `EC_TENENCIA_VIVIENDA_LISTA` SECURE LOOKUP
>
> Replaces the value with one of the 7 values in `ec-tenencia-vivienda.txt` , chosen by a keyed hash.

### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `Ocupación de hecho` | → | **`Anticresis`** |

PART 13

# Judicial record

Ecuador is the only country of the region whose law names the **pasado judicial** among the sensitive data of article 4. The group is separate only because the shapes are: everything in it is sensitive data, and article 26 forbids processing it unless one of the grounds it lists applies.

## 13.1 Judicial record

EC_PENAL_ANTECEDENTES

COLUMN NAME | REGEX DECOMPOSE

### WHAT IT IS

The **pasado judicial** article 4 lists among the sensitive data: the record certificate, the type of offence, the procedural situation, the arrest warrant, being deprived of liberty and the SNAI record.

### HOW IT IS FOUND

COLUMN NAME — EC_PENAL_ANTECEDENTES — Path

| | |
|---|---|
| 95% | `antecedentes?(_?(penales\|policiales\|judiciales))?` `pasado_?judicial` `certificado_?(de_?)?(antecedentes\|pasado_?judicial)` `r[eé]cord_?policial` `reincidencia` `condena(_?(penal\|tipo))?` `delito(_?(tipo\|cometido))?` `tipo_?(de_?)?delito` `situaci[oó]n_?jur[ií]dica` `boleta_?(de_?)?(captura\|excarcelaci[oó]n)` `privad[oa]_?(de_?)?libertad` `ppl` `snai` `centro_?(de_?)?rehabilitaci[oó]n_?social` `criminal_?record` |
| 55% | `estado_?(del_?)?proceso` `sentencia` `fallo` |

### HOW IT IS MASKED

Replaced by a generic status from the list — no record, sentenced, acquitted —, whatever the outcome; a flag stays a flag.

`EC_ANTECEDENTES` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_ANTECEDENTES_LISTA`.

> `EC_BANDERA` REGEX DECOMPOSE
>
> Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `EC_BANDERA_SI_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-si-no.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-true-false.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-yes-no.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_S_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-s-n.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-y-n.txt`, chosen by a keyed hash.
> >
> > `EC_BANDERA_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `ec-bandera-0-1.txt`, chosen by a keyed hash.
>
> `EC_CM_ALFANUM` CHARACTER MAPPING
>
> Maps each character to another of its group (7 groups), keeping length and format.
>
> `EC_ANTECEDENTES_LISTA` SECURE LOOKUP
>
> Replaces the value with one of the 7 values in `ec-antecedentes.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Sentenciado por hurto en 2019 | → | **Absuelto** |

## 13.2 Court file

EC_PENAL_EXPEDIENTE

COLUMN NAME | VALUE FORMAT | REGEX DECOMPOSE

WHAT IT IS

The number the Función Judicial gives a case — judicial unit, year and correlative, `17294-2024-00123` — and the file of the Fiscalía.

HOW IT IS FOUND

COLUMN NAME EC_PENAL_EXPEDIENTE – Path

| | |
|---|---|
| 90% | `n(o\|ro\|um\|umero)_?(de_?)?(expediente_?(judicial\|penal\|fiscal)\|causa\|proceso_?(judicial\|penal)\|denuncia\|noticia_?del_?delito\|referencia_?(judicial\|fiscal))` `expediente_?(judicial\|penal\|fiscal)` `n(o\|ro\|um\|umero)_?(de_?)?juicio` `juicio_?n(o\|ro\|um\|umero)` `causa_?penal` `noticia_?del_?delito` `satje` |
| 50% | `expediente` `causa` `proceso` `juicio` |

VALUE FORMAT EC_PENAL_EXPEDIENTE – Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `0*(?:\d{5}-\d{4}-\d{5})` | 90% | |
| `0*(?:\d{1,6}-[A-Z]{2,10}-\d{4})` | 50% | |

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

The judicial unit and the correlative are mapped digit by digit; the year stays. Other numbers are masked character by character.

`EC_EXPEDIENTE_JUDICIAL` REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: `EC_CM_ALFANUM` .

> `EC_CM_DIGITOS` CHARACTER MAPPING
>
> Maps each character to another of its group (1 groups), keeping length and format.
>
> `EC_CM_ALFANUM` CHARACTER MAPPING
>
> Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 17294-2024-00123 | → | **23323-2024-00957** |
| 1234/2023 | → | **9707/2023** |

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# The cédula and RUC check digits

The **cédula** is ten digits. Each of the first nine is multiplied by the series 2 1 2 1 2 1 2 1 2, every product over nine is reduced by nine, and the tenth digit is ten minus the remainder of the sum modulo ten. That is the **Luhn algorithm**, the same one a payment card uses — which is what makes the masking a single step.

Delphix's **Payment Card** framework keeps the Luhn residue of the value it is given: a valid cédula comes out as another valid cédula. Its `preserve` parameter fixes how many leading digits stay, and three of them keep the **province of issue** and the digit that says the holder is a natural person. The province stays for the same reason the provincia column stays: the policy generalizes below the province, not at it.

The **RUC** is thirteen digits, and its third one decides the rule:

- **0 to 5 — a natural person**: the RUC is the cédula plus the establishment code `001`. The policy masks the ten digits as a cédula and keeps the code, so the cédula and the RUC of the same person still agree.
- **9 — a private company**: the check digit is in the tenth position, computed modulo 11 with the coefficients 4 3 2 7 6 5 4 3 2. **6 — a public body**: the check digit is in the ninth, with the coefficients 3 2 7 6 5 4 3 2. Neither is a natural person, so neither is personal data: the policy leaves both exactly as they are, and says so.

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# Where people live: parroquias, DPA code and postal code

With the population the 2022 census counted and the location the DPA layer publishes, the policy knows the size, cantón, provincia and position of about 1,040 parroquias and 224 cantones.

- A **parroquia** with fewer than 20,000 inhabitants becomes the nearest parroquia of its **cantón** with at least 20,000; when its cantón has none, the nearest of its provincia. There are 913 of them, with 4.3 million inhabitants.
- The **93 cantones** under 20,000 follow the same rule inside their provincia. The 24 provincias stay.
- Three provincias have no parroquia of 20,000 at all — Galápagos, Zamora Chinchipe and the zona no delimitada. There the value stays inside the provincia and becomes its most populous parroquia: a smaller group than elsewhere, and the documentation says so rather than sending a Galápagos resident to the mainland.
- Distance is measured between the centroids of the DPA layer, in a straight line.
- Names repeat — San José, Santa Rosa, San Vicente, Bellavista. A name is generalized only when every place bearing it is small, and then follows the most populous of them; when that bearer is a cantón, the replacement is a cantón. Written with the cantón or the provincia ( `Facundo Vela – Bolívar` ) the same rule applies inside it.
- The **DPA code** follows the same table, so a person's parroquia name and code still agree — and because the postal code *is* the DPA code, the **ciudadela** is replaced by another and the **postal code** keeps only the provincia.

Parroquia names that are also given names or surnames, and provincia names, are left out of the detection list, so that a surname column is not taken for parroquias.

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# Limits

- **Binary columns** — cédula photos, fingerprints, validation selfies, clinical PDFs in BLOBs — cannot be masked by these algorithms. Inventory them and report them as residual risk.
- **The province of issue of the cédula stays**, and so does the provincia: the policy generalizes below the province, not at it. If the province itself is too revealing for your data, add a step that maps it — remembering that any change to those two digits breaks the Luhn digit, which then has to be recomputed.
- **A company RUC is not masked**: article 4 makes only a natural person a data subject. If your rule set needs it masked anyway, give the column a data class of its own.
- **The DPA layer publishes the place names without their acute accents.** The policy restores them with a dictionary, and writes both spellings into the tables, but a name it could not restore comes out unaccented.
- **Data Cleansing leaves unknown values as they are**: a parroquia spelled in a way that is not in the table — abbreviated, misspelled — passes unchanged.
- **Free text** is redacted word by word: a phone written with spaces inside a sentence may not be found, and a name that is also a common word ( `Rosa` , `Cruz` , `Flores` ) is redacted wherever it appears.
- **Children and adolescents** are a special category of their own (art. 25 b). A birth date moves at most 60 days, so few people cross their 18th birthday — but some do. Where that line matters, split the rule by age group with Multi-Column Condition.
- **Addresses** are replaced as a whole line. Street, ciudadela, parroquia and postal code in separate columns are masked independently and no longer form a coherent address.
- **The engine key** links masked values back to production for whoever holds both: the copy is **pseudonymized** in the sense of article 4, not anonymized, and remains personal data. Keep the key apart and under technical and organizational measures.

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# Verification

Before publication, the policy was checked outside Delphix:

- The cédula rule was checked against published valid numbers, and the two RUC rules against RUCs published by companies and public bodies.
- Every classifier configuration was reviewed against what Delphix accepts.
- Discovery was simulated on 142 columns with the whole policy at its assignment threshold — well named, badly named and 29 columns that are not personal data — and again with only the classifiers of the essential pack.
- The table that generalizes places was audited for all 1,040 parroquias and 224 cantones: by name, by name with the provincia, and by DPA code.
- The algorithms, run with the Delphix algorithm plugin, masked each data class's sample values and placeholders such as `N/A` , checking every output against its vocabulary; 200 cédulas for validity, distinctness and the province and type they keep, 25 cédula and RUC pairs to confirm they still agree, and 40 company RUCs to confirm they come back untouched. Every algorithm also masked its own sample value.

Every column was assigned as expected and every value masked as expected. Before using the policy on real data, run a discovery job and a masking job on a representative copy.

# Sources

- Ley Orgánica de Protección de Datos Personales (Fifth Supplement of Official Register 459, 26 May 2021) and its Reglamento General (Official Register Supplement 435, 13 November 2023).
- Constitution of the Republic (indigenous nationalities and pueblos, union of fact). Ley Orgánica de Salud. Ley de Prevención y Atención Integral del VIH/SIDA. Ley Orgánica de Discapacidades. Ley Orgánica Integral para Prevenir y Erradicar la Violencia contra las Mujeres. Código Orgánico Integral Penal.
- Instituto Nacional de Estadística y Censos: VIII Censo de Población y VII de Vivienda 2022, population by parroquia, and the División Política Administrativa with the code and location of every parroquia and cantón.
- Registro Civil (cédula and its check digit). SRI (RUC and its two rules). Superintendencia de Bancos (risk categories A-1 to E). IESS, ISSFA and ISSPOL. CONADIS. MIES (Registro Social and the social programmes). CNE (register of political organizations). Ministry of Labour (unified basic wage 2025). ANT (Placa). Correos del Ecuador (six-digit postal code). SENESCYT.

APPENDIX A

# Algorithms

Every algorithm of the policy in alphabetical order, with its configuration. The one a data class names is the top of its tree; the others are building blocks.

## EC_ANIO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: `EC_L2_ANIO_NACIMIENTO`

### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(0*?)(?:(\d{3})(\d))` | keep · keep · `EC_CM_DIGITOS` |
| otherwise | keep |

### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

## EC_ANTECEDENTES REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_ANTECEDENTES_LISTA`.

Used by: `EC_PENAL_ANTECEDENTES`

### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(s[ií]\|no\|true\|false\|yes\|[snyx01])` | `EC_BANDERA` |
| `(0*?)(?:(\d{1,6}))` | keep · `EC_CM_ALFANUM` |
| otherwise | `EC_ANTECEDENTES_LISTA` |

### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

## EC_ANTECEDENTES_LISTA SECURE LOOKUP

Replaces the value with one of the 7 values in `ec-antecedentes.txt`, chosen by a keyed hash.

Used by: `EC_ANTECEDENTES`

### CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-antecedentes.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

## EC_APELLIDO NAME

Replaces the name with one of the 366 names in `ec-apellidos.txt`.

Used by: `EC_PALABRA_APELLIDO`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | ec-apellidos.txt |
| maskedValueCase | PRESERVE_INPUT |
| filterAccent | yes |
| inputCaseSensitive | no |

---

### EC_APELLIDOS REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: EC_CM_ALFANUM .

Used by: EC_L1_APELLIDO , EC_NOMBRE_COMPLETO

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| (\S+) | EC_PALABRA_APELLIDO |
| (\S+)\s+(\S+) | EC_PALABRA_APELLIDO · EC_PALABRA_APELLIDO |
| (\S+)\s+(\S+)\s+(\S+) | EC_PALABRA_APELLIDO · EC_PALABRA_APELLIDO · EC_PALABRA_APELLIDO |
| (\S+)\s+(\S+)\s+(\S+)\s+(\S+) | EC_PALABRA_APELLIDO · EC_PALABRA_APELLIDO · EC_PALABRA_APELLIDO · EC_PALABRA_APELLIDO |
| otherwise | EC_CM_ALFANUM |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### EC_APTITUD_LABORAL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: EC_APTITUD_LABORAL_LISTA .

Used by: EC_SALUD_OCUPACIONAL

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| (?i)(s[ií]\|no\|true\|false\|yes\|[snyx01]) | EC_BANDERA |
| (0*?)(?:(\d{1,6})) | keep · EC_CM_ALFANUM |
| otherwise | EC_APTITUD_LABORAL_LISTA |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### EC_APTITUD_LABORAL_LISTA SECURE LOOKUP

Replaces the value with one of the 4 values in ec-aptitud-laboral.txt , chosen by a keyed hash.

Used by: EC_APTITUD_LABORAL

#### CONFIGURATION

| | |
|---|---|
| lookupFile | ec-aptitud-laboral.txt |
| hashMethod | SHA256 |
| maskedValueCase | PRESERVE_INPUT |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### EC_AUTOIDENTIFICACION REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_AUTOIDENTIFICACION_LISTA`.

Used by: `EC_L3_ETNIA`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(s[ií]\|no\|true\|false\|yes\|[snyx01])` | `EC_BANDERA` |
| `(0*)([1-9]\d{0,5}\|0)` | keep · `EC_AUTOIDENTIFICACION_CODIGO` |
| otherwise | `EC_AUTOIDENTIFICACION_LISTA` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### EC_AUTOIDENTIFICACION_CODIGO SECURE LOOKUP

Replaces the value with one of the 8 values in `ec-autoidentificacion-codigos.txt`, chosen by a keyed hash.

Used by: `EC_AUTOIDENTIFICACION`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-autoidentificacion-codigos.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### EC_AUTOIDENTIFICACION_LISTA SECURE LOOKUP

Replaces the value with one of the 8 values in `ec-autoidentificacion.txt`, chosen by a keyed hash.

Used by: `EC_AUTOIDENTIFICACION`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-autoidentificacion.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### EC_BANDERA REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `EC_CATEGORIA_SUPRIMIDA`, `EC_ESTADO_CIVIL`, `EC_NIVEL_INSTRUCCION`, `EC_AUTOIDENTIFICACION`, `EC_IDENTIDAD_CULTURAL`, `EC_LENGUA`, `EC_CONDICION_MIGRATORIA`, `EC_RELIGION`, `EC_IDEOLOGIA`, `EC_PARTIDO`, `EC_ORIENTACION_SEXUAL`, `EC_IDENTIDAD_GENERO`, `EC_COBERTURA_SALUD`, `EC_DIAGNOSTICO`, `EC_MEDICAMENTO`, `EC_DISCAPACIDAD`, `EC_APTITUD_LABORAL`, `EC_HISTORIAL_CREDITICIO`, `EC_PROGRAMA_SOCIAL`, `EC_REGISTRO_SOCIAL`, `EC_TENENCIA_VIVIENDA`, `EC_ANTECEDENTES`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(s[ií]\|no)` | `EC_BANDERA_SI_NO` |
| `(?i)(true\|false)` | `EC_BANDERA_TRUE_FALSE` |
| `(?i)(yes)` | `EC_BANDERA_YES_NO` |

| | |
|---|---|
| `(?i)([sn])` | `EC_BANDERA_S_N` |
| `(?i)([yx])` | `EC_BANDERA_Y_N` |
| `([01])` | `EC_BANDERA_0_1` |
| otherwise | keep |

CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### EC_BANDERA_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-0-1.txt`, chosen by a keyed hash.

Used by: `EC_BANDERA`

CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-bandera-0-1.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### EC_BANDERA_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-s-n.txt`, chosen by a keyed hash.

Used by: `EC_BANDERA`

CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-bandera-s-n.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### EC_BANDERA_SI_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-si-no.txt`, chosen by a keyed hash.

Used by: `EC_BANDERA`

CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-bandera-si-no.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### EC_BANDERA_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-true-false.txt`, chosen by a keyed hash.

Used by: `EC_BANDERA`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-bandera-true-false.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### EC_BANDERA_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-y-n.txt`, chosen by a keyed hash.

Used by: `EC_BANDERA`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-bandera-y-n.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### EC_BANDERA_YES_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-bandera-yes-no.txt`, chosen by a keyed hash.

Used by: `EC_BANDERA`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-bandera-yes-no.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### EC_CATEGORIA_SUPRIMIDA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_SUPRIMIR`.

Used by: `EC_L3_AFILIACION_SINDICAL`, `EC_L3_VICTIMA`, `EC_SALUD_VIH`, `EC_SALUD_REPRODUCTIVA`, `EC_SALUD_MENTAL`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(s[ií]\|no\|true\|false\|yes\|[snyx01])` | `EC_BANDERA` |
| `(0*?)(?:(0*?)(?:(0*?)(?:(0*?)(?:(0*?)(?:(\d{1,6})))))))` | keep · keep · keep · keep · keep · `EC_CM_ALFANUM` |
| otherwise | `EC_SUPRIMIR` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### EC_CEDULA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_CM_DIGITOS`.

Used by: `EC_DOCUMENTO`

PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `((?:[0-2]\d\|30)[0-5]\d{7})` | `EC_CEDULA_LUHN` |
| `((?:[0-2]\d\|30)[0-5]\d{6})([\s\-])(\d)` | `EC_CM_DIGITOS` · keep · `EC_CM_DIGITOS` |
| otherwise | `EC_CM_DIGITOS` |

CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

## EC_CEDULA_LUHN PAYMENT CARD

Replaces the card number with another that passes the Luhn check.

Used by: `EC_CEDULA` , `EC_RUC`

CONFIGURATION

| | |
|---|---|
| minMaskedPositions | `1` |
| preserve | `3` |

---

## EC_CIE10 SECURE LOOKUP

Replaces the value with one of the 37 values in `ec-cie10.txt` , chosen by a keyed hash.

Used by: `EC_DIAGNOSTICO`

CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-cie10.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_LOOKUP_FILE` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

## EC_CIE10_DECIMAL SECURE LOOKUP

Replaces the value with one of the 33 values in `ec-cie10-decimal.txt` , chosen by a keyed hash.

Used by: `EC_DIAGNOSTICO`

CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-cie10-decimal.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_LOOKUP_FILE` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

## EC_CIUDADELA SECURE LOOKUP

Replaces the value with one of the 108 values in `ec-ciudadelas.txt` , chosen by a keyed hash.

Used by: `EC_L2_CIUDADELA`

CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-ciudadelas.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_LOOKUP_FILE` |

| | |
|---|---|
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### EC_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

Used by: `EC_L1_DOCUMENTO_EXTRANJERO`, `EC_L1_PASAPORTE`, `EC_L1_ACTA`, `EC_L1_SEGURO_SOCIAL`, `EC_L1_CODIGO_ESTUDIANTE`, `EC_L1_REGISTRO_PROFESIONAL`, `EC_L1_DIRECCION_COMPLEMENTO`, `EC_L1_BILLETERA`, `EC_L1_COOKIE`, `EC_L1_VEHICULO`, `EC_L1_INMUEBLE`, `EC_L1_CONTRATO`, `EC_L1_USUARIO`, `EC_SALUD_IDENTIFICADOR`, `EC_SALUD_RESULTADO_EXAMEN`, `EC_CATEGORIA_SUPRIMIDA`, `EC_DOCUMENTO`, `EC_NOMBRES`, `EC_APELLIDOS`, `EC_NOMBRE_COMPLETO`, `EC_EMAIL`, `EC_IDENTIDAD_CULTURAL`, `EC_LENGUA`, `EC_CONDICION_MIGRATORIA`, `EC_RELIGION`, `EC_IDEOLOGIA`, `EC_PARTIDO`, `EC_ORIENTACION_SEXUAL`, `EC_IDENTIDAD_GENERO`, `EC_MEDICAMENTO`, `EC_DISCAPACIDAD`, `EC_APTITUD_LABORAL`, `EC_PROGRAMA_SOCIAL`, `EC_TENENCIA_VIVIENDA`, `EC_ANTECEDENTES`, `EC_EXPEDIENTE_JUDICIAL`

#### CONFIGURATION

| | |
|---|---|
| characterGroups | `0123456789` `AEIOU` `BCDFGHJKLMNPQRSTVWXYZ` `aeiou` `bcdfghjklmnpqrstvwxyz` `ÁÉÍÓÚÜ` `áéíóúü` |
| caseSensitive | yes |
| minMaskedPositions | `0` |
| preserveLeadingZeros | yes |

### EC_CM_DIGITOS CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

Used by: `EC_L1_CUENTA_BANCARIA`, `EC_CEDULA`, `EC_RUC`, `EC_TELEFONO`, `EC_COORDENADA`, `EC_ANIO`, `EC_EDAD`, `EC_OCUPACION_O_CODIGO`, `EC_COBERTURA_SALUD`, `EC_PROCEDIMIENTO`, `EC_HISTORIAL_CREDITICIO`, `EC_EXPEDIENTE_JUDICIAL`

#### CONFIGURATION

| | |
|---|---|
| characterGroups | `0123456789` |
| caseSensitive | yes |
| minMaskedPositions | `0` |
| preserveLeadingZeros | yes |

### EC_CM_DIGITOS_LETRAS CHARACTER MAPPING

Maps each character to another of its group (2 groups), keeping length and format.

Used by: `EC_L1_PLACA`

#### CONFIGURATION

| | |
|---|---|
| characterGroups | `0123456789` `ABCDEFGHIJKLMNOPQRSTUVWXYZ` |
| caseSensitive | yes |
| minMaskedPositions | `0` |
| preserveLeadingZeros | yes |

### EC_CM_HEX CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

Used by: `EC_L1_DISPOSITIVO`, `EC_IP`

#### CONFIGURATION

| | |
|---|---|
| characterGroups | `0123456789` `abcdef` `ABCDEF` `ghijklmnopqrstuvwxyz` `GHIJKLMNOPQRSTUVWXYZ` |
| caseSensitive | yes |
| minMaskedPositions | `0` |

| | |
|---|---|
| preserveLeadingZeros | yes |

### EC_COBERTURA SECURE LOOKUP

Replaces the value with one of the 8 values in `ec-cobertura-salud.txt`, chosen by a keyed hash.

Used by: `EC_COBERTURA_SALUD`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-cobertura-salud.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_LOOKUP_FILE` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### EC_COBERTURA_SALUD REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_COBERTURA`.

Used by: `EC_SALUD_COBERTURA`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(s[ií]\|no\|true\|false\|yes\|[snyx01])` | `EC_BANDERA` |
| `(0*?)(?:(\d{1,8}))` | keep · `EC_CM_DIGITOS` |
| otherwise | `EC_COBERTURA` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### EC_CODIGO_POSTAL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `EC_L2_CODIGO_POSTAL`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(0*?)(?:(\d{2})(\d{4}))` | keep · keep · replace with `0000` |
| `(0*?)(?:(\d{2})(\d{3}))` | keep · keep · replace with `000` |
| otherwise | keep |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### EC_CONDICION_MIGRATORIA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_CONDICION_MIGRATORIA_LISTA`.

Used by: `EC_L3_CONDICION_MIGRATORIA`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(s[ií]\|no\|true\|false\|yes\|[snyx01])` | `EC_BANDERA` |
| `(0*?)(?:(\d{1,6}))` | keep · `EC_CM_ALFANUM` |
| otherwise | `EC_CONDICION_MIGRATORIA_LISTA` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### EC_CONDICION_MIGRATORIA_LISTA SECURE LOOKUP

Replaces the value with one of the 10 values in `ec-condicion-migratoria.txt`, chosen by a keyed hash.

Used by: `EC_CONDICION_MIGRATORIA`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-condicion-migratoria.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### EC_CONTEO_LIMITADO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `EC_L2_PERSONAS_A_CARGO`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(0*?)(?:([0-5]))` | keep · keep |
| `(0*?)(?:(\d+))` | keep · replace with `5` |
| otherwise | keep |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### EC_COORDENADA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `EC_L1_GEOLOCALIZACION`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(-?\d{1,3}[.,]\d)(\d+)\s*([,;])\s*(-?\d{1,3}[.,]\d)(\d+)` | keep · `EC_CM_DIGITOS` · keep · keep · `EC_CM_DIGITOS` |
| `(-?\d{1,3}[.,]\d)(\d+)` | keep · `EC_CM_DIGITOS` |
| otherwise | keep |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### EC_DIAGNOSTICO REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: `EC_DIAGNOSTICO_TEXTO`.

Used by: `EC_SALUD_DIAGNOSTICO`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `([A-TV-Za-tv-z]\d{2}\.\d{1,2})` | `EC_CIE10_DECIMAL` |

| | |
|---|---|
| `([A-TV-Za-tv-z]\d{2,3}X?)` | `EC_CIE10` |
| `(?i)(s[ií]\|no\|true\|false\|yes\|[snyx01])` | `EC_BANDERA` |
| otherwise | `EC_DIAGNOSTICO_TEXTO` |

CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### EC_DIAGNOSTICO_TEXTO SECURE LOOKUP

Replaces the value with one of the 30 values in `ec-diagnosticos.txt`, chosen by a keyed hash.

Used by: `EC_DIAGNOSTICO`

CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-diagnosticos.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### EC_DIRECCION SECURE LOOKUP

Replaces the value with one of the 4,000 values in `ec-direcciones.txt`, chosen by a keyed hash.

Used by: `EC_L1_DIRECCION`

CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-direcciones.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_LOOKUP_FILE` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### EC_DISCAPACIDAD REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_DISCAPACIDAD_LISTA`.

Used by: `EC_SALUD_DISCAPACIDAD`

PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(s[ií]\|no\|true\|false\|yes\|[snyx01])` | `EC_BANDERA` |
| `(0*?)(?:(\d{1,6}))` | keep · `EC_CM_ALFANUM` |
| otherwise | `EC_DISCAPACIDAD_LISTA` |

CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### EC_DISCAPACIDAD_LISTA SECURE LOOKUP

Replaces the value with one of the 8 values in `ec-discapacidades.txt`, chosen by a keyed hash.

Used by: `EC_DISCAPACIDAD`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-discapacidades.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### EC_DOCUMENTO REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: `EC_CM_ALFANUM`.

Used by: `EC_L1_CEDULA`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(0*?)(?:((?:[0-2]\d\|30)[0-5]\d{7}00\d))` | keep · `EC_RUC` |
| `(0*?)(?:(\d{2}[69]\d{10}))` | keep · keep |
| `(0*?)(?:((?:[0-2]\d\|30)[0-5]\d{7}))` | keep · `EC_CEDULA` |
| `(0*?)(?:((?:[0-2]\d\|30)[0-5]\d{6}[\s\-]\d))` | keep · `EC_CEDULA` |
| otherwise | `EC_CM_ALFANUM` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### EC_DPA REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: `EC_L2_DPA`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(0*)([1-9]\d*)` | keep · `EC_DPA_TABLA` |
| otherwise | keep |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### EC_DPA_TABLA DATA CLEANSING

Replaces each value found in `ec-dpa-generalizado.txt` (1,441 pairs) by its counterpart; values not in the file stay as they are.

Used by: `EC_DPA`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-dpa-generalizado.txt` |
| delimiter | `\|` |
| caseSensitive | yes |
| trimWhitespace | yes |

---

### EC_EDAD REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `EC_L2_EDAD`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(0*?)(?:([1-9]\d{2,}))` | keep · replace with `90` |
| `(0*?)(?:([1-9])(\d)([.,]\d+))` | keep · keep · `EC_CM_DIGITOS` · keep |
| `(0*?)(?:([1-9])(\d))` | keep · keep · `EC_CM_DIGITOS` |
| `(0*?)(?:(\d))` | keep · `EC_CM_DIGITOS` |
| otherwise | keep |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### EC_EMAIL REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `EC_CM_ALFANUM`.

Used by: `EC_L1_EMAIL`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `([^@\s]+)@([^@\s]+)\.([A-Za-z]{2,24}(?:\.[A-Za-z]{2})?)` | `EC_CM_ALFANUM` · `EC_CM_ALFANUM` · replace with `test` |
| otherwise | `EC_CM_ALFANUM` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### EC_EMPLEADOR SECURE LOOKUP

Replaces the value with one of the 24 values in `ec-empleadores.txt`, chosen by a keyed hash.

Used by: `EC_L2_EMPLEADOR`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-empleadores.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_LOOKUP_FILE` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### EC_ESTADO_CIVIL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_ESTADO_CIVIL_LISTA`.

Used by: `EC_L2_ESTADO_CIVIL`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(s[ií]\|no\|true\|false\|yes\|[snyx01])` | `EC_BANDERA` |
| `(0*)([1-9]\d{0,5}\|0)` | keep · `EC_ESTADO_CIVIL_CODIGO` |
| otherwise | `EC_ESTADO_CIVIL_LISTA` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### EC_ESTADO_CIVIL_CODIGO SECURE LOOKUP

Replaces the value with one of the 6 values in `ec-estado-civil-codigos.txt`, chosen by a keyed hash.

Used by: `EC_ESTADO_CIVIL`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-estado-civil-codigos.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### EC_ESTADO_CIVIL_LISTA SECURE LOOKUP

Replaces the value with one of the 6 values in `ec-estado-civil.txt`, chosen by a keyed hash.

Used by: `EC_ESTADO_CIVIL`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-estado-civil.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### EC_ESTADO_CREDITO SECURE LOOKUP

Replaces the value with one of the 15 values in `ec-estados-credito.txt`, chosen by a keyed hash.

Used by: `EC_HISTORIAL_CREDITICIO`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-estados-credito.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### EC_EXPEDIENTE_JUDICIAL REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: `EC_CM_ALFANUM`.

Used by: `EC_PENAL_EXPEDIENTE`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(0*?)(?:(\d{5})(-)(\d{4})(-)(\d{5}))` | keep · `EC_CM_DIGITOS` · keep · keep · keep · `EC_CM_DIGITOS` |
| `(0*?)(?:(\d{1,6})([\s\-/])([A-Za-z]{2,10})([\s\-/])(\d{4}))` | keep · `EC_CM_DIGITOS` · keep · keep · keep · keep |
| `([A-Za-z]{1,6}[\s\-]?)(\d{1,6})([\s\-/])(\d{4})` | keep · `EC_CM_DIGITOS` · keep · keep |
| `(0*?)(?:(\d{1,6})([\s\-/])(\d{4}))` | keep · `EC_CM_DIGITOS` · keep · keep |
| otherwise | `EC_CM_ALFANUM` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### EC_FECHA_EVENTO DATE SHIFT

Moves the date by -30 to 30 days.

Used by: `EC_L2_FECHA_EVENTO`

#### CONFIGURATION

| | |
|---|---|
| minRange | `-30` |
| maxRange | `30` |
| unit | `DAYS` |
| roll | no |

---

### EC_FECHA_NACIMIENTO DATE SHIFT

Moves the date by -60 to 60 days.

Used by: `EC_L2_FECHA_NACIMIENTO`

#### CONFIGURATION

| | |
|---|---|
| minRange | `-60` |
| maxRange | `60` |
| unit | `DAYS` |
| roll | no |

---

### EC_HISTORIAL_CREDITICIO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_ESTADO_CREDITO`.

Used by: `EC_FIN_HISTORIAL_CREDITICIO`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(0*?)(?:(\d{1,4}))` | keep · `EC_CM_DIGITOS` |
| `(?i)(s[ií]\|no\|true\|false\|yes\|[snyx01])` | `EC_BANDERA` |
| otherwise | `EC_ESTADO_CREDITO` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### EC_IDENTIDAD_CULTURAL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_IDENTIDAD_CULTURAL_LISTA`.

Used by: `EC_L3_IDENTIDAD_CULTURAL`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(s[ií]\|no\|true\|false\|yes\|[snyx01])` | `EC_BANDERA` |
| `(0*?)(?:(\d{1,6}))` | keep · `EC_CM_ALFANUM` |
| otherwise | `EC_IDENTIDAD_CULTURAL_LISTA` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |

| | |
|---|---|
| requireMask | no |

### EC_IDENTIDAD_CULTURAL_LISTA SECURE LOOKUP

Replaces the value with one of the 30 values in `ec-identidad-cultural.txt`, chosen by a keyed hash.

Used by: `EC_IDENTIDAD_CULTURAL`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-identidad-cultural.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### EC_IDENTIDAD_GENERO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_IDENTIDAD_GENERO_LISTA`.

Used by: `EC_L3_IDENTIDAD_GENERO`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(s[ií]\|no\|true\|false\|yes\|[snyx01])` | `EC_BANDERA` |
| `(0*?)(?:(\d{1,6}))` | keep · `EC_CM_ALFANUM` |
| otherwise | `EC_IDENTIDAD_GENERO_LISTA` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### EC_IDENTIDAD_GENERO_LISTA SECURE LOOKUP

Replaces the value with one of the 8 values in `ec-identidades-genero.txt`, chosen by a keyed hash.

Used by: `EC_IDENTIDAD_GENERO`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-identidades-genero.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### EC_IDEOLOGIA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_IDEOLOGIA_LISTA`.

Used by: `EC_L3_IDEOLOGIA`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(s[ií]\|no\|true\|false\|yes\|[snyx01])` | `EC_BANDERA` |
| `(0*?)(?:(\d{1,6}))` | keep · `EC_CM_ALFANUM` |
| otherwise | `EC_IDEOLOGIA_LISTA` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### EC_IDEOLOGIA_LISTA SECURE LOOKUP

Replaces the value with one of the 7 values in `ec-ideologias.txt`, chosen by a keyed hash.

Used by: `EC_IDEOLOGIA`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-ideologias.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### EC_INSTITUCION_EDUCATIVA SECURE LOOKUP

Replaces the value with one of the 12 values in `ec-instituciones-educativas.txt`, chosen by a keyed hash.

Used by: `EC_L2_INSTITUCION_EDUCATIVA`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-instituciones-educativas.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### EC_IP REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `EC_CM_HEX`.

Used by: `EC_L1_IP`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(0*?)(?:(\d{1,3})\.(\d{1,3})\.(\d{1,3})\.(\d{1,3}))` | keep · `EC_OCTETO` · `EC_OCTETO` · `EC_OCTETO` · `EC_OCTETO` |
| otherwise | `EC_CM_HEX` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### EC_LENGUA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_LENGUA_LISTA`.

Used by: `EC_L3_LENGUA`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(s[ií]\|no\|true\|false\|yes\|[snyx01])` | `EC_BANDERA` |
| `(0*?)(?:(\d{1,6}))` | keep · `EC_CM_ALFANUM` |
| otherwise | `EC_LENGUA_LISTA` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### EC_LENGUA_LISTA SECURE LOOKUP

Replaces the value with one of the 16 values in `ec-lenguas.txt`, chosen by a keyed hash.

Used by: `EC_LENGUA`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-lenguas.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### EC_MEDICAMENTO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_MEDICAMENTO_LISTA`.

Used by: `EC_SALUD_MEDICAMENTO`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(s[ií]\|no\|true\|false\|yes\|[snyx01])` | `EC_BANDERA` |
| `(0*?)(?:(\d{1,6}))` | keep · `EC_CM_ALFANUM` |
| otherwise | `EC_MEDICAMENTO_LISTA` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### EC_MEDICAMENTO_LISTA SECURE LOOKUP

Replaces the value with one of the 22 values in `ec-medicamentos.txt`, chosen by a keyed hash.

Used by: `EC_MEDICAMENTO`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-medicamentos.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### EC_NACIONALIDAD_PAIS SECURE LOOKUP

Replaces the value with one of the 14 values in `ec-nacionalidades.txt`, chosen by a keyed hash.

Used by: `EC_L3_NACIONALIDAD`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-nacionalidades.txt` |
| hashMethod | `SHA256` |

| | |
|---|---|
| maskedValueCase | PRESERVE_INPUT |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### EC_NIVEL_INSTRUCCION REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: EC_NIVEL_INSTRUCCION_LISTA .

Used by: EC_L2_NIVEL_INSTRUCCION

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| (?i)(s[ií]\|no\|true\|false\|yes\|[snyx01]) | EC_BANDERA |
| (0*)([1-9]\d{0,5}\|0) | keep · EC_NIVEL_INSTRUCCION_CODIGO |
| otherwise | EC_NIVEL_INSTRUCCION_LISTA |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### EC_NIVEL_INSTRUCCION_CODIGO SECURE LOOKUP

Replaces the value with one of the 8 values in ec-nivel-instruccion-codigos.txt , chosen by a keyed hash.

Used by: EC_NIVEL_INSTRUCCION

#### CONFIGURATION

| | |
|---|---|
| lookupFile | ec-nivel-instruccion-codigos.txt |
| hashMethod | SHA256 |
| maskedValueCase | PRESERVE_INPUT |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### EC_NIVEL_INSTRUCCION_LISTA SECURE LOOKUP

Replaces the value with one of the 8 values in ec-nivel-instruccion.txt , chosen by a keyed hash.

Used by: EC_NIVEL_INSTRUCCION

#### CONFIGURATION

| | |
|---|---|
| lookupFile | ec-nivel-instruccion.txt |
| hashMethod | SHA256 |
| maskedValueCase | PRESERVE_INPUT |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### EC_NOMBRE NAME

Replaces the name with one of the 338 names in ec-nombres.txt .

Used by: EC_PALABRA_NOMBRE

#### CONFIGURATION

| | |
|---|---|
| lookupFile | ec-nombres.txt |

| | |
|---|---|
| maskedValueCase | PRESERVE_INPUT |
| filterAccent | yes |
| inputCaseSensitive | no |

## EC_NOMBRE_COMPLETO REGEX DECOMPOSE

Tries 7 patterns in order and gives each part of the value its own action; when none matches: EC_CM_ALFANUM .

Used by: EC_L1_NOMBRE_COMPLETO

### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| ([^,]+),\s*(.+) | EC_APELLIDOS · EC_NOMBRES |
| (\S+) | EC_PALABRA_NOMBRE |
| (\S+)\s+(\S+) | EC_PALABRA_NOMBRE · EC_PALABRA_APELLIDO |
| (\S+)\s+(\S+)\s+(\S+) | EC_PALABRA_NOMBRE · EC_PALABRA_APELLIDO · EC_PALABRA_APELLIDO |
| (\S+)\s+(\S+)\s+(\S+)\s+(\S+) | EC_PALABRA_NOMBRE · EC_PALABRA_NOMBRE · EC_PALABRA_APELLIDO · EC_PALABRA_APELLIDO |
| (\S+)\s+(\S+)\s+(\S+)\s+(\S+)\s+(\S+) | EC_PALABRA_NOMBRE · EC_PALABRA_NOMBRE · EC_PALABRA_APELLIDO · EC_PALABRA_APELLIDO · EC_PALABRA_APELLIDO |
| (\S+)\s+(\S+)\s+(\S+)\s+(\S+)\s+(\S+)\s+(\S+) | EC_PALABRA_NOMBRE · EC_PALABRA_NOMBRE · EC_PALABRA_NOMBRE · EC_PALABRA_APELLIDO · EC_PALABRA_APELLIDO · EC_PALABRA_APELLIDO |
| otherwise | EC_CM_ALFANUM |

### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

## EC_NOMBRES REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: EC_CM_ALFANUM .

Used by: EC_L1_NOMBRE , EC_NOMBRE_COMPLETO

### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| (\S+) | EC_PALABRA_NOMBRE |
| (\S+)\s+(\S+) | EC_PALABRA_NOMBRE · EC_PALABRA_NOMBRE |
| (\S+)\s+(\S+)\s+(\S+) | EC_PALABRA_NOMBRE · EC_PALABRA_NOMBRE · EC_PALABRA_NOMBRE |
| (\S+)\s+(\S+)\s+(\S+)\s+(\S+) | EC_PALABRA_NOMBRE · EC_PALABRA_NOMBRE · EC_PALABRA_NOMBRE · EC_PALABRA_NOMBRE |
| otherwise | EC_CM_ALFANUM |

### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

## EC_OCTETO NUMERIC MAPPING

Maps the number to another between 1 and 254.

Used by: EC_IP

### CONFIGURATION

| | |
|---|---|
| minValue | 1 |
| maxValue | 254 |

### EC_OCUPACION SECURE LOOKUP

Replaces the value with one of the 38 values in `ec-ocupaciones.txt`, chosen by a keyed hash.

Used by: `EC_OCUPACION_O_CODIGO`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-ocupaciones.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### EC_OCUPACION_O_CODIGO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `EC_OCUPACION`.

Used by: `EC_L2_OCUPACION`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(0*?)(?:(\d{2,5}))` | keep · `EC_CM_DIGITOS` |
| otherwise | `EC_OCUPACION` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### EC_ORIENTACION_SEXUAL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_ORIENTACION_SEXUAL_LISTA`.

Used by: `EC_L3_ORIENTACION_SEXUAL`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(s[ií]\|no\|true\|false\|yes\|[snyx01])` | `EC_BANDERA` |
| `(0*?)(?:(\d{1,6}))` | keep · `EC_CM_ALFANUM` |
| otherwise | `EC_ORIENTACION_SEXUAL_LISTA` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### EC_ORIENTACION_SEXUAL_LISTA SECURE LOOKUP

Replaces the value with one of the 8 values in `ec-orientaciones-sexuales.txt`, chosen by a keyed hash.

Used by: `EC_ORIENTACION_SEXUAL`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-orientaciones-sexuales.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |

| | |
|---|---|
| trimWhitespaceInLookupFile | yes |

### EC_PALABRA_APELLIDO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `EC_APELLIDO`.

Used by: `EC_APELLIDOS`, `EC_NOMBRE_COMPLETO`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(de\|del\|la\|las\|los\|y\|e\|san\|santa\|van\|von\|di\|da)` | keep |
| otherwise | `EC_APELLIDO` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### EC_PALABRA_NOMBRE REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `EC_NOMBRE`.

Used by: `EC_NOMBRES`, `EC_NOMBRE_COMPLETO`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(de\|del\|la\|las\|los\|y\|e\|san\|santa\|van\|von\|di\|da)` | keep |
| otherwise | `EC_NOMBRE` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### EC_PARROQUIA DATA CLEANSING

Replaces each value found in `ec-parroquias-generalizadas.txt` (25,214 pairs) by its counterpart; values not in the file stay as they are.

Used by: `EC_L2_PARROQUIA`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-parroquias-generalizadas.txt` |
| delimiter | `\|` |
| caseSensitive | yes |
| trimWhitespace | yes |

### EC_PARTIDO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_PARTIDO_LISTA`.

Used by: `EC_L3_FILIACION_POLITICA`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(s[ií]\|no\|true\|false\|yes\|[snyx01])` | `EC_BANDERA` |
| `(0*?)(?:(\d{1,6}))` | keep · `EC_CM_ALFANUM` |
| otherwise | `EC_PARTIDO_LISTA` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### EC_PARTIDO_LISTA SECURE LOOKUP

Replaces the value with one of the 13 values in `ec-partidos.txt`, chosen by a keyed hash.

Used by: `EC_PARTIDO`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-partidos.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### EC_PROCEDIMIENTO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `EC_PROCEDIMIENTO_TEXTO`.

Used by: `EC_SALUD_PROCEDIMIENTO`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(0*?)(?:(\d{4,6}))` | keep · `EC_CM_DIGITOS` |
| otherwise | `EC_PROCEDIMIENTO_TEXTO` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### EC_PROCEDIMIENTO_TEXTO SECURE LOOKUP

Replaces the value with one of the 10 values in `ec-procedimientos.txt`, chosen by a keyed hash.

Used by: `EC_PROCEDIMIENTO`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-procedimientos.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### EC_PROGRAMA_SOCIAL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_PROGRAMA_SOCIAL_LISTA`.

Used by: `EC_FIN_PROGRAMA_SOCIAL`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(s[ií]\|no\|true\|false\|yes\|[snyx01])` | `EC_BANDERA` |
| `(0*?)(?:(\d{1,6}))` | keep · `EC_CM_ALFANUM` |

| | |
|---|---|
| otherwise | `EC_PROGRAMA_SOCIAL_LISTA` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### EC_PROGRAMA_SOCIAL_LISTA SECURE LOOKUP

Replaces the value with one of the 7 values in `ec-programas-sociales.txt`, chosen by a keyed hash.

Used by: `EC_PROGRAMA_SOCIAL`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-programas-sociales.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### EC_RAZON_SOCIAL SECURE LOOKUP

Replaces the value with one of the 20 values in `ec-razones-sociales.txt`, chosen by a keyed hash.

Used by: `EC_L1_RAZON_SOCIAL`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-razones-sociales.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_LOOKUP_FILE` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### EC_REDACTAR REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: `EC_L1_CREDENCIAL`, `EC_L3_BIOMETRICO`, `EC_L3_GENETICO`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(.+)` | each character → `X` |
| otherwise | keep |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### EC_REGISTRO_SOCIAL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_REGISTRO_SOCIAL_LISTA`.

Used by: `EC_FIN_REGISTRO_SOCIAL`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(s[ií]\|no\|true\|false\|yes\|[snyx01])` | `EC_BANDERA` |

| | |
|---|---|
| `(0*)([1-9]\d{0,5}\|0)` | keep · `EC_REGISTRO_SOCIAL_CODIGO` |
| otherwise | `EC_REGISTRO_SOCIAL_LISTA` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### EC_REGISTRO_SOCIAL_CODIGO SECURE LOOKUP

Replaces the value with one of the 5 values in `ec-registro-social-codigos.txt`, chosen by a keyed hash.

Used by: `EC_REGISTRO_SOCIAL`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-registro-social-codigos.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### EC_REGISTRO_SOCIAL_LISTA SECURE LOOKUP

Replaces the value with one of the 9 values in `ec-registro-social.txt`, chosen by a keyed hash.

Used by: `EC_REGISTRO_SOCIAL`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-registro-social.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### EC_RELIGION REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_RELIGION_LISTA`.

Used by: `EC_L3_RELIGION`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(s[ií]\|no\|true\|false\|yes\|[snyx01])` | `EC_BANDERA` |
| `(0*?)(?:(\d{1,6}))` | keep · `EC_CM_ALFANUM` |
| otherwise | `EC_RELIGION_LISTA` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### EC_RELIGION_LISTA SECURE LOOKUP

Replaces the value with one of the 10 values in `ec-religiones.txt`, chosen by a keyed hash.

Used by: `EC_RELIGION`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-religiones.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### EC_RUC REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_CM_DIGITOS` .

Used by: `EC_L1_RUC` , `EC_DOCUMENTO`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(0*?)(?:((?:[0-2]\d\|30)[0-5]\d{7})(00\d))` | keep · `EC_CEDULA_LUHN` · keep |
| `(0*?)(?:(\d{2}[69]\d{10}))` | keep · keep |
| otherwise | `EC_CM_DIGITOS` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### EC_SEXO REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `EC_L2_SEXO`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(femenino\|masculino)` | `EC_SEXO_FEMENINO_MASCULINO` |
| `(?i)(mujer\|hombre)` | `EC_SEXO_MUJER_HOMBRE` |
| `(?i)(female\|male)` | `EC_SEXO_FEMALE_MALE` |
| `(?i)([fm])` | `EC_SEXO_F_M` |
| `(?i)([hm])` | `EC_SEXO_H_M` |
| `(0*?)(?:([12]))` | keep · `EC_SEXO_1_2` |
| otherwise | keep |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### EC_SEXO_1_2 SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-sexo-1-2.txt` , chosen by a keyed hash.

Used by: `EC_SEXO`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-sexo-1-2.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |

| | |
|---|---|
| trimWhitespaceInLookupFile | yes |

### EC_SEXO_F_M SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-sexo-f-m.txt`, chosen by a keyed hash.

Used by: `EC_SEXO`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-sexo-f-m.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### EC_SEXO_FEMALE_MALE SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-sexo-female-male.txt`, chosen by a keyed hash.

Used by: `EC_SEXO`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-sexo-female-male.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### EC_SEXO_FEMENINO_MASCULINO SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-sexo-femenino-masculino.txt`, chosen by a keyed hash.

Used by: `EC_SEXO`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-sexo-femenino-masculino.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### EC_SEXO_H_M SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-sexo-h-m.txt`, chosen by a keyed hash.

Used by: `EC_SEXO`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-sexo-h-m.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |

| | |
|---|---|
| trimWhitespaceInLookupFile | yes |

### EC_SEXO_MUJER_HOMBRE SECURE LOOKUP

Replaces the value with one of the 2 values in `ec-sexo-mujer-hombre.txt`, chosen by a keyed hash.

Used by: `EC_SEXO`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-sexo-mujer-hombre.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### EC_SUPRIMIR SECURE LOOKUP

Replaces the value with one of the 1 values in `ec-sin-informacion.txt`, chosen by a keyed hash.

Used by: `EC_CATEGORIA_SUPRIMIDA`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-sin-informacion.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_LOOKUP_FILE` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### EC_TARJETA REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: `EC_L1_TARJETA`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(0*?)(?:(\d[\d\s\-]{11,22}\d))` | keep · `dlpx-core:Credit Card` |
| otherwise | keep |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### EC_TELEFONO REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: `EC_CM_DIGITOS`.

Used by: `EC_L1_TELEFONO`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(0*?)(?:(\+?593[\s\-]?)(0?9)([\s\-]?)([\d\s\-]{8,12}))` | keep · keep · keep · keep · `EC_CM_DIGITOS` |
| `(0*?)(?:(0?9)([\s\-]?)([\d\s\-]{8,12}))` | keep · keep · keep · `EC_CM_DIGITOS` |
| `(0*?)(?:(\+?593[\s\-]?)?(\(?0?[2-7]\)?)([\s\-]?)([\d\s\-]{6,10}))` | keep · keep · keep · keep · `EC_CM_DIGITOS` |
| otherwise | `EC_CM_DIGITOS` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### EC_TENENCIA_VIVIENDA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `EC_TENENCIA_VIVIENDA_LISTA`.

Used by: `EC_FIN_VIVIENDA`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(s[ií]\|no\|true\|false\|yes\|[snyx01])` | `EC_BANDERA` |
| `(0*?)(?:(\d{1,6}))` | keep · `EC_CM_ALFANUM` |
| otherwise | `EC_TENENCIA_VIVIENDA_LISTA` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### EC_TENENCIA_VIVIENDA_LISTA SECURE LOOKUP

Replaces the value with one of the 7 values in `ec-tenencia-vivienda.txt`, chosen by a keyed hash.

Used by: `EC_TENENCIA_VIVIENDA`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `ec-tenencia-vivienda.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### EC_TEXTO_LIBRE FREE TEXT REDACTION

Replaces the text matching 8 patterns with `[DATO]`, and the 1,668 terms of `ec-texto-nombres.txt` with `[NOMBRE]`.

Used by: `EC_L1_TEXTO_LIBRE`, `EC_SALUD_TEXTO_CLINICO`

#### PATTERNS, TRIED IN ORDER

| |
|---|
| `(?<!\d)(0[1-9]\|1\d\|2[0-4]\|30)[0-5]\d{7}00\d(?!\d)` |
| `(?<!\d)(0[1-9]\|1\d\|2[0-4]\|30)[0-5]\d{7}(?!\d)` |
| `(?<!\d)\d{9}-\d(?!\d)` |
| `[\w.+\-]+@[\w\-]+(\.[\w\-]+)+` |
| `(?<![\d\-])(\+?593[\s\-]?)?0?9\d{2}[\s\-]?\d{3}[\s\-]?\d{3}(?![\d\-])` |
| `(?<!\d)\d{13,19}(?!\d)` |
| `(?<![A-Za-z])[A-Z]{3}[\s\-]?\d{3,4}(?![A-Za-z\d])` |
| `(?<!\d)(\d{1,3}\.){3}\d{1,3}(?!\d)` |

#### CONFIGURATION

| | |
|---|---|
| regExRedactValue | `[DATO]` |
| lookupFile | `ec-texto-nombres.txt` |
| lookupFileRedactValue | `[NOMBRE]` |

| | |
|---|---|
| isDenyList | yes |

APPENDIX B

# Files

The lookup and detection files the policy reads. In Delphix each one is uploaded with the object that reads it — the lookup file of a Secure Lookup or Data Cleansing algorithm, the value list of a List classifier — and referenced by that object's configuration.

| FILE | LINES | USED BY | FIRST LINES |
|---|---|---|---|
| `ec-antecedentes.txt` | 7 | `EC_ANTECEDENTES_LISTA` | No registra antecedentes penales · Registra antecedentes · Procesado · Sentenciado |
| `ec-apellidos.txt` | 366 | `EC_APELLIDO` | Zambrano · Vera · García · Mero |
| `ec-aptitud-laboral.txt` | 4 | `EC_APTITUD_LABORAL_LISTA` | Apto · Apto con restricciones · No apto · Pendiente |
| `ec-autoidentificacion-codigos.txt` | 8 | `EC_AUTOIDENTIFICACION_CODIGO` | 1 · 2 · 3 · 4 |
| `ec-autoidentificacion.txt` | 8 | `EC_AUTOIDENTIFICACION_LISTA` | Mestizo · Montubio · Afroecuatoriano · Negro |
| `ec-bandera-0-1.txt` | 2 | `EC_BANDERA_0_1` | 0 · 1 |
| `ec-bandera-s-n.txt` | 2 | `EC_BANDERA_S_N` | S · N |
| `ec-bandera-si-no.txt` | 2 | `EC_BANDERA_SI_NO` | Sí · No |
| `ec-bandera-true-false.txt` | 2 | `EC_BANDERA_TRUE_FALSE` | true · false |
| `ec-bandera-y-n.txt` | 2 | `EC_BANDERA_Y_N` | Y · N |
| `ec-bandera-yes-no.txt` | 2 | `EC_BANDERA_YES_NO` | Yes · No |
| `ec-cie10-decimal.txt` | 33 | `EC_CIE10_DECIMAL` | E11.9 · E78.5 · J45.9 · J06.9 |
| `ec-cie10.txt` | 37 | `EC_CIE10` | I10 · E11 · E78 · J45 |
| `ec-ciudadelas.txt` | 108 | `EC_CIUDADELA` | Cdla. Kennedy · Cdla. Urdesa Central · Cdla. Urdesa Norte · Cdla. Alborada |
| `ec-cobertura-salud.txt` | 8 | `EC_COBERTURA` | IESS (Seguro General) · Seguro Social Campesino · ISSFA · ISSPOL |
| `ec-condicion-migratoria.txt` | 10 | `EC_CONDICION_MIGRATORIA_LISTA` | Ecuatoriano · Residente permanente · Residente temporal · Visitante temporal |
| `ec-detectar-apellidos.txt` | 454 | `EC_L1_APELLIDO – List` `EC_L1_NOMBRE_COMPLETO – List` | Zambrano · Vera · García · Garcia |

| FILE | LINES | USED BY | FIRST LINES |
|---|---|---|---|
| ec-detectar-nacionalidades.txt | 32 | EC_L3_NACIONALIDAD – List | Ecuatoriana · Venezolana · Colombiana · Peruana |
| ec-detectar-nombres.txt | 395 | EC_L1_NOMBRE – List EC_L1_NOMBRE_COMPLETO – List | José · Jose · María · Maria |
| ec-detectar-sexo-letra.txt | 3 | EC_L2_SEXO – List | f · m · h |
| ec-detectar-sexo.txt | 6 | EC_L2_SEXO – List | femenino · masculino · mujer · hombre |
| ec-diagnosticos.txt | 30 | EC_DIAGNOSTICO_TEXTO | Hipertensión arterial · Diabetes mellitus tipo 2 · Asma · Lumbalgia |
| ec-direcciones.txt | 4,000 | EC_DIRECCION | Malecón Occidental 2067 y Simón Bolívar · Av. Fray Vicente Solano N52-293 y Mariscal Sucre · Barrio La Mariscal, Mz. M Villa 60 · Calle 10 de Agosto N38-214 y Brasil |
| ec-discapacidades.txt | 8 | EC_DISCAPACIDAD_LISTA | Física · Auditiva · Visual · Intelectual |
| ec-dpa-generalizado.txt | 1,441 | EC_DPA_TABLA | 020151\|020150 · 020153\|020150 · 020155\|020150 · 020156\|020150 |
| ec-empleadores.txt | 24 | EC_EMPLEADOR | Distribuidora Andina Cía. Ltda. · Constructora Pichincha S.A. · Transportes Mitad del Mundo S.A. · Supermercados del Litoral S.A. |
| ec-estado-civil-codigos.txt | 6 | EC_ESTADO_CIVIL_CODIGO | 1 · 2 · 3 · 4 |
| ec-estado-civil.txt | 6 | EC_ESTADO_CIVIL_LISTA | Soltero/a · Casado/a · Unión de hecho · Separado/a |
| ec-estados-credito.txt | 15 | EC_ESTADO_CREDITO | A-1 Riesgo normal · A-2 Riesgo normal · A-3 Riesgo normal · B-1 Riesgo potencial |
| ec-identidad-cultural.txt | 30 | EC_IDENTIDAD_CULTURAL_LISTA | Kichwa · Shuar · Achuar · Awá |
| ec-identidades-genero.txt | 8 | EC_IDENTIDAD_GENERO_LISTA | Mujer cisgénero · Hombre cisgénero · Mujer trans · Hombre trans |
| ec-ideologias.txt | 7 | EC_IDEOLOGIA_LISTA | Izquierda · Centroizquierda · Centro · Centroderecha |
| ec-instituciones-educativas.txt | 12 | EC_INSTITUCION_EDUCATIVA | Unidad Educativa Nuevo Amanecer · Unidad Educativa Fiscal 10 de Agosto · Escuela de Educación Básica Simón Bolívar · Colegio Nacional Mejía |
| ec-lenguas.txt | 16 | EC_LENGUA_LISTA | Castellano · Kichwa · Shuar chicham · Achuar chicham |
| ec-medicamentos.txt | 22 | EC_MEDICAMENTO_LISTA | Paracetamol · Ibuprofeno · Diclofenaco · Metformina |

| FILE | LINES | USED BY | FIRST LINES |
|---|---|---|---|
| ec-nacionalidades.txt | 14 | EC_NACIONALIDAD_PAIS | Ecuatoriana · Venezolana · Colombiana · Peruana |
| ec-nivel-instruccion-codigos.txt | 8 | EC_NIVEL_INSTRUCCION_CODIGO | 1 · 2 · 3 · 4 |
| ec-nivel-instruccion.txt | 8 | EC_NIVEL_INSTRUCCION_LISTA | Ninguno · Centro de alfabetización · Educación inicial · Educación general básica |
| ec-nombres.txt | 338 | EC_NOMBRE | José · María · Juan · Carlos |
| ec-ocupaciones.txt | 38 | EC_OCUPACION | Empleado administrativo · Vendedor · Cajero · Chofer |
| ec-orientaciones-sexuales.txt | 8 | EC_ORIENTACION_SEXUAL_LISTA | Heterosexual · Gay · Lesbiana · Bisexual |
| ec-parroquias-generalizadas.txt | 25,214 | EC_PARROQUIA | Facundo Vela\|Guaranda · Facundo Vela - Guaranda\|Guaranda - Guaranda · Facundo Vela, Guaranda\|Guaranda, Guaranda · Facundo Vela (Guaranda)\|Guaranda (Guaranda) |
| ec-partidos.txt | 13 | EC_PARTIDO_LISTA | Revolución Ciudadana · Acción Democrática Nacional · Movimiento CREO · Partido Social Cristiano |
| ec-procedimientos.txt | 10 | EC_PROCEDIMIENTO_TEXTO | Consulta médica general · Consulta de emergencia · Biometría hemática · Glucosa en sangre |
| ec-programas-sociales.txt | 7 | EC_PROGRAMA_SOCIAL_LISTA | Bono de Desarrollo Humano · Bono Joaquín Gallegos Lara · Pensión para Adultos Mayores · Pensión para Personas con Discapacidad |
| ec-razones-sociales.txt | 20 | EC_RAZON_SOCIAL | Comercial Andina Cía. Ltda. · Distribuidora del Pacífico S.A. · Importadora Los Andes Cía. Ltda. · Transportes Mitad del Mundo S.A. |
| ec-registro-social-codigos.txt | 5 | EC_REGISTRO_SOCIAL_CODIGO | 1 · 2 · 3 · 4 |
| ec-registro-social.txt | 9 | EC_REGISTRO_SOCIAL_LISTA | Quintil 1 · Quintil 2 · Quintil 3 · Quintil 4 |
| ec-religiones.txt | 10 | EC_RELIGION_LISTA | Católica · Evangélica · Testigo de Jehová · Iglesia de Jesucristo de los Santos de los Últimos Días |
| ec-sexo-1-2.txt | 2 | EC_SEXO_1_2 | 1 · 2 |
| ec-sexo-f-m.txt | 2 | EC_SEXO_F_M | F · M |
| ec-sexo-female-male.txt | 2 | EC_SEXO_FEMALE_MALE | Female · Male |
| ec-sexo-femenino-masculino.txt | 2 | EC_SEXO_FEMENINO_MASCULINO | Femenino · Masculino |
| ec-sexo-h-m.txt | 2 | EC_SEXO_H_M | H · M |

| FILE | LINES | USED BY | FIRST LINES |
|---|---|---|---|
| `ec-sexo-mujer-hombre.txt` | 2 | `EC_SEXO_MUJER_HOMBRE` | Mujer · Hombre |
| `ec-sin-informacion.txt` | 1 | `EC_SUPRIMIR` | Sin información |
| `ec-tenencia-vivienda.txt` | 7 | `EC_TENENCIA_VIVIENDA_LISTA` | Propia y totalmente pagada · Propia y la está pagando · Propia (regalada, donada, heredada) · Prestada o cedida |
| `ec-texto-nombres.txt` | 1,668 | `EC_TEXTO_LIBRE` | José · JOSÉ · Jose · JOSE |

APPENDIX C

# Exclusion lists

Words that keep a column-name rule from matching when they follow it — so that `NOMBRE_PRODUCTO` is not taken for a person's name.

**E1** `producto` `prod` `item` `articulo` `empresa` `emp` `razon` `social` `comercial` `archivo` `calle` `avenida` `pasaje` `ciudadela` `urbanizacion` `barrio` `parroquia` `canton` `provincia` `region` `depto` `dpto` `pais` `banco` `sucursal` `agencia` `plan` `campana` `proyecto` `servicio` `tabla` `columna` `campo` `usuario` `host` `servidor` `dominio` `marca` `modelo` `categoria` `tipo` `escuela` `colegio` `institucion` `universidad` `curso` `materia` `documento` `doc` `unidad` `clinica` `hospital` `iess` `seguro` `area` `dependencia` `cargo` `sistema` `app` `aplicacion` `grupo` `lista` `reporte` `informe` `proceso` `evento` `tarea` `perfil` `rol` `clase` `objeto` `cuenta` `medicamento` `diagnostico` `estudio` `programa` `beneficio` `partido` `sindicato` `religion` `pueblo` `nacionalidad` `lengua` `idioma` `lugar` `sede` `local` `tienda` `bodega` `proveedor` `convenio` `contrato` `poliza` `aseguradora` `entidad` `organismo` `organizacion` `parametro` `variable` `metodo` `funcion` `indice` `job` `script` `cola` `recurso` `red` `dispositivo` `plantilla` `imagen` `foto` `pagina` `sitio` `modulo` `menu` `opcion` `accion` `permiso` `concepto` `impuesto` `moneda` `emisor` `factura` `comprobante` `pago` `forma` `zona` `ruta` `linea` `actividad` `rubro` `sector` `file` `table` `column` `field` `user` `server` `product` `company` `category` `type` `school` `document` `department` `system` `group` `report` `role` `account` `place` `store` `supplier` `contract` `device` `image` `page` `module` `action` `key` `label` `code` `project` `city` `country` `street` `bank` `cliente` `cli` `paciente` `pac` `usuario` `afiliado` `asegurado` `beneficiario` `benef` `titular` `empleado` `trabajador` `colaborador` `alumno` `estudiante` `deudor` `codeudor` `garante` `solicitante` `propietario` `arrendatario` `inquilino` `conductor` `responsable` `representante` `apoderado` `persona` `ciudadano` `contribuyente` `madre` `padre` `conyuge` `conviviente` `contacto` `referencia` `victima` `denunciante` `procesado` `testigo` `medico` `docente` `socio` `heredero` `donante` `tutor` `jubilado` `pensionista` `derechohabiente` `customer` `patient` `employee` `member` `holder` `mother` `father` `spouse` `contact` `person` `student` `driver` `owner` `completo` `apellidos?` `usuario` `corto` `largo`

**E2** `producto` `prod` `item` `articulo` `empresa` `emp` `razon` `social` `comercial` `archivo` `calle` `avenida` `pasaje` `ciudadela` `urbanizacion` `barrio` `parroquia` `canton` `provincia` `region` `depto` `dpto` `pais` `banco` `sucursal` `agencia` `plan` `campana` `proyecto` `servicio` `tabla` `columna` `campo` `usuario` `host` `servidor` `dominio` `marca` `modelo` `categoria` `tipo` `escuela` `colegio` `institucion` `universidad` `curso` `materia` `documento` `doc` `unidad` `clinica` `hospital` `iess` `seguro` `area` `dependencia` `cargo` `sistema` `app` `aplicacion` `grupo` `lista` `reporte` `informe` `proceso` `evento` `tarea` `perfil` `rol` `clase` `objeto` `cuenta` `medicamento` `diagnostico` `estudio` `programa` `beneficio` `partido` `sindicato` `religion` `pueblo` `nacionalidad` `lengua` `idioma` `lugar` `sede` `local` `tienda` `bodega` `proveedor` `convenio` `contrato` `poliza` `aseguradora` `entidad` `organismo` `organizacion` `parametro` `variable` `metodo` `funcion` `indice` `job` `script` `cola` `recurso` `red` `dispositivo` `plantilla` `imagen` `foto` `pagina` `sitio` `modulo` `menu` `opcion` `accion` `permiso` `concepto` `impuesto` `moneda` `emisor` `factura` `comprobante` `pago` `forma` `zona` `ruta` `linea` `actividad` `rubro` `sector` `file` `table` `column` `field` `user` `server` `product` `company` `category` `type` `school` `document` `department` `system` `group` `report` `role` `account` `place` `store` `supplier` `contract` `device` `image` `page` `module` `action` `key` `label` `code` `project` `city` `country` `street` `bank` `cliente` `cli` `paciente` `pac` `usuario` `afiliado` `asegurado` `beneficiario` `benef` `titular` `empleado` `trabajador` `colaborador` `alumno` `estudiante` `deudor` `codeudor` `garante` `solicitante` `propietario` `arrendatario` `inquilino` `conductor` `responsable` `representante` `apoderado` `persona` `ciudadano` `contribuyente` `madre` `padre` `conyuge` `conviviente` `contacto` `referencia` `victima` `denunciante` `procesado` `testigo` `medico` `docente` `socio` `heredero` `donante` `tutor` `jubilado` `pensionista` `derechohabiente` `customer` `patient` `employee` `member` `holder` `mother` `father` `spouse` `contact` `person` `student` `driver` `owner`

| | |
|---|---|
| **E3** | `producto` `prod` `item` `articulo` `empresa` `emp` `razon` `social` `comercial` `archivo` `calle` `avenida` `pasaje` `ciudadela` `urbanizacion` `barrio` `parroquia` `canton` `provincia` `region` `depto` `dpto` `pais` `banco` `sucursal` `agencia` `plan` `campana` `proyecto` `servicio` `tabla` `columna` `campo` `usuario` `host` `servidor` `dominio` `marca` `modelo` `categoria` `tipo` `escuela` `colegio` `institucion` `universidad` `curso` `materia` `documento` `doc` `unidad` `clinica` `hospital` `iess` `seguro` `area` `dependencia` `cargo` `sistema` `app` `aplicacion` `grupo` `lista` `reporte` `informe` `proceso` `evento` `tarea` `perfil` `rol` `clase` `objeto` `cuenta` `medicamento` `diagnostico` `estudio` `programa` `beneficio` `partido` `sindicato` `religion` `pueblo` `nacionalidad` `lengua` `idioma` `lugar` `sede` `local` `tienda` `bodega` `proveedor` `convenio` `contrato` `poliza` `aseguradora` `entidad` `organismo` `organizacion` `parametro` `variable` `metodo` `funcion` `indice` `job` `script` `cola` `recurso` `red` `dispositivo` `plantilla` `imagen` `foto` `pagina` `sitio` `modulo` `menu` `opcion` `accion` `permiso` `concepto` `impuesto` `moneda` `emisor` `factura` `comprobante` `pago` `forma` `zona` `ruta` `linea` `actividad` `rubro` `sector` `file` `table` `column` `field` `user` `server` `product` `company` `category` `type` `school` `document` `department` `system` `group` `report` `role` `account` `place` `store` `supplier` `contract` `device` `image` `page` `module` `action` `key` `label` `code` `project` `city` `country` `street` `bank` `cliente` `cli` `paciente` `pac` `usuario` `afiliado` `asegurado` `beneficiario` `benef` `titular` `empleado` `trabajador` `colaborador` `alumno` `estudiante` `deudor` `codeudor` `garante` `solicitante` `propietario` `arrendatario` `inquilino` `conductor` `responsable` `representante` `apoderado` `persona` `ciudadano` `contribuyente` `madre` `padre` `conyuge` `conviviente` `contacto` `referencia` `victima` `denunciante` `procesado` `testigo` `medico` `docente` `socio` `heredero` `donante` `tutor` `jubilado` `pensionista` `derechohabiente` `customer` `patient` `employee` `member` `holder` `mother` `father` `spouse` `contact` `person` `student` `driver` `owner` `full` `complete` `last` `first` |
| **E4** | `ip` `mac` `email` `e_?mail` `correo` `electr[oó]nica` `web` `url` `tipo` `id` `general` `regional` |
| **E5** | `ip` `mac` `email` `e_?mail` `web` `url` `type` `id` |
| **E6** | `id` `cd` `cod` `codigo` `tipo` `fecha` `estado` `num` `nro` |
| **E7** | `id` `tipo` `codigo` `fecha` `estado` `cola` `log` |

APPENDIX D

# Discovery policy and classifiers

The discovery policy `EC – LOPDP – v3` runs these 143 classifiers, with an assignment threshold of 60%.

LOPDP - datos personales y sensibles

| CLASSIFIER | KIND | DATA CLASS |
|---|---|---|
| EC_L1_CEDULA – Path | COLUMN NAME | EC_L1_CEDULA |
| EC_L1_CEDULA – Type | DATA TYPE | EC_L1_CEDULA |
| EC_L1_CEDULA – Regex | VALUE FORMAT | EC_L1_CEDULA |
| EC_L1_RUC – Path | COLUMN NAME | EC_L1_RUC |
| EC_L1_RUC – Regex | VALUE FORMAT | EC_L1_RUC |
| EC_L1_DOCUMENTO_EXTRANJERO – Path | COLUMN NAME | EC_L1_DOCUMENTO_EXTRANJERO |
| EC_L1_PASAPORTE – Path | COLUMN NAME | EC_L1_PASAPORTE |
| EC_L1_PASAPORTE – Regex | VALUE FORMAT | EC_L1_PASAPORTE |
| EC_L1_ACTA – Path | COLUMN NAME | EC_L1_ACTA |
| EC_L1_SEGURO_SOCIAL – Path | COLUMN NAME | EC_L1_SEGURO_SOCIAL |
| EC_L1_CODIGO_ESTUDIANTE – Path | COLUMN NAME | EC_L1_CODIGO_ESTUDIANTE |
| EC_L1_REGISTRO_PROFESIONAL – Path | COLUMN NAME | EC_L1_REGISTRO_PROFESIONAL |
| EC_L1_NOMBRE – Path | COLUMN NAME | EC_L1_NOMBRE |
| EC_L1_NOMBRE – List | KNOWN VALUES | EC_L1_NOMBRE |
| EC_L1_APELLIDO – Path | COLUMN NAME | EC_L1_APELLIDO |
| EC_L1_APELLIDO – List | KNOWN VALUES | EC_L1_APELLIDO |
| EC_L1_NOMBRE_COMPLETO – Path | COLUMN NAME | EC_L1_NOMBRE_COMPLETO |
| EC_L1_NOMBRE_COMPLETO – List | KNOWN VALUES | EC_L1_NOMBRE_COMPLETO |
| EC_L1_RAZON_SOCIAL – Path | COLUMN NAME | EC_L1_RAZON_SOCIAL |
| EC_L1_EMAIL – Path | COLUMN NAME | EC_L1_EMAIL |
| EC_L1_EMAIL – Type | DATA TYPE | EC_L1_EMAIL |
| EC_L1_EMAIL – Regex | VALUE FORMAT | EC_L1_EMAIL |
| EC_L1_TELEFONO – Path | COLUMN NAME | EC_L1_TELEFONO |
| EC_L1_TELEFONO – Regex | VALUE FORMAT | EC_L1_TELEFONO |
| EC_L1_DIRECCION – Path | COLUMN NAME | EC_L1_DIRECCION |
| EC_L1_DIRECCION – Type | DATA TYPE | EC_L1_DIRECCION |
| EC_L1_DIRECCION – Regex | VALUE FORMAT | EC_L1_DIRECCION |
| EC_L1_DIRECCION_COMPLEMENTO – Path | COLUMN NAME | EC_L1_DIRECCION_COMPLEMENTO |
| EC_L1_CUENTA_BANCARIA – Path | COLUMN NAME | EC_L1_CUENTA_BANCARIA |
| EC_L1_TARJETA – Path | COLUMN NAME | EC_L1_TARJETA |
| EC_L1_TARJETA – Regex | VALUE FORMAT | EC_L1_TARJETA |
| EC_L1_BILLETERA – Path | COLUMN NAME | EC_L1_BILLETERA |

| CLASSIFIER | KIND | DATA CLASS |
|---|---|---|
| EC_L1_BILLETERA – Regex | VALUE FORMAT | EC_L1_BILLETERA |
| EC_L1_IP – Path | COLUMN NAME | EC_L1_IP |
| EC_L1_IP – Regex | VALUE FORMAT | EC_L1_IP |
| EC_L1_DISPOSITIVO – Path | COLUMN NAME | EC_L1_DISPOSITIVO |
| EC_L1_DISPOSITIVO – Regex | VALUE FORMAT | EC_L1_DISPOSITIVO |
| EC_L1_COOKIE – Path | COLUMN NAME | EC_L1_COOKIE |
| EC_L1_PLACA – Path | COLUMN NAME | EC_L1_PLACA |
| EC_L1_PLACA – Regex | VALUE FORMAT | EC_L1_PLACA |
| EC_L1_VEHICULO – Path | COLUMN NAME | EC_L1_VEHICULO |
| EC_L1_VEHICULO – Regex | VALUE FORMAT | EC_L1_VEHICULO |
| EC_L1_INMUEBLE – Path | COLUMN NAME | EC_L1_INMUEBLE |
| EC_L1_CONTRATO – Path | COLUMN NAME | EC_L1_CONTRATO |
| EC_L1_USUARIO – Path | COLUMN NAME | EC_L1_USUARIO |
| EC_L1_CREDENCIAL – Path | COLUMN NAME | EC_L1_CREDENCIAL |
| EC_L1_CREDENCIAL – Regex | VALUE FORMAT | EC_L1_CREDENCIAL |
| EC_L1_GEOLOCALIZACION – Path | COLUMN NAME | EC_L1_GEOLOCALIZACION |
| EC_L1_GEOLOCALIZACION – Regex | VALUE FORMAT | EC_L1_GEOLOCALIZACION |
| EC_L1_TEXTO_LIBRE – Path | COLUMN NAME | EC_L1_TEXTO_LIBRE |
| EC_L1_TEXTO_LIBRE – Type | DATA TYPE | EC_L1_TEXTO_LIBRE |
| EC_L1_TEXTO_LIBRE – Regex | VALUE FORMAT | EC_L1_TEXTO_LIBRE |
| EC_L2_FECHA_NACIMIENTO – Path | COLUMN NAME | EC_L2_FECHA_NACIMIENTO |
| EC_L2_FECHA_NACIMIENTO – Type | DATA TYPE | EC_L2_FECHA_NACIMIENTO |
| EC_L2_ANIO_NACIMIENTO – Path | COLUMN NAME | EC_L2_ANIO_NACIMIENTO |
| EC_L2_ANIO_NACIMIENTO – Type | DATA TYPE | EC_L2_ANIO_NACIMIENTO |
| EC_L2_EDAD – Path | COLUMN NAME | EC_L2_EDAD |
| EC_L2_EDAD – Type | DATA TYPE | EC_L2_EDAD |
| EC_L2_FECHA_EVENTO – Path | COLUMN NAME | EC_L2_FECHA_EVENTO |
| EC_L2_FECHA_EVENTO – Type | DATA TYPE | EC_L2_FECHA_EVENTO |
| EC_L2_SEXO – Path | COLUMN NAME | EC_L2_SEXO |
| EC_L2_SEXO – List | KNOWN VALUES | EC_L2_SEXO |
| EC_L2_PARROQUIA – Path | COLUMN NAME | EC_L2_PARROQUIA |
| EC_L2_PARROQUIA – Values | VALUE FORMAT | EC_L2_PARROQUIA |
| EC_L2_DPA – Path | COLUMN NAME | EC_L2_DPA |
| EC_L2_DPA – Regex | VALUE FORMAT | EC_L2_DPA |
| EC_L2_CIUDADELA – Path | COLUMN NAME | EC_L2_CIUDADELA |
| EC_L2_CODIGO_POSTAL – Path | COLUMN NAME | EC_L2_CODIGO_POSTAL |
| EC_L2_CODIGO_POSTAL – Type | DATA TYPE | EC_L2_CODIGO_POSTAL |

| CLASSIFIER | KIND | DATA CLASS |
|---|---|---|
| EC_L2_ESTADO_CIVIL – Path | COLUMN NAME | EC_L2_ESTADO_CIVIL |
| EC_L2_ESTADO_CIVIL – Values | VALUE FORMAT | EC_L2_ESTADO_CIVIL |
| EC_L2_OCUPACION – Path | COLUMN NAME | EC_L2_OCUPACION |
| EC_L2_EMPLEADOR – Path | COLUMN NAME | EC_L2_EMPLEADOR |
| EC_L2_NIVEL_INSTRUCCION – Path | COLUMN NAME | EC_L2_NIVEL_INSTRUCCION |
| EC_L2_NIVEL_INSTRUCCION – Values | VALUE FORMAT | EC_L2_NIVEL_INSTRUCCION |
| EC_L2_INSTITUCION_EDUCATIVA – Path | COLUMN NAME | EC_L2_INSTITUCION_EDUCATIVA |
| EC_L2_PERSONAS_A_CARGO – Path | COLUMN NAME | EC_L2_PERSONAS_A_CARGO |
| EC_L2_PERSONAS_A_CARGO – Type | DATA TYPE | EC_L2_PERSONAS_A_CARGO |
| EC_L3_ETNIA – Path | COLUMN NAME | EC_L3_ETNIA |
| EC_L3_ETNIA – Values | VALUE FORMAT | EC_L3_ETNIA |
| EC_L3_IDENTIDAD_CULTURAL – Path | COLUMN NAME | EC_L3_IDENTIDAD_CULTURAL |
| EC_L3_IDENTIDAD_CULTURAL – Values | VALUE FORMAT | EC_L3_IDENTIDAD_CULTURAL |
| EC_L3_LENGUA – Path | COLUMN NAME | EC_L3_LENGUA |
| EC_L3_LENGUA – Values | VALUE FORMAT | EC_L3_LENGUA |
| EC_L3_CONDICION_MIGRATORIA – Path | COLUMN NAME | EC_L3_CONDICION_MIGRATORIA |
| EC_L3_CONDICION_MIGRATORIA – Values | VALUE FORMAT | EC_L3_CONDICION_MIGRATORIA |
| EC_L3_NACIONALIDAD – Path | COLUMN NAME | EC_L3_NACIONALIDAD |
| EC_L3_NACIONALIDAD – List | KNOWN VALUES | EC_L3_NACIONALIDAD |
| EC_L3_RELIGION – Path | COLUMN NAME | EC_L3_RELIGION |
| EC_L3_RELIGION – Values | VALUE FORMAT | EC_L3_RELIGION |
| EC_L3_IDEOLOGIA – Path | COLUMN NAME | EC_L3_IDEOLOGIA |
| EC_L3_IDEOLOGIA – Values | VALUE FORMAT | EC_L3_IDEOLOGIA |
| EC_L3_FILIACION_POLITICA – Path | COLUMN NAME | EC_L3_FILIACION_POLITICA |
| EC_L3_FILIACION_POLITICA – Values | VALUE FORMAT | EC_L3_FILIACION_POLITICA |
| EC_L3_AFILIACION_SINDICAL – Path | COLUMN NAME | EC_L3_AFILIACION_SINDICAL |
| EC_L3_AFILIACION_SINDICAL – Values | VALUE FORMAT | EC_L3_AFILIACION_SINDICAL |
| EC_L3_BIOMETRICO – Path | COLUMN NAME | EC_L3_BIOMETRICO |
| EC_L3_BIOMETRICO – Type | DATA TYPE | EC_L3_BIOMETRICO |
| EC_L3_GENETICO – Path | COLUMN NAME | EC_L3_GENETICO |
| EC_L3_GENETICO – Regex | VALUE FORMAT | EC_L3_GENETICO |
| EC_L3_ORIENTACION_SEXUAL – Path | COLUMN NAME | EC_L3_ORIENTACION_SEXUAL |
| EC_L3_ORIENTACION_SEXUAL – Values | VALUE FORMAT | EC_L3_ORIENTACION_SEXUAL |
| EC_L3_IDENTIDAD_GENERO – Path | COLUMN NAME | EC_L3_IDENTIDAD_GENERO |
| EC_L3_IDENTIDAD_GENERO – Values | VALUE FORMAT | EC_L3_IDENTIDAD_GENERO |
| EC_L3_VICTIMA – Path | COLUMN NAME | EC_L3_VICTIMA |
| EC_SALUD_IDENTIFICADOR – Path | COLUMN NAME | EC_SALUD_IDENTIFICADOR |

| CLASSIFIER | KIND | DATA CLASS |
|---|---|---|
| EC_SALUD_COBERTURA – Path | COLUMN NAME | EC_SALUD_COBERTURA |
| EC_SALUD_COBERTURA – Values | VALUE FORMAT | EC_SALUD_COBERTURA |
| EC_SALUD_DIAGNOSTICO – Path | COLUMN NAME | EC_SALUD_DIAGNOSTICO |
| EC_SALUD_DIAGNOSTICO – Regex | VALUE FORMAT | EC_SALUD_DIAGNOSTICO |
| EC_SALUD_PROCEDIMIENTO – Path | COLUMN NAME | EC_SALUD_PROCEDIMIENTO |
| EC_SALUD_MEDICAMENTO – Path | COLUMN NAME | EC_SALUD_MEDICAMENTO |
| EC_SALUD_MEDICAMENTO – Values | VALUE FORMAT | EC_SALUD_MEDICAMENTO |
| EC_SALUD_TEXTO_CLINICO – Path | COLUMN NAME | EC_SALUD_TEXTO_CLINICO |
| EC_SALUD_TEXTO_CLINICO – Type | DATA TYPE | EC_SALUD_TEXTO_CLINICO |
| EC_SALUD_RESULTADO_EXAMEN – Path | COLUMN NAME | EC_SALUD_RESULTADO_EXAMEN |
| EC_SALUD_VIH – Path | COLUMN NAME | EC_SALUD_VIH |
| EC_SALUD_GRUPO_SANGUINEO – Path | COLUMN NAME | EC_SALUD_GRUPO_SANGUINEO |
| EC_SALUD_GRUPO_SANGUINEO – Values | VALUE FORMAT | EC_SALUD_GRUPO_SANGUINEO |
| EC_SALUD_DISCAPACIDAD – Path | COLUMN NAME | EC_SALUD_DISCAPACIDAD |
| EC_SALUD_DISCAPACIDAD – Values | VALUE FORMAT | EC_SALUD_DISCAPACIDAD |
| EC_SALUD_OCUPACIONAL – Path | COLUMN NAME | EC_SALUD_OCUPACIONAL |
| EC_SALUD_OCUPACIONAL – Values | VALUE FORMAT | EC_SALUD_OCUPACIONAL |
| EC_SALUD_REPRODUCTIVA – Path | COLUMN NAME | EC_SALUD_REPRODUCTIVA |
| EC_SALUD_MENTAL – Path | COLUMN NAME | EC_SALUD_MENTAL |
| EC_FIN_HISTORIAL_CREDITICIO – Path | COLUMN NAME | EC_FIN_HISTORIAL_CREDITICIO |
| EC_FIN_DEUDA – Path | COLUMN NAME | EC_FIN_DEUDA |
| EC_FIN_DEUDA – Type | DATA TYPE | EC_FIN_DEUDA |
| EC_FIN_INGRESOS – Path | COLUMN NAME | EC_FIN_INGRESOS |
| EC_FIN_INGRESOS – Type | DATA TYPE | EC_FIN_INGRESOS |
| EC_FIN_PATRIMONIO – Path | COLUMN NAME | EC_FIN_PATRIMONIO |
| EC_FIN_PATRIMONIO – Type | DATA TYPE | EC_FIN_PATRIMONIO |
| EC_FIN_PENSION – Path | COLUMN NAME | EC_FIN_PENSION |
| EC_FIN_PENSION – Type | DATA TYPE | EC_FIN_PENSION |
| EC_FIN_PROGRAMA_SOCIAL – Path | COLUMN NAME | EC_FIN_PROGRAMA_SOCIAL |
| EC_FIN_PROGRAMA_SOCIAL – Values | VALUE FORMAT | EC_FIN_PROGRAMA_SOCIAL |
| EC_FIN_REGISTRO_SOCIAL – Path | COLUMN NAME | EC_FIN_REGISTRO_SOCIAL |
| EC_FIN_REGISTRO_SOCIAL – Values | VALUE FORMAT | EC_FIN_REGISTRO_SOCIAL |
| EC_FIN_VIVIENDA – Path | COLUMN NAME | EC_FIN_VIVIENDA |
| EC_FIN_VIVIENDA – Values | VALUE FORMAT | EC_FIN_VIVIENDA |
| EC_PENAL_ANTECEDENTES – Path | COLUMN NAME | EC_PENAL_ANTECEDENTES |
| EC_PENAL_EXPEDIENTE – Path | COLUMN NAME | EC_PENAL_EXPEDIENTE |
| EC_PENAL_EXPEDIENTE – Regex | VALUE FORMAT | EC_PENAL_EXPEDIENTE |